SESTINA v1.0: Constraint-Based Inference and Empirical Evaluation in Six-Player Canadian Fish

Dylan Nguyen

September 21, 2026

Abstract

Six-player Canadian Fish is a decentralised imperfect-information team game in which card transfers and actions are public. We present SESTINA v1.0, an agent developed from the FishBot policy family, together with an exact dynamic program for the uniform distribution over deals satisfying observer-specific game constraints. The deployed agent uses approximate inference with behavioural weighting, a linear action score, public-history tie-breaking, a half-suit contestation feature, a deduction-state stall detector, and guarded determinized search.

Evaluation comprises 428 scored comparison cells over 4,322,400 games under a preregistered protocol. Against the registered baseline, `F-cheap`, SESTINA v1.0 achieves a win-rate edge of +3.33 percentage points relative to parity (95% CI $[+2.88, +3.78]$) over 48,000 games, with positive estimates on both holdout banks. Its comparison with an earlier development composite is unresolved: +0.15 points (95% CI $[-0.29, +0.59]$). Cross-play and rule-variant tests retain positive performance estimates, while the worst partner-substitution difference, −0.19 points (95% CI $[-1.04, +0.67]$), does not resolve the registered −1.00-point threshold. None of eight tested adversarial searches achieves a positive edge against SESTINA v1.0 at the evaluated budgets.

Four candidate modifications yield no demonstrated additional improvement under the tested conditions. Known cross-deal state persistence and the difference between scored and isolation-tested execution modes limit implementation-level conclusions. The results establish an improvement over the specified baseline, but do not establish near-optimality or an upper bound on exploitability.

Contents

1 Introduction . . . 4
1.1 Research questions . . . 4
1.2 Scope of comparative claims . . . 5
1.3 Contributions . . . 5
2 Six-player Canadian Fish . . . 5
2.1 Rules . . . 6
2.2 Evaluated rule variant . . . 6
2.3 Computational and informational structure . . . 7
3 Related work . . . 9
3.1 Prior work on Canadian Fish . . . 9
3.2 Determinization and perfect-information analysis . . . 10
3.3 Search under imperfect information . . . 10
3.4 Policy-aware inference . . . 11
3.5 Team games and common information . . . 12
3.6 Evaluation methodology . . . 12
4 FishBot development background . . . 13
4.1 Engine, inference, and policy foundations (v0.2–v0.4) . . . 13
4.2 Deadlock mitigation (v0.5) . . . 13
4.3 Policy and evaluation revisions (v0.6) . . . 13
4.4 Development objectives . . . 15
5 Exact constraint-conditioned deal inference . . . 15
5.1 Notation . . . 15
5.2 Engine architecture and information access . . . 15
5.3 Observer-specific deal constraints . . . 15
5.4 Half-suit factorisation and count dynamic programming . . . 17
5.5 Approximate inference and behavioural weighting . . . 19
5.6 Implementation verification . . . 19
6 SESTINA v1.0 architecture . . . 20
6.1 Action space and ask policy . . . 21
6.2 Public-history tie-breaking . . . 22
6.3 Half-suit contestation . . . 23
6.4 Declaration policy . . . 24
6.5 Urgency-escalation removal . . . 25
6.6 Deduction-state stall detection . . . 25
6.7 Guarded determinized search . . . 26
6.8 Coordination and information constraints . . . 27
6.9 Decision procedure . . . 28
7 Evaluation framework . . . 29
7.1 Outcome measures . . . 29
7.2 Duplicate deal blocks . . . 29
7.3 Uncertainty estimation . . . 30
7.4 Sample-size planning . . . 30
7.5 Prespecified sign-agreement criterion . . . 31
7.6 Holdout sealing and preregistration . . . 31
7.7 Selection effects and multiple comparisons . . . 32
7.8 Behavioural validity checks . . . 32

7.9 Adversarial-search calibration . . . 32
7.10 Information-access and isolation tests . . . 33
7.11 Worst-case performance and empirical regret . . . 34
8 Adversary model . . . 34
9 Exploratory development . . . 36
9.1 Baseline selection . . . 36
9.2 Detection calibration . . . 36
9.3 Adversary generation against baseline configurations . . . 37
9.4 Candidate-mechanism evaluation . . . 37
9.5 Configuration selection and freezing . . . 40
10 Experimental design . . . 40
10.1 Evaluated configurations . . . 40
10.2 Evaluation data and integrity checks . . . 40
10.3 Behavioural validation results . . . 40
10.4 Preregistered evaluation protocol . . . 42
10.5 Experiment inventory . . . 43
11 Results . . . 43
11.1 Baseline comparisons . . . 44
11.2 Secondary criteria and diagnostic comparisons . . . 45
11.3 Opponent-panel performance . . . 47
11.4 Component ablations . . . 49
11.5 Partner substitution and single-seat upgrades . . . 49
11.6 Cross-play across training runs . . . 51
11.7 Rule-variant sensitivity . . . 51
11.8 Adversarial-search results . . . 52
11.9 Evaluation controls . . . 53
11.10 Computational cost . . . 55
12 Discussion . . . 55
12.1 Observed performance gains . . . 55
12.2 Null results and attribution limits . . . 56
12.3 Scope of the policy-class findings . . . 56
12.4 Limits of adversarial evaluation . . . 57
12.5 External validity . . . 57
12.6 Implementation and evaluation limitations . . . 57
12.7 Future work . . . 58
13 Conclusion . . . 59
References . . . 60
A Rule specification . . . 65
B Configuration grammar and parameters . . . 66
C Preregistered criteria and recorded verdicts . . . 69
D Complete opponent-panel results . . . 71
E Corrections to prior reports . . . 72
E.1 Corrections to earlier FishBot reports . . . 72
E.2 Corrections to development and evaluation records . . . 73
E.3 Cross-version comparability . . . 75
F Protocol deviations and amendments . . . 75
G Reproducibility and artifact availability . . . 79

1 Introduction

Six-player Canadian Fish, also called Literature, is an imperfect-information game played by two teams of three. Players observe their own cards and the public action history, but not other players' hands. Because every card transfer is public, a candidate initial deal and the observed history determine subsequent card ownership. This structure permits exact enumeration of deals satisfying observer-specific game constraints, providing a reference calculation for constraint-based inference.

Strategic action choices can convey information beyond legality constraints. We therefore distinguish the uniform constraint-conditioned distribution from a policy-aware posterior. SESTINA v1.0 uses an approximate belief model with behavioural weighting and combines it with a linear action policy and guarded search. The study evaluates this configuration against specified internal baselines, examines component contributions, and assesses sensitivity to partners, rule variants, and tested adversarial responses.

Computational work on the six-player game appears to be limited. In the literature searches conducted during this project's development cycles we found no academic work on it, no public repository containing search, planning or a working learner for it, and no coverage in RLCard [9] (§3.1); the scope of those searches is stated there. The nearest published agent plays a four-player, 48-card variant that does not raise the six-player allocation problem.

This paper presents **SESTINA v1.0**, a configuration developed from FishBot v0.6, and evaluates it using the protocol described in §7. Earlier versions established the engine, the constraint-based inference machinery and a sequence of progressively stronger policies, but their development also exposed evaluation problems: comparisons that could not resolve a one-point effect, an optimiser that could not move a policy, and at least two published conclusions that turned out to be artefacts of the measurement rather than properties of the agent (§4). The question this paper addresses is whether the tested modifications improve performance under the revised evaluation protocol.

Version nomenclature. SESTINA v1.0 was developed and evaluated under the identifier FishBot v0.7. The released agent retains the frozen configuration; source paths, experiment identifiers, seeds, deal banks, digests and configuration strings retain their original `v07` names for reproducibility, including the frozen configuration `v07:r12=25,...` of §6 and the sealed banks of §7. Throughout, *SESTINA v1.0* names the agent and *the v0.7 cycle* names the programme that produced and measured it. Appendix G gives the detailed mapping.

1.1 Research questions

RQ1 Does SESTINA v1.0 achieve a replicated improvement over `F-cheap`, the registered baseline (§9.1), measured on material sealed before SESTINA v1.0 existed?

RQ2 Does that improvement survive changes of partner, cross-play against independently trained policies, alternative rule variants, and adversarial responses fitted against it?

RQ3 Which of SESTINA v1.0's changes account for the improvement, and which candidate mechanisms fail to produce measurable strength?

RQ4 What do the results imply about the headroom remaining in the current policy class?

§11 answers RQ1 and RQ2 directly, §9 and §11.4 answer RQ3, and §12 takes up RQ4.

1.2 Scope of comparative claims

This paper makes comparative claims of three different kinds, and they are kept separate throughout.

Within-project comparisons. SESTINA v1.0 is compared with its own predecessors and with the *v0.6 frontier*, by which this paper means the set of v0.6-era configurations available at the end of that cycle: the deployed non-searching policy and the searching configurations `F-cheap` and `F-mid`, ordered by search cost (§9.1). The paper establishes an improvement over the registered baseline `F-cheap`. It does not establish that SESTINA v1.0 is the strongest configuration in the lineage on every metric: its comparison with a development-time composite is unresolved, and on empirical minimax regret over a shared panel it ranks third of four (§11.3).

Robustness evaluation. Whether the improvement persists against a fixed opponent panel, under substituted partners, across rule variants, and against adversarial responses fitted to the agent. The paper evaluates these and reports mixed results.

External comparisons. Whether SESTINA v1.0 is the strongest Canadian Fish agent anyone has built. The paper does not establish this and does not claim it. The comparison class contains only agents this project wrote, and every adversarial-response figure here is the observed edge of a particular search rather than a bound on what a search could find.

Accordingly the paper does not describe SESTINA v1.0 as near-optimal, unexploitable, or the strongest known Fish-playing team. §12.7 states what evidence those claims would require.

1.3 Contributions

1. **Constraint-based inference** (§5): a half-suit factorisation and count dynamic program for the normalisation constant, card-location marginals, and declaration-allocation probabilities under the uniform feasible-deal model.
2. **Agent specification** (§6): a reproducible configuration combining approximate inference with behavioural weighting, linear action scoring, public-history tie-breaking, half-suit contestation, stall detection, and guarded search.
3. **Evaluation design** (§7): duplicate-deal comparisons, held-out deal banks, preregistered criteria, uncertainty estimation, adversarial-search calibration, and information-access tests.
4. **Empirical findings** (§11, §9.4): a positive comparison with the registered baseline, an unresolved comparison with the development composite, component ablations, sensitivity analyses with explicitly stated limitations, and four candidate modifications that did not demonstrate an additional improvement under the tested conditions.

2 Six-player Canadian Fish

This section specifies the evaluated game variant and then describes the computational and informational structure that motivates the study. Readers who know the game should still read §2.2, because published descriptions differ on several consequential points and the rule set studied here is one explicit selection among them. Appendix A lists the rule choices against the engine switches that change them.

2.1 Rules

Six players form two teams of three, seated alternately so that no two teammates are adjacent. The deck is 54 cards: the standard 52 plus two jokers. It is partitioned into nine **half-suits** of six cards each, being for each suit the low half 2 through 7 and the high half 9 through A, plus a ninth half-suit made of the four eights and the two jokers. Every card belongs to exactly one half-suit, and each player is dealt nine. The team that takes at least five of the nine half-suits wins.

Play proceeds by **asks**. On their turn a player names one opponent and one card. The ask is legal only if the named card lies in a half-suit that has not yet been claimed, the asker holds at least one *other* card of that half-suit, the asker does *not* hold the named card, and the target is an opponent who still holds at least one card. If the target holds the card they must surrender it and the asker keeps the turn; otherwise the turn passes to the target. The first turn belongs to the seat after a randomly chosen dealer. When the turn reaches a seat that holds no cards, that seat names a teammate who still holds cards to receive it; the engine substitutes the lowest-index such teammate if the named one is ineligible, and if no teammate holds cards the opposing team enters the forced endgame described in §2.2. A seat whose turn it is and that has no legal ask must declare, and does so on the lowest-index unclaimed half-suit in which it holds a card. The requirement that an asker already hold a card of the half-suit makes each legal ask informative about the asker's holdings, because it turns every ask into a public declaration of partial ownership.

A half-suit is claimed by a **declaration**. A declaring player names, for each of the six cards of a half-suit, which member of their *own* team holds it. The claim is all-or-nothing: if every assignment is correct the declaring team scores the half-suit, and if any assignment is wrong the opposing team scores it. A declaration is legal only while the half-suit is unclaimed. In the evaluated variant any seat may declare, in or out of turn and whether or not it holds cards, at any point between two public events; when several seats propose at the same point, the lowest seat index across both teams declares first and the others are polled again once that declaration has resolved. A declaration does not change whose turn it is.

Actions, outcomes, transfers, declarations and hand sizes are public; the identities of the cards in a player's hand remain private unless a transfer or declaration reveals them. Every ask, its answer, every card transfer, every declaration and its outcome, and every player's card count are observed by all six players, so the only hidden information is the initial deal, and the public record determines every later card position given that deal.

2.2 Evaluated rule variant

Published descriptions of this game document mutually incompatible variants [1], so the rule set below is one specified variant among the published rule sets. It matches the strategy corpus this line of work draws on [2] and the rule set adopted for this implementation. Six of the eight contested choices Appendix A lists are engine switches, so their effect can be measured, and §11.7 measures the principal comparison over eight rule-variant rows: the default and seven variations. The misdeclaration award is hardcoded and was not varied; the forced-endgame ladder has no flag of its own inside the scored harness and moves only inside the `--legacy` bundle, which is why §11.7 can estimate it only as a residual. The consequential choices are a 54-card deck in nine half-suits with nine cards per player; declaration legal at any moment, in or out of turn, by a seat holding cards or not; a misdeclaration awarding the half-suit to the opposing team; lowest seat index first among simultaneous declarers; an eight-level willingness ladder in the **forced endgame** that arises when one whole team is cardless; and a 400-ask action cap whose residue is adjudicated by physical majority. In the forced endgame the team that still holds cards must declare every remaining

half-suit: the engine polls that team's seats down a ladder of correctness-probability thresholds, 0.995, 0.98, 0.95, 0.90, 0.80, 0.65 and 0.50, followed by a compulsory level at which the first polled seat declares its best guess; at each level every unclaimed half-suit and every seat of the team is polled in index order, the first willing seat declares, and the sweep restarts from the top level for the next half-suit. At the action cap each unclaimed half-suit is awarded to the team physically holding more of its six cards, and a 3–3 split is awarded to the team holding the lowest-index card of the half-suit; every half-suit is awarded whole, so a nine-half-suit game cannot tie. In the 48-card, eight-half-suit variant of §11.7 each player is dealt eight cards and a 4–4 game is recorded by the harness as a win for the team seated at index 0; because every deal is played once in each seating, this is symmetric in expectation across the two configurations, but a tie is not scored as a half win. No prearranged signalling conventions are available to any policy, and team communication is limited to the forced-endgame willingness bit and to the successor a cardless turn-holder names. Appendix A lists each choice against the switch that changes it.

Two rules are places where the engine is deliberately *more* restrictive than the adopted rule set allows, and both close coordination channels a policy could otherwise exploit. We evaluated the two restrictions using the interventions described below. When the turn reaches a cardless seat whose teammates both hold cards, the rules permit open negotiation over who receives it; the engine instead has the cardless seat choose unilaterally. A genuine multi-candidate transfer decision arises 0.148 times per game. Replacing the unilateral rule on one team with a chooser that, given the true deal, hands the turn to the teammate with the most immediately certain pulls, paired on common deals, changed the mean half-suit score by -0.0007 ± 0.0024 half-suits per game (mean and 95% half-width) over 6,000 paired v0.4 mirror games, in which the two rules diverged in 0.48% of games. The tested intervention therefore produced no resolved change in mean half-suit score. This is an oracle-assisted diagnostic rather than a bound on every legal mechanism: the clairvoyant chooser is a greedy heuristic that does not optimise game value, so the interval prices the channel for that class of chooser, and it was measured on an earlier policy rather than on SESTINA v1.0. Similarly, the engine arbitrates simultaneous declarations by lowest seat index rather than by stated confidence, because a confidence comparison would move a statistic of one seat's hand to five others. That restriction costs +0.37 pp of win rate pooled over 30,000 games on five seeds, and an arbitrator that, given the true deal, selects a correct declaration from those proposed is worth +0.35 pp pooled over three of those seeds. Both figures bound the value of choosing among a fixed set of proposals; neither bounds a policy that changes what it proposes. A half-suit differential and a win-rate edge are different units and are not converted into one another here.

2.3 Computational and informational structure

Hidden-state representation. Because a card changes hands only through an observed ask, the only thing any player does not know is the initial deal, and the public record determines every subsequent card position given that deal. In many hidden-information games the object that must be filtered is the set of reachable *histories*, for which constructing even one consistent history is FNP-complete in the worst case, with efficient enumeration available exactly when the public tree is sparse [23]. The representation used here reduces hidden-state reconstruction to constraints on the initial deal: there is exactly one object to reconstruct, and the constraints the rules impose on it are logical rather than probabilistic. A player told "I do not hold the seven of hearts" knows it, permanently and with certainty. Throughout, a *certificate* is such a hard constraint on card location derived from a public event.

The research consequence is that Fish permits the constraint-based part of inference to be *separated* from the policy. Exact enumeration of the deals consistent with the certificates (§5)

provides a reference for the constraint-based inference computation. It does not remove uncertainty about the behavioural model used to interpret strategic actions: a strategically chosen ask can make some feasible deals more likely than others, and that weighting is a modelling choice the reference calculation does not make (§5.5). The separation isolates, for this paper's negative results, the uncertainty that lies in the policy and the behavioural model rather than in the certificate arithmetic. It also motivates the belief-model comparison reported in §4.3, in which the exact constraint-based model is a worse predictor of card location than the deployed approximation with behavioural weighting.

Decentralised team decision-making. Three teammates share a payoff, hold different private information, and have no communication channel beyond their public actions. This is the decentralised control problem the common-information approach formalises, in which a notional coordinator who knows the common information issues *prescriptions* mapping each controller's private information to an action [52]. In Fish the common information is the entire public history, so the coordinator's information state is exactly the belief object the engine already maintains. The coordinator's problem is not solvable here, since a prescription maps every possible nine-card hand to an ask, so the reduction serves as a correctness lens rather than an algorithm. What it does yield cheaply is a property the cooperative-search literature has to work for [28]: the mapping implemented by a *deterministic* procedure reading only public state and its own hand is common knowledge among the three teammates, with no shared seed and no communication, although the action it produces at another seat is not, since that seat's hand is private. The determinism qualifier matters, and §6.2 is where it becomes an engine switch.

Public policy specification. In the extensive-form team-game models this paper cites, a team coordinates through an ex-ante correlation device: a joint strategy is sampled before play, the opponent may know the distribution it is sampled from, and the realised sample is private to the team [53, 54, 55]. A team of three identical, deterministic, published agents differs from that model in two respects: the policy mapping is public, and there is no private sampled signal. Knowledge of the policy does not reveal an action determined by a private hand, but it lets an opponent compute the distribution of a seat's action from the public record (§6.8). §8 adopts that as the threat model, and §11.8 reports what an adversary that reads the target's policy and inverts its transcript achieved. In the literature and repositories we reviewed we found no prior work covering this case.

Computational cost. A deal resolves in roughly a hundred public events, and on one fifteen-thread desktop the engine plays a six-seat mirror of the *blueprint* policy, the non-searching base policy that the test-time search of §6.7 modifies, at 326 complete games per second, with the deployed belief running in every seat; the evaluated configuration, which searches, shows an observed whole-match slowdown of 4.52× relative to the non-searching blueprint (§11.10). This is what makes the evaluation programme of §7 possible: 48,000 paired games for a single headline comparison, two disjoint holdout banks for every claim, a 31-member opponent panel scored identically for four configurations, and eight independent adversarial searches all run on one desktop. In a domain where one evaluation cell costs GPU-hours, the version of this study that reports its own negative results could not have been afforded. The corresponding liability appears in §12.5: low simulation cost does not provide independent validation.

Planning limitations. Two limitations bear on how later results should be read. First, the perfect-information version of this game offers little to a clairvoyant planner. A player who sees every hand names only cards the target holds, so never fails an ask, never loses the turn, and declares each half-suit as soon as their team holds all of it; under such play the value of a position is determined by the current card distribution, so a clairvoyant rollout distinguishes little between the asks available at a decision. We have not characterised this formally, and the claim is about the evaluated variant under clairvoyant play rather than a theorem about the game. Determinization, the standard recipe for hidden-hand card games [11, 12], therefore reduces here largely to choosing the ask most likely to hit, blind to the information an ask leaks, the information a refusal supplies, and the option value of postponing a declaration. §3.2 places this against the literature. Second, Fish has no passive continuation: a player must ask or declare, so there is no equivalent of the check-and-call rollout on which the standard cheap exploitability probe is built [38]. §12.7 names this as the largest single obstacle between this work and a defensible near-optimality claim. Second, a deterministic procedure reading only the public state and its own hand can be evaluated by a teammate on any hypothesised hand; §6.8 states the conditions under which that supports coordination without communication.

3 Related work

We review computational work on Canadian Fish and related methods for hidden-state inference, imperfect-information search, team coordination, and agent evaluation. For each group, we distinguish methods used in this study from alternatives requiring assumptions or computational resources not adopted here.

3.1 Prior work on Canadian Fish

The published material on this game is human and informal, and neither of its two substantial sources is quantitative or describes a program. McLeod's Pagat entry [1] is the canonical rules source and records several points on which published variants differ; ours is specified in §2.2. Develin's chapter [2] is the most detailed strategy writing we located, describes the same variant, and supplies two observations this design uses: an ask teaches the opponents as much as it teaches a teammate, and a player is better off not declaring a half-suit it holds entirely, since holding one carries no risk. The second is a tactic and not a rule: the engine imposes no such condition, and §2.1 states what it does impose. Secondary strategy pages [3, 4, 5] restate blackballing, asking the asker, and an endgame delegation heuristic; they are evidence about how the game is played and talked about, not sources of quantitative claims.

Computational prior work. Literature searches conducted during this project's development cycles, covering academic indexes, public code repositories and the RLCard toolkit [9], found no academic work on the six-player game and no public repository containing search, planning or a working learner for it. The one agent located, Somani's `literature` [6, 7], plays the four-player, 48-card variant, which does not raise the six-player allocation problem, and publishes no strength numbers; a closed-source mobile application advertises bots with no stated methodology [8]. Every novelty statement in this paper therefore takes the form "we found no prior published work that ..." and is a statement about those searches rather than a demonstrated absence. The Valet testbed [10] packages twenty-one traditional imperfect-information card games in a common description language with measured branching factors and durations; a shared testbed of that kind would

supply the external calibration this line of work lacks, but no comparison against it is attempted here.

3.2 Determinization and perfect-information analysis

Determinization, or perfect-information Monte Carlo (sample hidden cards consistently with the public history, solve the resulting perfect-information game, and play the move with the best average value), is the standard recipe for hidden-hand card games [11, 12], and is unsound however many worlds are drawn. Frank and Basin named the two mechanisms: *strategy fusion*, in which the searcher plays a different strategy in each sampled world although a real player must commit to one strategy per information set, and *non-locality*, in which a node's value depends on parts of the tree outside its own subtree [13, 14]. Long, Sturtevant, Buro and Furtak characterise when the method nevertheless performs well, through leaf correlation, bias and a disambiguation factor [15]; Information Set Monte Carlo Tree Search replaces independent trees with a single tree over information sets [16, 17]; the $\alpha\mu$ family backs up Pareto fronts of world-vectors instead of scalar averages, repairing both mechanisms, and reports Bridge 3NT declarer play at 60.2% to 63.0% with twenty worlds [18]; and knowledge-based paranoia search reasons about the worst case over deduced-consistent worlds rather than the average, reporting over 1,000 points on the extended Seeger scale in Skat [19].

The evaluated variant presents an additional limitation for clairvoyant rollout evaluation, stated in §2.3: under clairvoyant play the available asks are scarcely distinguished in value, so repairing strategy fusion and non-locality, as $\alpha\mu$ does, would not by itself make a clairvoyant rollout informative here. We have not tested $\alpha\mu$ or its relatives on this game, and the statement is about the evaluated variant rather than about those methods.

What this corpus adds is that the binding problem is not the abstract failure but *the statistic*. One determinization's value is the final half-suit differential, whose spread is an order of magnitude larger than the differences between the leading candidates, so an argmax over sampled means selects on residual noise, and does so more confidently the more candidates it is offered. That is the optimiser's curse rather than strategy fusion, and §4.3 reports it measured at 35.69 pp by a control that forces the same code to return the blueprint's own pick. These experiments motivate the guarded action-selection rule in §6.7.

3.3 Search under imperfect information

States in an imperfect-information game do not have well-defined values, so depth-limited search needs something at the leaf that represents the opponent's ability to choose. Brown, Sandholm and Amos supply it: let the opponent select among k continuation strategies at the depth limit, each yielding its own leaf-value vector, and require the searcher to be good against that choice [24]. The soundness argument is stated for two-player zero-sum games, and Fish's team-level game presents three uncorrelated opponent seats at the leaf, so nothing is inherited; it is adopted here as a robustification heuristic with a measurable consequence, and audited rather than assumed.

Two lines address what a search may prune. Zhang and Sandholm solve subgames over low-order knowledge rather than the full common-knowledge closure, and prove that the naive form can *increase* exploitability [25]. That warning applies directly to the live-ask gate this policy family ships, which is a knowledge-limited pruning rule, and it is why the side-channel and pathology gates of §7.8 were built before any search work rather than after it. Subgame solving in adversarial

team games builds a gadget over the team belief DAG and runs in polynomial time given a best-response oracle under a sparsity condition on the blueprint [26]; the cited sparsity condition is not established for the blueprint used here, and we did not attempt the construction.

In cooperative games the constraint transfers verbatim. SPARTA improves a blueprint purely at test time and distinguishes single-agent search, which assumes every other agent strictly follows the blueprint, from multi-agent search, in which all agents run the *same* common-knowledge procedure [28]. The moment a second teammate also searches, the first teammate's belief over its partner is computed under a policy that partner is no longer playing, and the improvement guarantee is void. Fish satisfies the requirement cheaply, for the reason §2.3 gives; the qualifier about randomisation is what §4.3 turns into an engine switch.

The full-solver line [37, 35, 36] established that test-time re-solving over public belief states is supported by the formal models and assumptions in those studies. We do not implement these algorithms, because each requires iterating or learning over a closure whose cardinality at deal time is $45!/(9!)^5 \approx 1.9 \times 10^{28}$ deals from one seat. Update equivalence recasts search as replicating a last-iterate-convergent update locally, needing no public-information enumeration, and reports substantially lower search time than public-belief-state search in Hanabi [27]; its improvement guarantee is stated for common-payoff games, which Fish is not. Learned Belief Search [29] learns a belief model for settings in which the exact belief is unavailable; we do not use it because the constraint-conditioned belief is computed exactly here and the behavioural weighting is deliberately simple, not because a learned belief could not add information. Online planning by policy fine-tuning [30] requires gradient updates at decision time and was not attempted in this CPU-only engine with no network. The depth-limited counter-strategy line [31, 32] carries one warning this study honours, robust-response methods do not compose with limited-look-ahead solving off the shelf, and the learned-model and gadget-refinement work [33, 34] is recorded for its meta-lesson rather than its machinery: when a search subproblem has many near-equivalent optima, which one is selected is not a tie-break. That is the same observation §4.3 makes about an argmax resolved by array order.

The result that a search of this kind can *increase* exploitability is Zhang and Sandholm's [25]. Where the other work above carries a guarantee at all, its conditions and this game's fit to them are recorded above, method by method. What none of them supports is reading a win rate as a bound on exploitability, which is why this paper measures the two separately. Local best response gives an approximate lower bound on best-response value: a small greedy search looking one action ahead and assuming checks thereafter showed each of the 2016 competition entrants it evaluated to be more than 3,180 mBB/h exploitable [38], the competition's winning submission was not among them, and the authors say why. Its two documented pitfalls are carried into §11.8: restricting where the response may begin deviating understates exploitability, and a negative score proves nothing. Accordingly every adversarial-response figure in this paper is reported as the observed win-rate edge of a particular response within its class, which is a lower bound on the best-response value over that class only up to sampling error, and never as an upper bound.

3.4 Policy-aware inference

Skat engines enumerate the deals consistent with the public record and reweight them by a model of what each player would have done, a supervised per-card location model [21], or reach probabilities taken from a policy [22], and a dedicated inference module was reported worth +217 tournament points per 36 games to the engine Kermit on its own [20]. The policy prior of §5.5 is the same idea in a much simpler form: a fitted soft reweighting of the rules posterior, with two parameters and no learned opponent model.

This is the literature that has measured the same trade-off, and it is where §4.3's belief-model comparison belongs. The finding there is that in this game the constraint-conditioned distribution is computable exactly by a tractable factorised dynamic program and is cheap, so in this implementation computational cost is not the only motivation for approximation, and yet substituting the exact constraint-conditioned distribution into an otherwise identical policy loses to the approximation with behavioural weighting, on two independent seed banks.

Two adjacent lines are named. Belief modelling in cooperative games, the Bayesian Action Decoder's public-belief formulation [40] and its factorised-prescription successor [44], is aimed at games whose exact belief is intractable, which is not the situation here. The permanent-like counts that a constraint system of this shape generally induces are the subject of exact formulae, Ryser's inclusion–exclusion formula [58] and Glynn's sign-vector formula [59], and of approximation algorithms, the Jerrum–Sinclair–Vigoda chain [60], matrix scaling [61] and sequential importance sampling [62]; the block structure of §5 makes the exact count tractable directly, so none of the five is needed for it. Matrix scaling appears in the deployed path for a different reason: the approximate belief replaces the exact count join with a Sinkhorn iteration of that kind, not because the exact count is unaffordable but to incorporate the two behavioural weighting parameters (§5.5). No learned belief model appears anywhere in this corpus; that is a stated non-goal of the project.

3.5 Team games and common information

The formal statement of three teammates sharing a payoff, holding different information and unable to communicate is the common-information approach [52], discussed in §2.3. The team-game equilibrium literature is cited to delimit scope: the target concept for a Fish team is a team-maxmin equilibrium with correlation [53], connected to extensive-form correlation by Farina and coauthors [54] and made computable through the team belief DAG and the two-player conversion [55, 56], with subsequent work on hidden-role settings [57]. None of it is computed here. We have not established whether a Fish team satisfies the recall conditions those constructions require; the evident obstacle is that each seat privately observes nine cards its teammates do not.

The cooperative-play line is cited for its evaluation discipline rather than its algorithms. Hanabi was proposed as a benchmark precisely because self-play scores overstate coordination [39], and the ad-hoc-teamplay results that followed [41, 42, 43, 45, 48, 49, 51] are the reason the cross-play measurement of §11.6 exists at all: a convention that collapses when partners are swapped is brittleness, and it is invisible to self-play evaluation. Two neighbouring results are cited for what they show about porting rather than as ad-hoc teamplay: joint policy search is the bridge-bidding analogue, scoring simultaneous changes at several information sets [47], and the Hanabi construction that most clearly fails to port is the hat-guessing code [46], which works because each receiver sees every other receiver's hand, and in Fish no player sees any other player's hand. The same lesson from a different direction is Cicero's use of a human-regularised policy [50].

3.6 Evaluation methodology

The evaluation framework of §7 draws on duplicate-match and variance-reduction methods. Every cell in this paper is a paired duplicate rather than an independent sample, on the model of duplicate bridge and of the Annual Computer Poker Competition's duplicate matches and common seeds [63], whose protocols are also the model for a fixed opponent panel scored identically for every arm. The estimator-side alternatives, DIVAT [64] and AIVAT [65], subtract a learned or hand-built value function from a single sample instead; neither is used here, and the pairing is what makes the design competitive with them without introducing a fitted object. Abstraction pathology [66] is

the standing warning that a model can be improved on its own metric and made worse in play; §9.4.2 illustrates a related mismatch between a surrogate metric and playing performance. Re-evaluation of agent populations [67] and the rating literature [68, 69, 70, 71] inform why this paper reports a worst case and a minimax regret over a shared panel rather than a single rating number. Recent work on gold-standard evaluation practice [72] is consistent with the preregistration and sealing discipline of §7.6, which this project adopted independently.

4 FishBot development background

SESTINA v1.0 is the seventh configuration in a lineage, and several of its components and most of its evaluation machinery exist because of what earlier versions found. This section is deliberately short. Its purpose is to explain why SESTINA v1.0 looks the way it does, not to re-report earlier studies; the detailed corrections those studies require are collected in Appendix E.

4.1 Engine, inference, and policy foundations (v0.2–v0.4)

The first versions established the domain and the inference. A count-conditioned, empirically tuned linear policy in a scripting-language simulator became a C++ engine implementing the specified rules and carrying an exact constraint-conditioned reference distribution, a faster deployed approximation, a twenty-feature ask score and a value-based declaration rule. The durable contribution of that period is the inference object of §5: an exactly computable observer-conditioned constraint distribution over deals, cheap enough to run in all six seats at once.

Earlier evaluations used smaller samples, unpaired ablations, and point-estimate comparisons: cells were a few thousand games against a handful of opponents.

4.2 Deadlock mitigation (v0.5)

FishBot v0.5 addressed a mirror-play deadlock in which two agents repeatedly requested unavailable cards until a forced declaration ended the sequence. Removing this behaviour did not produce a statistically resolved win-rate improvement in the reported evaluation. The result motivated separate checks of behavioural validity and playing strength, as well as reassessment of the evaluation's precision.

4.3 Policy and evaluation revisions (v0.6)

The v0.6 cycle set out to recover several points of headroom that the accumulated design register described. It found little of it, and found in the process that much of the measurement machinery lacked the precision its conclusions required. Four results from that cycle bear directly on SESTINA v1.0.

Tied action scores. Measured in the v0.5 mirror, the policy that cycle set out to diagnose, 54.74% of *contested* ask decisions end with two or more candidates scoring bit-for-bit identically, and the winner is whichever the enumerator emitted first. 93.77% of those ties are two cards of one half-suit at one target, and the exact posterior *separates* them in 0.00% of cases, meaning it assigns them equal probability in every case. The same measurement on the v0.6 mirror gives 53.80%, and that is the figure §6.2 and §9.4.4 use, because SESTINA v1.0 modifies v0.6 rather than v0.5. Every rule that ranks tied candidates by that probability therefore realises the same hit rate, and so does enumeration order. The largest item on the inherited design register was not a defect but

a property of the information set. What remains addressable is the dependence on enumeration order itself, which is a correctness problem, and §6.2 is how SESTINA v1.0 addresses it.

Belief-model comparison. Scored as predictors of card location on identical states in the v0.5 mirror, the ranking is:

Posterior	Log loss (nats)	Argmax hit rate
Exact, uniform deal prior	1.42246	47.94%
Approximation, certificate half of the prior deleted ($\theta = 0$)	1.39083	49.99%
v0.5's deployed approximation	**1.38218**	**51.49%**

The exact constraint-based model has the highest log loss among the compared models, and it remains highest with the behavioural weighting deleted outright as well ($\theta = \varphi = 0$: 1.39339 nats and 49.14%), so most of the log-loss gap has no behavioural content, though most of the hit-rate gap does. The certificate half of the weighting is the gap between the second and third rows; the weighting as a whole is the gap from the $\theta = \varphi = 0$ figures to the third. In the implementation, the block construction of §5 has no parameter through which an opponent's tendencies could enter; the behavioural weighting enters only through the deployed paths, as the initialiser of the Sinkhorn fit (§5.5) and, in the search, as an importance weight on exact samples (§6.7). Whether an exact constraint computation could be combined with a behavioural model in one object is not addressed here. This closed an experiment three studies had listed as outstanding, and it established a methodological rule the project has applied since, namely that exact constraint-based inference is an assumption to be tested rather than a property to be assumed. §9.4.1 reports the second comparison of that kind.

Search-selection bias. v0.6 built a search that samples full deals from the public posterior and plays each forward with all six players reconstructed at their own information sets, rather than made clairvoyant, which §2.3 explains would be degenerate. Chosen by an unguarded argmax over sampled means, that search is 35.69 pp *worse* than the blueprint it searches from, measured against a control that forces the same code to return the blueprint's own choice at identical budget, seed and sample size. The cause is the statistic rather than the search: one determinization's return is the final half-suit differential, whose spread is an order of magnitude larger than genuine differences between candidates, so the argmax selects on residual noise and does so more confidently the more candidates it is offered. A paired lower-confidence-bound deviation rule removed the measured deficit in that experiment. The guarded search, which v0.6 measured and then deployed switched off, is a component of SESTINA v1.0 (§6.7).

Evaluation resolution and optimiser design. Pooling every same-pair cell in the v0.6 battery gave 50.53% with a naive z of 1.13, a null that an earlier draft of that report published as one. Re-run at high power across 7 disjoint banks, the same comparison is 50.89% over 126,000 games $[50.61, 51.16]$, above parity on all 7 banks. The effect had always been present; the instrument could not see it. In the same cycle, five mechanisms cleared a 95% paired interval at the per-cell budget the project had used since v0.3 and returned exactly zero at three times that budget. Separately, the fitting harness was found to be incapable of moving a policy at all, using one step size for thirty-four coordinates spanning two and a half orders of magnitude, an unpaired objective, and an objective documented as a worst case that was in fact a weighted mean.

4.4 Development objectives

The v0.6 cycle left a policy roughly nine tenths of a point stronger than its predecessor, a design register most of whose entries had not produced a resolved gain, a repaired optimiser, and the open question of whether, at the budget the project had used, the absence of resolved gains reflected the policy class or the precision of the instrument.

The v0.6 results motivated five development stages: evaluation calibration, adversary generation, candidate-mechanism testing, configuration selection and freezing, and holdout evaluation under a preregistered protocol. §9 and §11 report the exploratory and holdout stages, respectively. The protocol separated configuration selection from final evaluation to reduce selection-related bias; the organisational details, including which briefs, software processes and information restrictions separated the stages, are in Appendix G and the protocol document it cites.

5 Exact constraint-conditioned deal inference

Every agent in this lineage rests on an inference object that this section defines formally. The method uses exact enumeration through a factorised dynamic program. The object it computes is the uniform distribution over deals satisfying the acting player's hand information and the hard constraints induced by the public history; §5.5 distinguishes that object from a policy-aware posterior and describes the approximation the agent deploys.

5.1 Notation

Table 1 fixes the symbols used here and in §6.

5.2 Engine architecture and information access

The engine is a single-header C++20 core. A hand is one 64-bit word over the 54 card indices $c = 6h + i$ for half-suit h and offset i, so hand arithmetic is bit arithmetic. Random streams are counter-based and keyed by the triple (seed, seat, purpose). This gives the property the duplicate-block design of §7.2 requires: a deal is reproducible from its seed, and the play of a deal between deterministic agents reproduces from the seed and the agents' configurations. Complete reproducibility of a scored run is a stronger property, and §6.8 and Appendix E.2 record where searching agents fall short of it.

An acting policy never receives the game state. It receives its own knowledge object, its own hand and the public event stream. There is no reference from a policy to the deal, to another seat's hand or to the referee. This is an architectural barrier rather than a convention, and it is what allows the side-channel audit of §7.10 to be read as a test of the policy: a policy that cannot reach the deal cannot read it, so any decision the audit fails to reproduce from public information indicates state carried across deals rather than information taken from the referee.

5.3 Observer-specific deal constraints

Because a card changes hands only through an observed ask, the public record determines $\mathrm{own}_t(\cdot)$ given D. The only hidden object is therefore D itself. Write $\mathcal{D}$ for the set of all deals; from one seat's viewpoint at deal time $|\mathcal{D}| = 45!/(9!)^5 \approx 1.9 \times 10^{28}$.

Each public event imposes a *hard* constraint, meaning a logical restriction on $\mathcal{D}$ rather than a probabilistic tilt. The engine maintains them as a per-card candidate set $\mathcal{M}_t(c)$, initialised to all six seats. For every card that remains unresolved the set only shrinks, $\mathcal{M}_{t+1}(c) \subseteq \mathcal{M}_t(c)$; a

Table 1: Notation used in §5 and §6.

Symbol	Meaning
The game and the hidden state	
$N = 6$, $M = 9$, $\|H\| = 6$	seats, half-suits, cards per half-suit; the deck is $M\|H\| = 54$ cards
s	the acting seat; $\mathcal{T}_s$ its two teammates, $\mathcal{O}_s$ its three opponents
D	the hidden state: the initial deal, an assignment of all 54 cards to seats, nine each
$\mathcal{I}_s^t$	seat s's information set after t public events
$\mathcal{C}_s^t$	the hard constraints (C1)–(C5) seat s has accumulated after t events
$\mathcal{P}_t$	the running hash of the public event stream
$\mathrm{own}_t(c)$, $H(c)$	the seat holding card c; the half-suit containing c
$\mathcal{H}_t$	the half-suits not yet claimed
Beliefs	
$\mathcal{M}_t(c)$	the candidate set of card c: seats not yet excluded by a certificate
U_t, n_p	the unresolved cards; the unresolved cards seat p holds
B_h, $\mathcal{F}_h$	the block of half-suit h; its assignments satisfying every certificate
$\sigma(\alpha)$, $T_h(\mathbf{v})$	an assignment's per-seat count signature; the block count table
F_k, G_k	the forward and backward partial convolutions of the count join
$q_s^t(D)$, Z	the constraint-conditioned distribution of (1); the number of consistent deals
$\mu(c, p)$	the exact marginal of q_s^t: the probability that seat p holds card c
$\hat{\mu}(c, p)$	the deployed approximation to $\mu(c, p)$, with behavioural weighting
θ, φ	the two behavioural-weighting parameters of (8)
Acting	
$\mathcal{A}_t$, $a = (c, q)$	the legal ask set; an ask naming card c and opponent q
p_a	$\hat{\mu}(c, q)$, the modelled hit probability of ask a
$\phi(a)$, w, $u(a)$	the ask feature vector, its fitted weights, and the resulting score
λ, β	the linear and value scale parameters of (11)
M_H, q_H	opponent-held mass and residual ambiguity in half-suit H, in (18)
$\mathcal{E}_t$, ε	the bit-for-bit tied candidate set and its tolerance
π_{alloc}, π_{team}	under q_s^t: the probability a named allocation is exactly correct; that the team owns a half-suit outright
$\hat{\pi}_{\mathrm{alloc}}$, $\hat{\pi}_{\mathrm{team}}$	the deployed approximations to those two quantities (§6.4)
δ, τ	the declaration value margin and the forced-endgame threshold
χ_t, ν_t, K	the deduction-state digest, the no-progress counter, its threshold
Φ_t	the potential function of the termination argument in §6.6
D_{det}, w_d	the number of sampled worlds and their importance weights
κ, η	the search's deviation threshold and its blueprint blend, $\eta = 0$ here

card whose holder becomes public through a transfer leaves the unresolved set U_t with its holder recorded, so the monotonicity statement is about unresolved cards:

(C1) Own hand. For each c in seat s's hand, $\mathcal{M}(c) \leftarrow \{s\}$; for every other card, s is removed from $\mathcal{M}(c)$.

(C2) Card counts. Every seat's current hand size is public, so with the transfer record each seat's count of unresolved cards, n_p, is known. This is the only constraint that couples different half-suits.

(C3) Transfers. A successful ask makes own(c) public from then on, so $\mathcal{M}(c)$ collapses to a singleton.

(C4) Failed asks. If q was asked for c at time r and did not hold it, then $q \notin \mathcal{M}(c)$ unless q has acquired c since, which is public.

(C5) Ask legality. An asker of c held another card of $H(c)$ at that moment and did not hold c. The first is a disjunction over the other five cards of the half-suit and is the one constraint that is not a simple exclusion.

Write $\mathcal{C}_s^t$ for the constraints seat s has accumulated. The object this section computes is the uniform distribution over the deals that satisfy them:

$$q_s^t(D) \;=\; \frac{\mathbb{1}\big[D \models \mathcal{C}_s^t\big]}{Z}, \qquad Z \;=\; \big|\{D' \in \mathcal{D} : D' \models \mathcal{C}_s^t\}\big|. \tag{1}$$

This distribution is exact for the stated constraint model. It is not, in general, the posterior $\Pr[D \mid \mathcal{I}_s^t]$: a uniform prior over initial deals does not imply a uniform posterior after strategically chosen actions, because different feasible deals can make the observed action sequence differently likely under the players' policies. A policy-aware posterior would weight each feasible deal by the likelihood of the observed actions under a specified behavioural model, which is the computation the transcript-inversion adversary of §11.8.1 approximates and (25) states. The modelling content of (1) is entirely in which constraints are imposed; nothing in it refers to how anyone plays. §5.5 describes the behavioural weighting the deployed approximation adds.

5.4 Half-suit factorisation and count dynamic programming

Computing Z by enumerating $\mathcal{D}$ is infeasible, and a constraint system of this shape generally induces a permanent-like count. For such counts the literature supplies exact formulae that are exponential in the general case, inclusion–exclusion [58] and the sign-vector formula [59], and approximation algorithms, Markov chains [60], matrix scaling [61] and sequential importance sampling [62].

Fish avoids that, because every constraint except (C2) is confined to a single half-suit. Constraints (C3), (C4) and (C5) all speak about one card or one half-suit; only the per-seat capacities n_p couple half-suits, and they couple them only through counts.

Partition the unresolved cards by half-suit, $U_t = \bigcup_h B_h$, where B_h is the *block* of half-suit h. For a block, let an *assignment* be a map $\alpha : B_h \to \{0, \dots, 5\}$ with $\alpha(c) \in \mathcal{M}(c)$, and let $\mathcal{F}_h$ be the assignments that additionally satisfy every (C5) disjunction touching h. Write the *count signature* of an assignment as the vector

$$\sigma(\alpha) \;=\; \big(|\alpha^{-1}(0)|,\; \dots,\; |\alpha^{-1}(5)|\big) \in \mathbb{Z}_{\geq 0}^{6}.$$

Block enumeration. For each half-suit independently, enumerate $\mathcal{F}_h$ and accumulate a table indexed by count signature,

$$T_h(\mathbf{v}) \;=\; \big|\{\alpha \in \mathcal{F}_h : \sigma(\alpha) = \mathbf{v}\}\big|, \qquad T_h^{c \to p}(\mathbf{v}) \;=\; \big|\{\alpha \in \mathcal{F}_h : \sigma(\alpha) = \mathbf{v},\, \alpha(c) = p\}\big|. \tag{2}$$

A half-suit holds six cards and resolved ones are removed, so $|B_h| \leq 6$ and the enumeration is small.

Count convolution. Constraint (C2) says the signatures must sum to the vector of per-seat capacities $\mathbf{n} = (n_0, \dots, n_5)$. Order the blocks arbitrarily as $h_1, \dots, h_M$ and define the partial convolution. A half-suit carrying no live (C5) disjunction factorises further into one independent

unit per card, so the implementation runs one layer per unit rather than one per half-suit; the recursion below is the same object either way.

$$F_0(\mathbf{v}) = \mathbb{1}[\mathbf{v} = \mathbf{0}], \qquad F_k(\mathbf{v}) \;=\; \sum_{\mathbf{u}} F_{k-1}(\mathbf{v} - \mathbf{u})\, T_{h_k}(\mathbf{u}), \tag{3}$$

with every table taken as zero outside the box $\mathbf{0} \le \mathbf{v} \le \mathbf{n}$ (componentwise), so that

$$Z \;=\; F_M(\mathbf{n}). \tag{4}$$

By induction on k, $F_k(\mathbf{v})$ counts the joint assignments of the first k blocks that satisfy their own certificates and have total signature $\mathbf{v}$; since the blocks partition U_t and (C2) is the only constraint that couples them, $F_M(\mathbf{n})$ counts exactly the assignments satisfying (C1)–(C5), which is Z. Running the same recursion backwards, with $G_{M+1}(\mathbf{v}) = \mathbb{1}[\mathbf{v} = \mathbf{0}]$ and $G_k(\mathbf{v}) = \sum_{\mathbf{u}} T_{h_k}(\mathbf{u})\, G_{k+1}(\mathbf{v}-\mathbf{u})$, gives suffix tables G_k, and the marginals follow by substituting one block's card-resolved table for its plain one:

$$\mu(c, p) \;=\; \frac{1}{Z} \sum_{\mathbf{u},\,\mathbf{v}} F_{k-1}(\mathbf{v})\; T_{h_k}^{c \to p}(\mathbf{u})\; G_{k+1}(\mathbf{n} - \mathbf{v} - \mathbf{u}), \qquad c \in B_{h_k}. \tag{5}$$

Three quantities the policy needs are *defined* by this machinery, and §5.5 records that the deployed configuration substitutes a cheaper approximation for each of them. The per-card marginals are (5). The probability under q_s^t that the acting team owns half-suit h outright is the total weight of the block assignments whose image lies in the team,

$$\pi_{\text{team}}(h) \;=\; \frac{1}{Z} \sum_{\substack{\alpha \in \mathcal{F}_h \\ \alpha(B_h) \subseteq \mathcal{T}_s \cup \{s\}}} \; \sum_{\mathbf{v}} F_{k-1}(\mathbf{v})\; G_{k+1}(\mathbf{n} - \mathbf{v} - \sigma(\alpha)), \qquad h = h_k. \tag{6}$$

And the probability of a fully named allocation, which is exactly what a declaration asserts, is

$$\pi_{\text{alloc}}(h, \alpha^\star) \;=\; \frac{\mathbb{1}[\alpha^\star \in \mathcal{F}_h]}{Z} \sum_{\mathbf{u},\mathbf{v}} F_{k-1}(\mathbf{v})\, \mathbb{1}[\sigma(\alpha^\star) = \mathbf{u}]\, G_{k+1}(\mathbf{n} - \mathbf{v} - \mathbf{u}), \tag{7}$$

for the single assignment $\alpha^\star$ the declaration names. The leading indicator is not cosmetic: an assignment with a surviving count signature need not itself satisfy the half-suit's certificates, so feasibility is checked explicitly. No sampling is involved in any of the three.

Cost. The signature space is bounded by the capacities, each block table has at most $6^{|B_h|}$ entries before pruning, and the join is M convolutions over that space. An exact rebuild was timed at a mean of 0.4 ms in the v0.6 cycle's measurement, taken over the exact-program builds at contested ask decisions in 150 mirror self-play games on the desktop described in Appendix G; the cost is strongly state-dependent and the figure is a mixed-regime mean. The counts are exact as a formulation; the implementation evaluates the block tables in single precision and the convolutions, Z and the marginals in double precision with no rescaling, so the values are floating-point evaluations of exact integer quantities, and Z at deal time exceeds the range in which a double represents integers exactly. The verified calculation is the constraint-conditioned count and its marginals; it is not a policy-aware posterior, and the project's decision to spend its cycles on policy rests on the former being cheap rather than on inference being solved in the latter sense.

5.5 Approximate inference and behavioural weighting

Equation (1) is exact *for the constraint model*: it is the uniform distribution over constraint-consistent deals. It contains no model of behaviour, because the construction above consumes the deduction state alone and has no parameter through which an opponent's tendencies could enter.

The path the agent deploys adds two such parameters and gives up exactness for them. The history-dependent weighting it introduces is called the *policy prior* in the code and in earlier reports; this paper calls it a behavioural weighting model, since it is a reweighting of the constraint-consistent deals by the observed ask history rather than a prior in the Bayesian sense. It replaces the count join by iterative proportional fitting, better known here as Sinkhorn scaling, on the $|U_t| \times 6$ matrix of admissible card–seat pairs. Initialise

$$\hat{\mu}^{(0)}(c,p) \;=\; \mathbb{1}[p \in \mathcal{M}(c)] \cdot \exp\Big(\mathrm{clip}\big(\theta\, a_{p,H(c)} - \varphi\,(A_p - a_{p,H(c)}),\ \pm 2.6\big)\Big), \tag{8}$$

where $a_{p,h}$ is the number of asks seat p has made in half-suit h and A_p its total asks, and then alternate the two normalisations

$$\hat{\mu}(c,\cdot) \leftarrow \frac{\hat{\mu}(c,\cdot)}{\sum_p \hat{\mu}(c,p)}, \qquad \hat{\mu}(\cdot,p) \leftarrow \hat{\mu}(\cdot,p) \cdot \frac{n_p}{\sum_c \hat{\mu}(c,p)}, \tag{9}$$

to a fixed iteration count. The row step makes each card's distribution a probability; the column step enforces the capacity constraint (C2) that the exact join enforces combinatorially.

Between outer sweeps the routine also applies an independence conditioning step for each live (C5) disjunction: for a certificate saying that seat j held *some* card of a half-suit, each candidate card c' of that half-suit is rescaled to $\hat{\mu}(c',j)/(1-\prod_{c''}(1-\hat{\mu}(c'',j)))$, the product running over the half-suit's cards c''. So (C5) is neither enforced exactly nor discarded in the deployed path: it is imposed approximately, under an independence assumption the exact enumeration of §5.4 does not need. The deployed parameters are $\theta = 0.37062$ and $\varphi = 0.14525$.

Equation (8) is the entire behavioural content of the agent's beliefs: a seat that has asked repeatedly in a half-suit is treated as more likely to hold cards of it, because asking requires already holding one. It is a soft version of the (C5) certificate, in the spirit of the Skat inference line [21, 22, 20] but far simpler, with two parameters and no learned opponent model.

§4.3 reports the model comparison: scored as predictors of card location on identical states, $\hat{\mu}$ has lower log loss and a higher argmax hit rate than μ. In this implementation, exactness for the constraint model and behavioural weighting are in different objects: the block program carries no behavioural parameter, and the deployed paths add the weighting either as the initialiser of the Sinkhorn fit or, in the search, as an importance weight on exact samples (§6.7). In the comparison, the weighting contributed more predictive accuracy than exactness did. That is why the paper treats exact constraint-based inference as an assumption to test rather than a property to assume; §9.4.1 reports a second comparison of the same kind.

5.6 Implementation verification

Three checks run over the engine, and all three ran as preconditions of the evaluation in §11.

Soundness audit. Every seat's candidate set is checked against the true deal at every decision: $\mathrm{own}_t(c) \in \mathcal{M}_t(c)$ must hold always, since no agent's deduction may exclude the truth. The verification run for this evaluation recorded 0 violations in 6,737,436 checks, and the preceding cycle 0 in 23,594,580.

Cross-engine agreement. The block dynamic program, an independent card-counting path and an exact rejection sampler must agree on (5). The self-test reports 424,986 checks passing, a check being one (card, observer, event) row. Its tolerances are 10^{-9} for block-versus-card-program agreement, which is checked on states with no live (C5) disjunction and holds there at the limit of double precision, and 0.05 for block-versus-rejection-sampling agreement on states with live disjunctions, which is Monte Carlo precision at the sample sizes the test draws. A separate brute-force oracle checks Z, the marginals, and the team and allocation probabilities against exhaustive enumeration on small states to 10^{-9}.

Identity control. The current policy, carrying its predecessor's parameter vector with every new mechanism disabled, must produce a transcript identical to its predecessor's under md5. It does (PASS). This control checks whether refactoring changes behaviour before the new components are enabled, so that a measured ablation difference is not an artefact of refactoring.

Two defects these checks found are recorded here because both changed previously published numbers, and both survived for the same reason: they were invisible to the path the agent actually deploys. The block dynamic program parked every instance's tables in a shared per-thread pool, so a second construction on the same thread silently repointed the first instance's tables, giving 285 mismatches in 294 checks; any earlier number computed with the exact belief in a configuration where more than one seat held such an object is affected, and the deployed approximate-belief numbers are not. Separately, an audit that verifies the declaration pre-gate never rejects a correct declaration had been parsed only in an older branch and so had never run; once enabled it reports 0 false negatives over 5,472,906 declaration opportunities. Appendix E carries the full register.

6 SESTINA v1.0 architecture

SESTINA v1.0 is a configuration of the v0.6 policy family rather than a new architecture. This section describes the agent conceptually: Figure 1 shows the decision pipeline, and Table 2 sets its components against v0.6's. The literal configuration string, its option map, the parameter vector layout and the freeze artifact's hashes are in Appendix B.

Table 2: Components of SESTINA v1.0 against v0.6. Harness identifiers appear in parentheses and are used in the paper only where a command must be reproduced.

Component	v0.6	SESTINA v1.0	Section
Deal inference	Sinkhorn fit to the deal posterior, from a fitted policy prior	unchanged	§5
Ask and declaration policy	fitted linear score, 37 coordinates	unchanged; no refit in this cycle	§6.1
Tie-breaking among equal-scoring asks	enumeration order (unstable sort)	hash of the public event stream (`rtie`)	§6.2
Half-suit contestation	absent	contested-half-suit weight (`r12`)	§6.3
Declaration urgency escalation	event-count clock at 220 events	disabled (`pool`, `oppfloor`, `force`, `askfloor`)	§6.5
Termination safeguard	the urgency clock	deduction-state stall detector (`stall`)	§6.6
Test-time search	built and measured, deployed disabled	enabled, endgame-truncated, paired LCB guard	§6.7

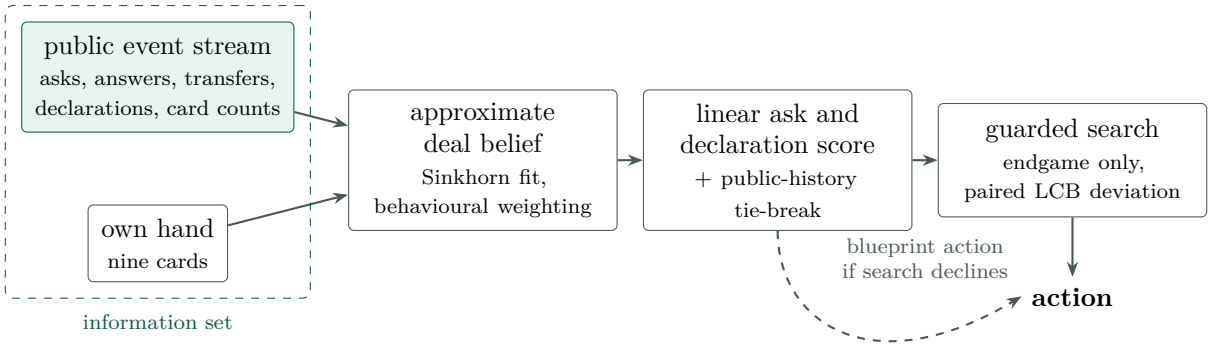


Figure 1: The information structure of a Fish seat and the SESTINA v1.0 decision pipeline. A seat observes the public event stream and its own hand, and nothing else; there is no path from a policy to the deal or to another seat's hand (§5.2). Because every card movement is public, that information set determines an exact constraint-conditioned distribution over the initial deal (§5). The deployed path does not construct it: it runs approximate constraint-based inference with behavioural weighting, a Sinkhorn fit that starts from the ask-history weighting and imposes the same constraints iteratively (§5.5). The first two stages together, the belief and the linear ask and declaration score with its tie-break, form the *blueprint*, the non-searching policy. The search runs only in the endgame and departs from the blueprint's choice only when the guarded selection rule of §6.7 favours the departure.

6.1 Action space and ask policy

At its turn a seat faces one decision with two branches: declare a half-suit, or ask. Declaration is treated first in the code and is described in §6.4; this subsection covers the ask.

The legal action set. With $\mathcal{O}_s$ the opponents and $\mathcal{H}_t$ the half-suits not yet claimed, the rules of §2.1 give

$$\mathcal{A}_t \;=\; \{(c,q) \;:\; q \in \mathcal{O}_s,\; \mathrm{hand}_t(q) \neq \emptyset,\; H(c) \in \mathcal{H}_t,\; c \notin \mathrm{hand}_t(s),\; \mathrm{hand}_t(s) \cap H(c) \neq \emptyset\}. \tag{10}$$

The last condition encodes the ownership condition that makes an ask informative: the asker must already hold a card of the half-suit, so asking publishes partial ownership.

What the policy actually enumerates is a strict subset of (10). A *live-ask filter* first removes candidates the seat can already prove are misses, that is, those whose target is excluded by a certificate, $q \notin \mathcal{M}(c)$. It is this filter, rather than any weighting inside (11), that keeps a provably unsuccessful ask off the score. It binds: it removes what would otherwise have been the argmax at roughly 1.4% of 269,423 ask decisions captured from SESTINA v1.0's own seats. If no candidate survives the filter the full legal set is restored, so the guarantee is not absolute.

The score. Each candidate is scored by a fitted linear function of a 20-dimensional feature vector $\phi(a)$, plus a learned value term and, in SESTINA v1.0, three additional ownership terms:

$$u(a) \;=\; \lambda\, w^{\top}\phi(a) \;+\; \beta\, V_{\mathrm{ask}}(a) \;+\; \lambda\big(w_{\mathrm{void}}\, g_1(a) + w_{\mathrm{team}}\, g_2(a) + w_{\mathrm{last}}\, g_3(a)\big), \tag{11}$$

with $\lambda = 0.78874$ and β scale parameters of the fit. SESTINA v1.0 adds a fourth block of 18 extended coordinates, of which 1 is non-zero; that one is (18) and §6.3 gives it. Writing $p_a = \hat{\mu}(c,q)$ for the modelled hit probability, the features ϕ are functions of the belief and the public state. They include p_a and p_a^2; an indicator that the card is publicly certain; the acting seat's holding in

$H(c)$ and the posterior mass its team already carries; the best continuation probability elsewhere in the half-suit; the reply threat and exposure a miss would create, both weighted by $1 - p_a$; the target's hand size and an indicator that this ask would empty it; the binary entropy $\mathrm{H}_2(p_a)$; and a score-margin term. Appendix B lists all twenty with their fitted weights.

The three extra terms are the ones v0.6 added, and each expresses an effect the twenty could not:

$$g_1(a) = p_a \prod_{c' \in H(c) \setminus \{c\}} (1 - \hat{\mu}(c', q)), \tag{12}$$

$$g_2(a) = \Pr\big[\,\mathrm{own}(c) \in \mathcal{T}_s \cup \{s\}\,\big], \tag{13}$$

$$g_3(a) = p_a \cdot \mathbb{1}\big[\,q \text{ is the last opponent holding cards, and holds one}\,\big]. \tag{14}$$

Equation (12) is an independence-based proxy for the probability that the ask takes the target's *last* card of the half-suit: it multiplies the marginals as if the target's holdings of the other cards were independent, which the count-constrained distribution does not imply. Equation (13) is the team-ownership probability of the named card, the belief mass the acting team already carries, and (14) is an indicator that the ask would trigger the forced endgame. Equation (12) expresses the loss of the target's access to that half-suit: because a legal ask requires holding a card of the half-suit, removing a target's last card of $H(c)$ permanently removes that target's right to ask in it.

Selection. The policy takes the argmax, with the tie handling of §6.2 and, when enabled, the search of §6.7:

$$a^\star \;=\; \arg\max_{a \in \mathcal{A}_t} u(a). \tag{15}$$

SESTINA v1.0 inherits w, λ, β and the three extra weights from v0.6 unchanged. **No gradient or population fit was run in this cycle**: the one coordinate that is not inherited, (18), had its value chosen as the argmax of a measured dose response, which is a one-dimensional selection on win rate and is reported as such in §11.2.3. The vector was found by v0.6's repaired optimiser under a minimax-regret objective over a four-member opponent panel. Of the 55 coordinates in the extended vector a number are known to be inert, and Appendix B lists which and why.

6.2 Public-history tie-breaking

In the v0.6 mirror, 53.80% of contested ask decisions end with two or more candidates scoring bit-for-bit identically under (11) (§4.3), and the winner was whichever the enumerator emitted first, under an unstable sort. That is a specification problem rather than a strength problem. The choice is arbitrary, and it is determined by an implementation detail: the result of an unstable sort is reproducible given the same implementation and inputs, but the policy does not specify it, so a teammate modelling the seat would have to model the sort rather than the rule. The defensible benefit of the rule below is a specified, reproducible selection rule that reads only the public record.

Order the candidates by score and let the tied set be

$$\mathcal{E}_t \;=\; \{a \in \mathcal{A}_t \;:\; u(a) \geq u(a_{(1)}) - \varepsilon\}, \tag{16}$$

where $a_{(1)}$ is the top-ranked candidate and ε is a fixed tolerance. SESTINA v1.0 replaces enumeration order with a deterministic draw keyed by the public record:

$$a^\star \;=\; \mathcal{E}_t\big[\,\iota(\rho,\, |\mathcal{E}_t|)\,\big], \qquad \rho \;=\; \mathrm{hash}(\mathcal{P}_t,\; s,\; t), \tag{17}$$

with $\mathcal{P}_t$ the running hash of the public event stream, s the seat index, t the event count, and ι a uniform index map from the hash into the tied set. The tolerance ε is 10^{-9} in absolute score units, a fixed engine constant rather than a configuration option, so $\mathcal{E}_t$ is a tolerance-based tied set rather than necessarily a bit-identical one. Every seat observes all three arguments of ρ, so every teammate can recompute it; no seat's private hand enters it. The index is applied to the tied block in the order the score sort leaves it, and the arrangement of equal keys under that sort is fixed by the implementation rather than by the rule, so what (17) specifies is the index into the block; an identical build reproduces the arrangement, as the isolation audit of §11.9 confirms for every non-searching decision.

Two scope conditions belong with that. (17) governs the non-search path only: when the endgame search fires it supplies its own selection (§6.7) and index 0 is whatever the sort left first. And the search draws its determinizations from a private per-seat stream seeded from the harness, so the policy is a deterministic function of (own hand, public record, reset seed) rather than of the first two alone. That third argument is exactly what the side-channel audit of §7.10 replays, which is why the audit is a test of seat isolation and not of determinism.

The engine also offers a variant that draws ρ from a private per-seat stream. That variant is *not* used: it would break the reconstructibility that makes (17) a coordination device. §9.4.4 reports what happened when a later experiment tried to *select* within $\mathcal{E}_t$ rather than randomise over it.

6.3 Half-suit contestation

The component with the largest observed leave-one-out drop in the holdout ablations (§11.4) is a single additional coordinate, one non-zero entry out of 18 in an extended block of the parameter vector. Let M_H be the expected number of cards of half-suit H held by the opposing team, and q_H the number whose holder this seat cannot place. The term is

$$r(a) \;=\; (1 - p_a) \cdot \frac{M_{H(c)}}{6} \cdot \frac{q_{H(c)}}{6}, \tag{18}$$

entering (11) with a positive coefficient of `25`, so a large value *encourages* the ask. It selects asks that are likely to miss, into half-suits the opponents dominate, whose ownership is still ambiguous.

Feature interpretation. The feature was designed to price *information denial*: a failed ask publishes a negative certificate, and the intent was that the fitted coefficient would come out negative, meaning refuse to hand that certificate over. It came out positive. The negative branch was swept separately and is flat to negative throughout, reaching −1.47 pp at the mirror-image dose, so the measured dose response does not support the design intent.

What the positive branch does was measured on the target rather than inferred. Over ∼261,000 target ask decisions, the target's ask accuracy falls from 0.540 to 0.521, while its declaration accuracy *rises* and it becomes *less* often urgent, which is inconsistent with a declaration-channel mechanism, where this phase expected to find the effect. The reading these measurements are consistent with is contestation: asking into half-suits the opponents are trying to complete degrades the quality of their asks, and the acting team pays part of the same cost, its own ask accuracy falling from 0.543 to 0.534. Three sibling coordinates written to express the same denial idea were measured at the same dose and are inert or harmful, which is consistent with a benefit specific to this feature combination in the tested ablations rather than to opponent mass in general; a positive coefficient and the associated changes in opponent accuracy suggest a mechanism without establishing it as the unique explanation.

Two facts about its provenance matter for §11.4. It was found by a coordinate sweep during adversary generation rather than by architecture work, and it is hand-set: the value `25` is the argmax of a dose response measured at 48,000 games a dose, not the output of a gradient or population fit. And it is not uniformly good. Against the then-deployed policy it scored +2.71 pp, but its worst cell over the v0.6 frontier is −1.69 pp against `Fsearch`, which is negative, so it is a strength gain with a worst case rather than an exploit (§9.3).

6.4 Declaration policy

A declaration names, for each of the six cards of a half-suit, which member of the declaring team holds it. It is all-or-nothing: correct, and the team scores the half-suit; wrong, and the opponents do. The decision therefore has two parts, choosing what to claim and choosing when.

Allocation selection. For half-suit h the reference objective is the most probable allocation among the acting team under the constraint-conditioned distribution,

$$\alpha^\star(h) \;=\; \operatorname*{arg\,max}_{\alpha:\, H \to \mathcal{T}_s \cup \{s\}} \; \pi_{\text{alloc}}(h, \alpha), \tag{19}$$

with π_{alloc} from (7). Since (7) depends on $\alpha^\star$ only through its count signature, the argmax in (19) is generally not unique; the reference implementation takes the lexicographically first feasible assignment among the maximisers. Equation (19) defines the reference objective; the deployed approximation is described here. Three procedures must be kept distinct. *Marginal-product selection*: SESTINA v1.0 enumerates the capacity-feasible assignments of the unresolved cards of h to its own team and takes the argmax of the product of per-card approximate marginals, $\prod_c \hat{\mu}(c, \alpha(c))$; that product is the selection score only. *Sequential approximate joint scoring*: the selected allocation is then priced separately, by sequential conditioning, in which the belief is refit after each card of the allocation is fixed in turn, and the resulting chain-rule product is what the policy reports as $\hat{\pi}^\star_{\text{alloc}}$ and thresholds on. *Exact constraint-joint selection*: the exact object (7) is available from the block dynamic program but sits on no decision path. §9.4.1 reports the relevant measurements: the marginal-product selection and the sequential approximate joint's argmax selected the same allocation in approximately 210,000 recorded voluntary declarations, counted over capture cells that include re-runs of the same seeds, an agreement that is empirical and does not establish equivalence to the exact joint maximiser; and substituting the exact constraint-joint selection changed a small fraction of decisions and performed worse on the declaration decision.

Declaration timing. Write $\hat{\pi}^\star_{\text{alloc}}$ for the sequentially estimated probability of the selected allocation and $\hat{\pi}_{\text{team}}$ for the deployed estimate of the probability the team owns h outright, which is the larger of $\hat{\pi}^\star_{\text{alloc}}$ and the product over the half-suit's unresolved cards of the team's summed marginals, $\prod_c \sum_{p \in \mathcal{T}_s \cup \{s\}} \hat{\mu}(c, p)$, an independence product rather than (6). The value function is written on the assumption that waiting is free, on the ground that an opponent cannot claim a half-suit the acting team provably holds. That ground binds strictly only in the locked case, and the rule is applied down to a team-ownership probability well below one, so the assumption is an approximation rather than an identity. Under it the decision is an approximate comparison between immediate declaration value and the value of waiting: declaring scores a point now with probability $\hat{\pi}^\star_{\text{alloc}}$, and publishes the team's holdings either way. The agent evaluates that with its learned state-value function V:

$$\Delta \;=\; \underbrace{\hat{\pi}^\star_{\text{alloc}}\, V(\text{state} \mid \text{correct}) \;+\; (1 - \hat{\pi}^\star_{\text{alloc}})\, V(\text{state} \mid \text{wrong})}_{\text{declare}} \;-\; \underbrace{V(\text{state})}_{\text{wait}}, \tag{20}$$

and declares when $\Delta > \delta$ for a fitted margin δ. Both conditioned states differ from the waiting state in having the *unresolved* cards of h removed from the unresolved set, typically fewer than six, and in the deltas of the four half-suit-control features the value function reads, all of which are functions of the team's expected share of h before the claim. They differ from *each other* only in the score margin, by $+1$ and -1.

A gate precedes (20): a half-suit is priced by the value rule only if its deployed team-ownership estimate $\hat{\pi}_{\text{team}}$ clears a fitted floor, which is lowered under pressure stage 1 and removed under stage 2. Two overrides sit above (20). Under the forced endgame, when the opposing team is cardless and the acting team must resolve the remaining half-suits among itself, the engine polls the team down the willingness ladder of §2.2, and at each level the agent is willing to declare a half-suit when $\hat{\pi}^{\star}_{\text{alloc}} \geq \tau$ for that level's threshold τ; the final level is compulsory. Under *pressure*, the escalation described next, stage 1 lowers the thresholds, declaring when $\hat{\pi}^{\star}_{\text{alloc}} \geq 0.5$, and stage 2 removes them: the best available allocation is declared unconditionally, whatever its probability. SESTINA v1.0 removes the mechanism that used to raise pressure in ordinary play, and §6.6 is what now supplies stage 2.

6.5 Urgency-escalation removal

v0.6 inherited a declaration-urgency escalation keyed to an event counter. Past a threshold of 220 public events the declaration rule lowers its team-ownership floor and permits declarations with estimated correctness probabilities near 0.5, which costs 15.18 pp to a policy held in that regime. No adversary reaches it, since the longest game any adversarial search produced against an unmodified opponent is 149 events, but the machinery is defective in ordinary play as well: §9.4.1 reports that removing it eliminates 71% of the entire declaration allocation-error class. SESTINA v1.0 disables it (`pool`, `oppfloor`, `force`, `askfloor`).

The escalation was, however, also the policy's termination safeguard, so removing this mechanism requires an alternative stall safeguard.

6.6 Deduction-state stall detection

The escalation SESTINA v1.0 removes was keyed to a clock. The condition the clock stood in for is that the monitored deduction state has not changed, and that condition is directly observable from the seat's own deduction state.

Let $\chi_t = \text{hash}\big(\text{own}_t(\cdot), \mathcal{M}_t(\cdot), U_t, (\text{hand sizes})_t, (\text{live half-suits})_t\big)$, the digest of the hard certificates the seat holds. It deliberately *excludes* the list of outstanding (C5) disjunctions: a repeated ask appends a duplicate disjunction carrying no information, so counting that list as progress would blind the detector to the one loop it exists to break. Define the no-progress counter

$$\nu_t \;=\; \begin{cases} 0, & \chi_t \neq \chi_{t-1}, \\ \nu_{t-1} + 1, & \chi_t = \chi_{t-1}, \end{cases} \tag{21}$$

and escalate to pressure stage 1 when $\nu_t \geq K$ and to stage 2 when $\nu_t \geq 2K$, with $K = 12$ and $2K = 24$ in SESTINA v1.0.

Scope of the termination safeguard. Consider $\Phi_t = \sum_{c \in U_t} \big(|\mathcal{M}_t(c)| - 1\big) + |U_t| + |\mathcal{H}_t|$. Certificates only accumulate: a seat excluded from a card's candidate set is never restored, a resolved card is never unresolved again, and a claimed half-suit never returns. Every event that changes χ by one of those routes therefore strictly decreases Φ_t, which is a non-negative integer, so there can be only finitely many of them. This potential-function argument does not establish termination,

because transfers can change the monitored state without decreasing the proposed potential: the digest also contains hand sizes, so a successful ask always counts as progress, and a card moving between two seats is not a monotone gain; nothing above forbids a card being won back later. The rule therefore bounds the mode of non-termination that actually occurs, a run of asks producing no new certificate, and not every mode. That is why this paper calls it a safeguard, and why §12.6 records termination as empirical.

The threshold was set from measurement. Over 800 mirror games v0.6 never produces a no-progress run longer than 6, while the deliberately deadlocked negative control of §7.8 produces 326. Any $K \in [12, 60]$ was therefore not reached in the sampled ordinary games and was reached in the tested deadlock control, after K and then $2K$ no-progress events. Armed, the rule's mirror transcript digest is byte identical to the same configuration without it: in those mirror games its effect on play is identically zero, because ν_t never reaches K. §11.4 confirms that on holdout material. Note also that ν_t is a function of the seat's own hand and the public record alone, so the rule opens no information channel.

6.7 Guarded determinized search

SESTINA v1.0 enables the search v0.6 built, measured, and then deployed switched off. It runs only in the endgame, when at most `26` cards remain unresolved, and it revises the blueprint's choice only when a standard-error-penalised paired comparison favours the revision.

Deal sampling. The search draws $D_{\text{det}} = \texttt{12}$ determinizations from the constraint-conditioned distribution (1), using a per-card capacity dynamic program as a uniform sampler over deals satisfying (C1)–(C4), with the (C5) disjunctions enforced by rejection, so an accepted draw is an exact sample from (1). This sampler is a distinct object from the block program of §5.4, which enumerates (C5) inside its blocks instead. D_{det} is a target rather than a guarantee: draws are attempted up to a cap and the rule proceeds with however many survive rejection. If no draw survives, or the sampler cannot be built, the decision falls back to the v0.5 scoring path rather than to the blueprint's own choice; that path scores with the twenty-feature vector alone and breaks ties in enumeration order. No artifact of this cycle records how often the fallback fired for the frozen configuration. The behavioural weighting of (8) is then applied as an importance weight rather than baked into the sampler, which keeps the sampler exact and the weighting auditable:

$$\tilde{w}_d \;=\; \prod_{c \in U_t} \exp\Big(\mathrm{clip}\big(\theta\, a_{o_d(c),H(c)} - \varphi\,(A_{o_d(c)} - a_{o_d(c),H(c)}),\, \pm 2.6\big)\Big), \qquad w_d = \frac{\tilde{w}_d}{\sum_{d'} \tilde{w}_{d'}}, \tag{22}$$

where $o_d(c)$ is the seat holding card c in world d.

Rollout evaluation. The candidate set is the top `4` blueprint candidates by score, extended through any tie at the cut up to a cap of twenty, with index 0 reserved for the blueprint's own choice $a_{(1)}$, the first candidate after the score sort rather than the output of (17). For each candidate r and world d, all six seats are reconstructed *at their own information sets* under d, each playing a cheaper blueprint than the acting agent's own: the v0.5 policy with an independent-marginal belief and no re-scoring pass and the game is played to a terminal state or until `12` further public events have elapsed. A run reaching the end returns the acting team's half-suit differential; a run cut at the limit returns a *material* evaluation of the position instead, being the scored half-suit differential plus the team's expected control of the live half-suits. That is v0.6's own hand-written leaf, not a learned one: §9.4.2 reports what happened when a fitted replacement was built. Write

$V_{r,d}$ for either. This is the point at which the method departs from standard determinization: the continuation players are not made clairvoyant. §2.3 explains why that matters here, namely that clairvoyant play in Fish is degenerate and every action would score alike. All candidates share the same D_{det} worlds, so the comparison is paired and the noise that matters is the noise in the difference.

Guarded action selection. For each candidate $r \geq 1$ form the paired difference against the blueprint,

$$m_r \;=\; \sum_d w_d \,(V_{r,d} - V_{0,d}), \qquad \text{var}_r \;=\; \sum_d w_d \,(V_{r,d} - V_{0,d} - m_r)^2, \tag{23}$$

and with the Kish effective sample size $n_{\text{eff}} = 1/\sum_d w_d^2$ take

$$\text{se}_r = \sqrt{\frac{\text{var}_r}{n_{\text{eff}} - 1}}, \qquad \text{LCB}_r = m_r - \kappa_r \,\text{se}_r + \eta\,(u(a_{(r+1)}) - u(a_{(1)})). \tag{24}$$

The search deviates to $\arg\max_r \text{LCB}_r$ if that maximum is strictly positive, and otherwise plays the blueprint's choice. When the effective sample size is at most 1.5, which includes the case of a single surviving world or one dominant weight, se_r is set to an effectively infinite value, so no candidate outside the tied set can deviate and within it the raw weighted paired mean decides. The shrinkage is $\kappa = 2.5$. The term η blends the blueprint's own ranking back in and is **zero in this configuration**, so (24) reduces to a pure paired lower confidence bound.

Equation (24) is the action-selection guard used in the evaluated configuration. Chosen instead by an unguarded $\arg\max_r m_r$, the same search is 35.69 pp *worse* than the blueprint it searches from (§4.3): one world's return is a final half-suit differential whose spread is an order of magnitude larger than genuine differences between candidates, so the maximum selects on residual noise, and does so more confidently the more candidates it is offered. Requiring κ standard errors of a *paired* difference removed that measured deficit in the v0.6 experiment. The rule is a lower-confidence-bound heuristic rather than a test with established finite-sample coverage: at most D_{det} weighted worlds enter it, several candidates are compared against the same reference, and the effective sample size can be small. What the record establishes is that the guard reduced the measured deficit, not that it guarantees an improvement over the blueprint.

One detail is worth stating because §9.4.4 turns on it. Within the tied set $\mathcal{E}_t$ of (16) the rule applies $\kappa_r = 0$, that is, no shrinkage at all, which is precisely where its selection signal is most noise-dominated.

6.8 Coordination and information constraints

SESTINA v1.0 deploys as three identical agents. They hold different private hands, they may not signal, and the only channels the rule variant leaves are the forced-endgame willingness bit and the successor a cardless turn-holder names (§2.2), for both of which §2.2 reports no resolved effect from the tested interventions. The coordination assumptions are therefore the following.

They rest on determinism plus a shared observation. Consider first the idealised non-searching policy, and let Π denote it: a map from (own hand, public record) to an action. Every teammate runs the same Π and observes the same public record $\mathcal{P}_t$. Teammate j cannot compute Π's output at seat i, because it does not know i's hand. But it knows Π itself, and it knows $\mathcal{P}_t$, so it can compute the *distribution* of i's action induced by its own posterior over i's hand,

$$\Pr\big[\,a_i = a \mid \mathcal{I}_j^t\,\big] \;=\; \sum_{D \models \mathcal{I}_j^t} \Pr\big[D \mid \mathcal{I}_j^t\big]\; \mathbb{1}\big[\Pi(\text{hand}_i(D),\, \mathcal{P}_t) = a\big]\,, \tag{25}$$

and, having observed a_i, condition on it. That is exactly the (C5) reasoning of §5.3 carried one step further: a legality certificate says the asker held *some* card of the half-suit, while (25) says the asker held a hand on which Π preferred that ask. The deployed approximation does not evaluate (25) in full, which would require enumerating hands; it captures the first-order part of it through the ask-tally prior (8). The white-box adversary of §11.8.1 is the construction that approximates (25) against SESTINA v1.0 rather than for it: it evaluates the one-step action likelihood on exact samples and accumulates per-card evidence additively rather than carrying the joint conditioning across the game, and §9.2 reports what that inversion is worth.

Two conditions make this work, and both are engineering constraints rather than assumptions. Π must be *specified*: a teammate reasoning about seat i must be able to evaluate $\Pi(\cdot, \mathcal{P}_t)$ on a hypothesised hand from the rule alone. An unstable sort does not break determinism, since it is deterministic for a fixed input and build, but its result is fixed by an implementation detail rather than by the policy specification, so a teammate would have to model the implementation; (17) replaces it with a specified rule on the non-search path. And Π must be common knowledge, which for three copies of one published agent it is. No shared secret and no pre-play randomisation are used, which is the sense in which the team's coordination is public (§8).

Equation (25) is stated for the idealised non-searching policy. The deployed agent departs from it in two ways. When the search of §6.7 fires, the action also depends on the seat's reset seed, which is private to the seat, so the deployed policy is a function of (own hand, $\mathcal{P}_t$, reset seed) on that path (§6.2); a teammate can still compute the distribution of the action given a hypothesised hand only by modelling that randomness. And the cross-deal state persistence described next introduces a further dependence that neither a teammate nor (25) models.

Cross-deal state persistence. Agents are constructed once per worker thread and reused across the deals that thread is given, and some per-agent state survives the inter-deal reset, reachable only under the search. This violates the premise that Π depends on (own hand, $\mathcal{P}_t$) alone: a decision can depend on which deals the thread happened to process earlier. The carrier was identified as the rollout engine, which the reset never touches; the specific field was not. It was deliberately not repaired, because the one-line fix changes the play of every searching configuration by roughly one deal in 800, and applying it after the configuration was frozen would defeat the purpose of freezing. §11.9 measures its magnitude on holdout material, and §12.6 states what it costs the argument above.

6.9 Decision procedure

Collecting the pieces, one turn of SESTINA v1.0 at seat s and time t runs as follows. The steps describe the deployed approximate path, not the reference objects of §5.

1. **Update beliefs.** Fold the new public events into the candidate sets $\mathcal{M}_t(\cdot)$ by (C1)–(C5), then compute $\hat{\mu}$ by (8) and (9). Update the no-progress counter ν_t by (21).
2. **Consider declaring.** For each live half-suit passing the cheap ownership gates, enumerate the capacity-feasible allocations of its unresolved cards to the acting team and select the allocation maximising the product of approximate card marginals. Estimate its probability $\hat{\pi}^{\star}_{\text{alloc}}$ by sequential conditioning and belief refitting. If the team-ownership estimate clears its floor, declare when $\Delta > \delta$ in (20); declare regardless if a pressure override fires, and answer the forced-endgame ladder by its thresholds (§6.4).
3. **Otherwise ask.** Enumerate the legal set $\mathcal{A}_t$ by (10) and apply the live-ask filter of §6.1, removing candidates whose target is excluded by a certificate; if the filter removes every candidate,

restore the legal set. Score the resulting candidates by (11), including the contestation term (18).

4. **Resolve ties.** Form $\mathcal{E}_t$ by (16). If the endgame condition does not hold, return (17) and stop.
5. **Search.** Otherwise sample D_{det} worlds with weights (22), roll out each candidate in each world, and return the candidate selected by (24). This is the blueprint's own choice, the first candidate after the score sort, unless a candidate's paired lower confidence bound is positive; the bound uses $\kappa_r = \kappa$ standard errors for candidates outside the tied set $\mathcal{E}_t$ and $\kappa_r = 0$ within it, and if no world survives rejection the v0.5 scoring path decides (§6.7).

Steps 1–4 read only the acting seat's own hand and the public record; step 5 also reads the seat's own reset seed. §6.8 records that, in the evaluated implementation, a searching agent's decisions can additionally depend on state carried across deals, so the property that every decision is a function of the listed information holds for the non-searching path and is an approximation for the searching one. The side-channel audit of §7.10 tests the property mechanically rather than by inspection, and §11.9 reports where it fails.

7 Evaluation framework

This section describes the evaluation protocol. Each component was adopted in response to a specific measured failure in an earlier cycle, and §4.3 describes those failures.

7.1 Outcome measures

The primary outcome is the team win rate, the proportion of games in which the evaluated configuration's team takes at least five of the nine half-suits. Comparisons are reported as an **edge in percentage points**,

$$\text{edge} \;=\; 100 \times (\text{win rate} - 0.5),$$

so that zero denotes a 50% win rate in the specified comparison and the sign gives the direction. The paper writes *pp* for percentage points of this quantity, and a bracketed pair $[\ell, u]$ after an estimate is its two-sided 95% confidence interval (CI) as constructed in §7.3. A comparison "against" a named configuration is a head-to-head comparison: an edge of +3 pp means a 53% win rate against that configuration in the paired design of §7.2, not a three-point change from a separately measured baseline win rate. Three other quantities appear and are never conflated with it: the mean number of half-suits a team takes, used for the search's internal return; predictive log loss and argmax hit rate, used to score belief models as predictors (§4.3); and the *adversarial response win-rate edge* of §11.8, which is an edge in the same units but measured for a fitted opponent against the evaluated configuration.

7.2 Duplicate deal blocks

A *cell* compares two configurations A and B over a *bank* of deals identified by one integer seed. Every deal in the bank is played twice: once with A in the odd seats and B in the even seats, and once with the seats exchanged. Both configurations therefore play the identical deal from both sides.

Fish has large per-deal variance, since a deal can be won or lost on the cards, and an unpaired comparison spends most of its sample resolving that variance rather than the difference between policies. The rotation balances team assignments within each deal and can reduce the variance

associated with deal strength, in the spirit of duplicate bridge and of the variance-reduction estimators developed for poker [64, 65]. The design requires reproducibility from the seed alone (§5.2); without it the two rotations are not the same deal and the design degrades silently into an unpaired one.

7.3 Uncertainty estimation

The two rotations of one deal are not independent observations: the two games share an initial deal and are therefore treated as a single resampling block. Resampling blocks preserves their within-deal dependence, whatever its sign.

Three interval constructions are used, and they are labelled by their construction rather than described as one method.

Per-bank intervals. Every cell measured on one bank carries a two-sided 95% **deal-clustered percentile bootstrap interval**. The resampling unit is the *deal*, carrying both of its rotations together; 20,000 bootstrap resamples are drawn with replacement from the deals of the cell, the edge is recomputed on each resample, and the interval is the empirical 2.5th and 97.5th percentiles of that distribution. No Wilson or normal-approximation interval on the game count is quoted for a single bank. The harness records both intervals for every cell, and in this corpus the clustered interval is typically no wider than, and often slightly narrower than, the interval that treats games as independent, which is the direction a negative within-deal correlation produces.

Cross-bank pooling. Two banks of equal size pool as the mean of their two edges, with standard error $\mathrm{se}_{\mathrm{pooled}} = \sqrt{\mathrm{se}_1^2 + \mathrm{se}_2^2}/2$, where each se is recovered from that bank's bootstrap interval as half-width divided by 1.96. The pooled interval is the estimate plus or minus $1.96\,\mathrm{se}_{\mathrm{pooled}}$. This is a normal approximation built on the per-bank bootstrap standard errors, not itself a bootstrap interval. Both per-bank values are printed beside every pooled figure in this paper.

Between-cell contrasts. A difference between two separately measured cells, such as an ablation's leave-one-out drop, is the difference of the estimates with the two half-widths combined in quadrature, omitting the covariance between the two cells. The configurations are paired on deals, so the two cells' estimation errors are generally correlated; the omission overstates the width only when that covariance is non-negative, which is expected when both cells are measured on the same deals against the same opponent but is not verified cell by cell. The harness produces no paired difference *across* two separately executed cells, and the scored artifacts record per-cell summaries rather than per-deal outcomes, so a joint bootstrap of the actual contrast would require re-running the cells with per-deal capture; that is listed in §12.7. Where a contrast combines cells that share banks and deals with unit coefficients, as the rule-variant residual of §11.7 does, the propagated interval is not a valid 95% interval and is labelled as such where it appears.

7.4 Sample-size planning

At parity, the 95% half-width of a cell's edge in percentage points is approximately $98/\sqrt{N}$ for N games, the half-width of a normal interval for an independent Bernoulli mean at $p = 0.5$. The harness computes this planning approximation for every cell and emits it with the result. It treats the N games as independent and ignores the duplicate-block covariance, so it is a planning figure rather than a power calculation: at $N = 9,604$ it gives a one-point half-width, which does not mean that a one-point effect would be detected with any specified probability. No formal

power calculation, with a specified test, effect size, significance level, desired power and block-level variance, was performed. The arithmetic is applied in advance because it determines what may be attempted: a one-point half-width requires roughly 9,604 games, and a quarter-point half-width requires sixteen times that.

The failure this planning step addresses is the one §4.3 describes, in which five mechanisms cleared a 95% paired interval at one budget and returned exactly zero at three times it. Sample sizes in this paper are chosen from the arithmetic before a cell is run, never after seeing its result.

7.5 Prespecified sign-agreement criterion

Every claim is measured on two disjoint deal banks, and the criterion was fixed before the data were seen:

> A claim whose **sign** does not agree on both banks is reported as not replicated, whatever its pooled interval says.

This is stricter than a pooled significance test, deliberately. A pooled interval can exclude zero while the two banks disagree, and that configuration has occurred in this corpus and has previously been published as a positive result. The criterion binds against this paper's own findings in §11.4 and again in §11.3, and both instances are reported.

7.6 Holdout sealing and preregistration

A programme that generates candidate configurations, scores them and selects the best is performing a maximisation, and the selected configuration's score is upward biased by the amount a maximum over K draws exceeds the mean. No care taken in the final measurement removes that bias if the final measurement uses material the selection has seen.

Before any candidate existed, deal banks were reserved by seed and their digests committed, a digest being a rolling hash over the six dealt hands and the dealer of every deal in the bank, computed without constructing a policy or playing a game. An adversary population was split in half by a rule fixed in advance, and one half was encrypted with the plaintext's SHA-256 committed, so that it could not be trained against even accidentally. The engine's match command enforces the seal: a run naming a sealed seed fails unless an explicit environment variable is set, which was set once, at the start of the final evaluation, and recorded. The guard's scope is the match command; the fitting, ablation and probe drivers derive their own seeds and bypass it, as Appendix F (D6) records, and a committed digest is a commitment to content rather than a proof that the content was not accessed. The documented access procedure is the one recorded there.

The entire final battery, comprising every cell, its sample size, its threshold, the arithmetic behind the sample size and the conditions that would count as non-confirmation, was written down and committed before any holdout bank had been played and before the sealed half had been decoded. The evaluation was then executed from a cleared context, reading that document and nothing else from the project's records. Where it required a training figure for comparison it took it from the registered protocol, which states such expectations in advance for exactly this purpose, so that agreement constitutes a check and disagreement constitutes a finding.

Anything the evaluation did that the protocol does not specify is recorded as a deviation, next to the affected result. There are 27, listed in Appendix F.

7.7 Selection effects and multiple comparisons

This study evaluates many candidate mechanisms, opponents, ablations, adversarial searches and configurations, and the inferential status of that needs stating precisely.

No formal family-wise or false-discovery correction was applied, and no interval reported here is adjusted for multiplicity. What the design provides instead is structural. *Registration* fixes the primary comparison, its threshold and its interpretation before the holdout outcomes were examined, so the primary claim is a single pre-specified test rather than the best of many. *Sealing* makes the holdout material unavailable to the selection process by construction rather than by discipline. *Disjoint banks with a sign criterion* require a claim to appear twice before it is reported. *Calibration* of the detection floor (§7.9) establishes what the instrument can resolve, so that a null is interpretable. And an explicit *selection check* (§11.2.3) computes how much of the measured gain a maximum over K candidates would produce under the null and compares advance-stated training figures against their holdout counterparts.

Exploratory comparisons in §9 are reported as exploratory: they were run on training material during the development phase, none is treated as confirmatory, and none is used to support the paper's primary claim.

7.8 Behavioural validity checks

Win rate and behavioural validity are dissociable in this policy family. The corpus contains a configuration that scores 56.60% against an earlier version while making 44.83% provably unsuccessful asks, asks whose target it can already prove does not hold the card (§6.1), and losing 14% of its games to the action cap. It improves win rate against the specified opponent while failing the chosen behavioural criteria.

A configuration is therefore gated *before* any strength number is computed for it, on eight rules covering provably unsuccessful asks, runs of them, action-limit games, the mirror game-length tail, late declarations, and the side-channel tests below. The protocol calls this the *soundness gate*, and the paper keeps that name for the artifact; it is an empirical test suite encoding behavioural criteria, not a proof. A provably unsuccessful ask is legal and could in principle serve an information or turn-transfer purpose, so a rejection expresses a design requirement rather than the invalidity of every rejected strategy. A configuration that fails is rejected regardless of its win rate. The gate runs on *training* material by design, being a check on self-play behaviour rather than a strength claim. It is accompanied by a negative control, a configuration constructed to be rejected, because a gate that cannot reject it would establish nothing about the configuration it accepts.

7.9 Adversarial-search calibration

An adversarial-response figure is produced by a search, and a search that finds nothing admits two explanations: there is nothing to find within the class and budget, or the search is too weak. Planted-handicap experiments characterise sensitivity to selected failure modes. An artificial handicap of known size is planted in a target, a responder is fitted against the handicapped target, and the *excess* of that responder's edge over its edge against the unhandicapped target is measured. The smallest planted edge the procedure recovers, for a given search class, budget and handicap family, is that class's **detection floor**.

Three quantities that this number is sometimes made to stand for are kept separate in this paper. The first is the observed planted-handicap recovery threshold for a particular search class, budget and handicap family, which is what the calibration measures. The second is the sampling uncertainty of a fixed policy comparison, which is a property of the cell's sample size and is described

by its interval (§7.3), not by the floor. The third is the preregistered decision threshold of 1.53 pp for declaring an advance (§10.4), which the protocol set equal to the phase-2 calibration floor of the principal responder class. The protocol itself describes that use as a pre-committed reference bar rather than a detection floor for a directly measured paired difference; it is a historical decision rule, not a resolution limit for every comparison in the paper.

The 1.53 pp figure has a specific origin. The phase-2 calibration planted a ladder of handicaps of increasing size in a v0.6 target and fitted a responder of the principal class at a fixed budget against each; 1.53 pp is the planted cost of the smallest rung whose recovered excess excluded zero on both banks, and the next rung down, below one point, was not recovered. The floor is therefore bracketed by the ladder's rungs rather than measured as a boundary, and it is specific to that responder class, budget, handicap family and target.

Two findings from the calibration shape the adversarial statements in this paper. First, within the tested class and budgets, the floor did not decrease with evaluation games: an early prediction that it would fall as (evaluation games)$^{-1/2}$ was not borne out at its own predicted value, since at four times the games the same rung, 1.53 pp, was the smallest recovered and the next rung down remained unrecovered, with a phase-2 excess whose interval was centred near zero rather than merely wide. This is one comparison, of two evaluation sizes for one responder class on nested banks. For the tested responder class at these budgets, then, more evaluation games and more fitting computation address different sources of uncertainty, and a planted effect below about a point and a half was not recovered by this class. Failure to recover a small planted effect does not establish its absence, and recovery of one handicap family does not establish sensitivity to every exploit of the same size. Second, floors are family-specific: the declaration channel has its own and higher floor of 2.13 pp, measured as a win-rate response to a declaration handicap, and reading a declaration-objective result against the general floor would overstate it. That figure is in win-rate units and is not applied to declaration accuracy without a justified mapping of units.

An adversary class contributes to an adversarial-response statement only after passing a planted-edge calibration at or below the effect size in question. That precondition is why the negative controls of §11.9 are run: an instrument that cannot recover a planted edge of a given size cannot report the absence of a real one of that size.

7.10 Information-access and isolation tests

The evaluated team is three identical agents with no shared secret. That their policy specification is common knowledge is legitimate (§2.3), but it also means an implementation detail could become an unintended signalling channel. Before this cycle the constraint had only ever been checked by reading the source.

Four mechanical tests now run over all four engine decision types. **S3, listening substitution** asks whether a seat's behaviour changes more when a teammate's hidden action changes than when an opponent's does, beyond what the rules explain. **S4, stream independence** asks whether the seat's private random stream is drawn independently of the deal. **S5, posterior invariance** asks whether the seat's action depends on true card locations beyond what its own posterior supports. **S6, seat isolation** replays every decision from own hand, public event stream, rules and reset seed alone, and counts the decisions the reconstruction cannot reproduce, at zero tolerance.

The tests are calibrated by three positive controls, each a policy carrying one planted violation, and each must fail exactly the tests the protocol names for it in advance and no others. §10.2 reports the calibration on holdout material. Without those controls a clean report from the gate would be uninformative.

One limitation is stated wherever these tests are used. The zero-tolerance form of S6 requires a single worker thread, because the audit depends on the deal-to-thread partition and work stealing does not fix that partition at higher counts, and it requires agents rebuilt for each deal. Rebuilding agents per deal *changes play*. The fresh-agent, single-thread configuration is therefore the one that passed the specified isolation tests. The scored configuration, with reused agents and 13 worker threads, retained a cross-deal state residual in searching agents, so these test results do not establish strict seat isolation or exact reproducibility for the scored mode, and every strength number in this paper is measured under that mode (§11.9).

7.11 Worst-case performance and empirical regret

The project's standing reporting rule is that the worst case leads and an aggregate never does, since a mean over an opponent panel rewards beating weak opponents by larger margins, which is not the quantity of interest. Two statistics are reported over a panel scored identically for every configuration. Write $e(a, o)$ for the pooled edge of configuration a against panel member o, over the four evaluated configurations a and the fixed panel $\mathcal{O}$. The **worst cell** of a is $\min_{o \in \mathcal{O}} e(a, o)$, and its **empirical minimax regret** is

$$R(a) \;=\; \max_{o \in \mathcal{O}} \Big[\max_{b} e(b, o) \;-\; e(a, o) \Big],$$

the largest shortfall from the best of the four configurations on any member. Both are defined over the evaluated configurations and the fixed panel; neither is a global minimax guarantee. They can disagree, and in §11.3 they do. We report both statistics because they assess different aspects of panel performance, and report the panel mean last and labelled as a diagnostic. Extrema and rankings computed from point estimates are descriptive: the interval printed beside a selected worst cell is that cell's own interval, not a confidence interval for the minimum over the panel, and no simultaneous or joint-bootstrap inference over the panel is supplied.

Every configuration must carry a cell for every member or the regret is undefined, which is why the panel is fixed by name in advance and scored for all configurations on the same banks. The 31 named members are 28 distinct policies, since three sealed aliases duplicate members that also appear under their own names. Three of the evaluated configurations are themselves panel members and meet themselves; a self-cell is a mirror match with an edge of exactly zero and no sampling variance. Under the higher-is-better convention a zero cell can only lower a configuration's worst cell, which it does exactly when every other cell is positive, and it can only raise a configuration's regret, so §11.3 prints the worst case both with and without self-cells so that the two can be compared; the reported regrets are unaffected because none is attained at a self-cell.

8 Adversary model

Robustness claims are only meaningful against a stated adversary, and the adversary appropriate to this domain is unusual enough to define explicitly. The definition below was written before any v0.7 measurement was taken, and it constrains how §11.8 may be read.

Evaluation unit. What is evaluated is the joint policy $\pi = (\pi_1, \pi_2, \pi_3)$ of three identical agents. The utility is the team win rate of §7.1, and parity is the benchmark: the rotation of §7.2 gives both teams every deal from both sides, so under the symmetric seating of the evaluated variant a team of three copies of the target has a 50% win rate against itself, and a response's edge is measured relative to that. The principal adversary controls all three opposing seats. A one-seat

deviation column is reported alongside it on the same deals, because a three-seat upgrade and a one-seat upgrade are different quantities that this corpus has previously conflated (§11.5).

Adversary information access. The adversary is white-box, with an explicit exclusion list. It is given the target's source, its parameter vector, and unlimited offline access to it. Each adversary seat observes its own hand and the public record, as the rules require; it is not given the deal, the deal seed, any other seat's hand, or the engine's confidence field. The exclusions are the point of the construction. Because the target team is three identical published deterministic agents, its policy specification is common knowledge and it carries no private correlation signal, which differs from the team-game models in which a sampled correlation signal is private to the team [53, 54] (§2.3). The adversary class built to exploit that property is the transcript inverter of §11.8.1.

Team composition and channel restrictions. The evaluated team is *homogeneous* in the sense that its three seats run identical policy specifications. Separately, two channel prohibitions apply: the team receives no shared secret pre-play seed, and no seat may use an illegal side channel, defined mechanically by the four tests of §7.10 over all four engine decision types. Everything else a seat-conditioned policy can do is permitted.

Adversary correlation regimes. The three adversary seats may be fitted independently, may be three copies of one vector, or may share a pre-play signal secret from the target team. The first is the *independent* regime, the second the *synchronised* regime, and the third the *ex-ante correlated* regime; the last is the only one whose result would bear on the target rather than on the search alone, and even there it remains a response within a class. It is the registered principal case. The synchronised regime is retained as the continuity column, because it is what earlier cycles of this corpus measured. §12.7 records which of the three this study actually reached.

Scope of response-value estimates. Every adversarial figure in this paper is the observed *adversarial response win-rate edge* of a particular fitted response against the target, reported with the adversary class, the number of controlled seats, the correlation regime and the search budget attached. A response's expected edge is no greater than the best-response value over any class containing it, so such a figure is informative about best-response value only as a lower bound, and only up to its sampling error: an observed point estimate is not itself a certified bound. The tested responses do not provide an upper bound on best-response value, and computing the true value is intractable even white-box. An estimated negative response edge is not evidence that a conventional non-negative exploitability quantity is negative. The consequence is stated once and observed throughout: a search that fails to find an exploit is evidence about the search, not a property of the target. The v0.4 record in this corpus contains exactly that reading error, having reported a null result as a demonstration of robustness, and §E carries the correction rather than repeating it.

Adversary calibration. An adversary class is admitted to an adversarial-response statement only once it has passed a planted-edge calibration at or below the effect size being claimed (§7.9).

Interpretation limits. The definition does not license the claim that a policy passing every test is unexploitable, and it does not license reading the opponent panel's worst cell as an exploitability figure. The panel measures robustness across playstyles; the adversary search measures what a fitted responder achieves. They are different objects and this paper keeps them in different subsections.

9 Exploratory development

This section reports the development phases: what the instrument was shown to be capable of detecting, what an open adversary search found, and which candidate mechanisms were tested. All measurements here are on *training* material and are exploratory in the sense of §7.7. None supports the paper's primary claim, and every pooled interval reported in this section has a lower bound below the registered 1.53 pp reference bar, so each is a measured effect rather than a confirmed one.

9.1 Baseline selection

v0.6 deployed its policy with test-time search disabled, while also having measured that search working. The configurations available at the end of that cycle therefore form what this paper calls the *v0.6 frontier*: the set comprising the deployed non-searching policy and the searching configurations `F-cheap` and `F-mid`, ordered by search cost. The term denotes that set of evaluated configurations, not a measured Pareto frontier in cost and performance; no dominance criterion was applied.

This distinction drove the most consequential methodological choice in the programme. Several available configurations already exceeded the deployed policy, including unfitted single-switch deviations of it, which motivated a searching configuration as the primary baseline. The comparison target was fixed as `F-cheap`: the v0.6 policy with its endgame-truncated search enabled at the same search settings SESTINA v1.0 uses, and the least expensive searching member of the frontier. §10.4 records that the choice was registered in advance. Table 3 defines every configuration this paper compares.

9.2 Detection calibration

The calibration procedure of §7.9 was built in this phase, which measured the principal responder class's floor at 1.68 pp and predicted that it would fall as (evaluation games)$^{-1/2}$. The phase of §9.3 re-measured the same handicap ladder at four times the games: the smallest recovered rung was then the 1.53 pp rung, and the next rung down remained unrecovered, so the prediction was not borne out within the ladder's resolution. That is the first of the two findings §7.9 reports, the second being that floors are family-specific, with the declaration channel's at 2.13 pp. Two further results shape later sections.

Transcript-inversion diagnostics. Because the target team is deterministic and published, an adversary can invert its transcript, asking for each observed ask which deals would have produced it and contracting the posterior accordingly. Measured, this contracts the deal posterior by approximately 2.0 bits per observed ask beyond the certificates the rules already supply. Converting those bits into play is worth +1.52 pp unfitted against the deployed policy.

The gain was not target-specific. The excess tracks planted-edge size to within ±0.4 pp on all eight rungs of the calibration ladder, which is consistent with a general strength gain rather than with a response that depends on a particular weakness of the target; a broadly stronger policy can still be a valid profitable response, so this is a statement about specificity rather than about exploitability. And no linear handicap adds more than 0.07 bits per ask, so within this policy family no readability handicap was found for a stochastic policy to remove. A proposed stochastic policy conditional on the inverter succeeding was therefore not pursued.

Revised search-cost estimates. The v0.6 paper priced the searching configuration at "three orders of magnitude" the blueprint, a ratio that mixed an all-threads timing with a single-thread one. Re-timed here on one basis, it is 242×. But that figure describes the *unrestricted* search, which is not the frontier point anything was measured at, and phase 3 re-measured the endgame-restricted operating point at approximately 3.2×. §11.10 corrects that correction in turn.

9.3 Adversary generation against baseline configurations

The constraint on this phase was that the searches must not share a bias, since repeated restarts of one method do not provide methodological diversity. Every exploiter search in the corpus before it had been the same search: a cross-entropy fit of a linear vector maximising win rate from one seed against one target. This phase constructed six adversary classes in the synchronised regime, plus one fitted search in the ex-ante-correlated regime, and measured channel ceilings before fitting anything against them.

Adversarial-search outcome. Nothing constructed produced a resolved adversarial edge against the v0.6 frontier beyond the class floor. The largest adversarial response edge found against the deployed policy is an in-class figure of +0.79 pp [+0.48, +1.10] over 96,000 games, which is below the class floor of 1.53. Seven independently fitted searches gave the same picture and none cleared its class floor.

Every configuration that beat the deployed policy by more than the floor proved on measurement not to be target-specific: the target's own test-time search at +1.89 pp, and a single out-of-class coordinate at +2.71 pp whose gain survives against opponents sharing none of the target's machinery. The second of those is the contestation weight of §6.3; §11.4 measures on holdout what it contributes and finds it the largest observed leave-one-out drop among the five components SESTINA v1.0 carries, rather than the whole of the gain.

Two further results from this phase bear on SESTINA v1.0's design. The declaration-urgency channel that the phase expected to be its headline has a measured ceiling of +0.38 pp [−0.06, +0.81] and did not yield a resolved adversarial advantage at the tested budget, but the same channel is worth +1.23 to +1.62 pp to a configuration that simply disables it: the adversary owns the smaller half of a two-point defect in the target. And the event-count cliff of §6.5 was not reached by any adversary in the tested searches, while a configuration that removes the counter inherits a self-play game-length tail of 405 events, which is the obligation §6.6 discharges.

9.4 Candidate-mechanism evaluation

Candidate modifications were evaluated on development data before configuration freezing. Five were developed in isolated branches, gated before being scored, and checked against the side-channel definition by the harness of §7.10 rather than by inspection. One was retained, and it is the configuration change described in §6 rather than an architecture. Four did not demonstrate an additional performance benefit under the tested settings: alternative declaration allocation, learned leaf evaluation, per-decision training objectives, and learned action ranking. The following subsections report their measured outcomes and limitations. These results concern the tested implementations and budgets, not all methods in the corresponding classes.

9.4.1 Declaration-allocation comparison

The design register's second-ranked hypothesis, at priority 0.48, held that v0.6's entire margin lives in the declaration channel and that roughly two points of it remained. Its proposed mechanism

was that the deployed allocator, a product of per-card marginals, selects a different allocation from the joint posterior's argmax often enough to matter.

The measurement did not support that mechanism. Over approximately 210,000 recorded voluntary declarations, across two banks and both urgency settings, and counted over capture cells that include re-runs of the same seeds, the sequential approximate joint's argmax and the marginal-product argmax selected the same allocation at every one, with the joint score holding a strict opinion on 95.6% to 100% of ambiguous cases. This agreement is empirical: the two procedures are not equivalent in general, since the six cards of a half-suit are coupled through the capacity constraint, and no argument that the sequential conditioning preserves the marginal-product ranking is supplied.

Two further measurements bear on the direction rather than only the mechanism. First, the exact constraint-joint maximiser (19) performed worse than the deployed approximation on the declaration decision itself: replayed on one capture cell of 17,978 voluntary declarations, it corrects 218 declarations and breaks 454, a net -1.313 pp of declaration accuracy on that cell. At the design register's fitted conversion of about 1.2 win-rate points per point of declaration accuracy (range 1.0 to 1.4 across banks), this corresponds to approximately -1.58 pp of win rate; that figure is inferred from the declaration-level difference, not measured as a game outcome. This is the second comparison of the kind §4.3 reports. Second, 72.4% to 74.4% of the error class occupies states in which the exact constraint-conditioned distribution assigns the competing allocations equal probability, so most of the apparent headroom is an information limit under that model rather than a policy defect.

The channel is also largely downstream of a defect that SESTINA v1.0 removes for other reasons. Disabling the urgency escalation eliminates 71% of the entire allocation-error class and drops the oracle ceiling for a perfect allocator from 2.45 to 0.84 pp, which is below the reference bar. The consequence is a composition constraint this paper observes: a declaration-channel gain and the urgency-off gain should not be added, because they overlap in the same error class.

9.4.2 Learned leaf evaluation

An inherited hypothesis held that the test-time search cannot be judged until its leaf evaluator is rebuilt, the deployed one being algebraically close to a rescaling of the hit probability. A leaf was fitted for the purpose, and the statistic that matters, between-candidate R^2, which is the only component the paired deviation rule can consume, improved from 0.00108 to 0.07976 in sample and from 0.00664 to 0.08424 out of sample; the in-sample ratio is 74-fold, the out-of-sample ratio is much smaller, and the absolute R^2 remains modest in both, so the direction replicates and the multiple does not. The fit was scored over approximately 1.3M leaves across two search regimes and three banks.

In play it is worth $+1.84$ pp against the material control's $+2.01$ pp, losing on both banks individually. The diagnosis is conversion failure rather than fit failure: the rule's deviation rate is 0.3205 for the deployed material leaf, 0.3217 for the fitted one and 0.3237 for a third arm that merely rescales the material leaf's constant, so at the deployed shrinkage the deviation rule is nearly leaf-invariant. The tested leaf replacement did not improve playing strength under the deployed selection rule. This illustrates a mismatch between a surrogate metric and playing performance related to the abstraction pathology the poker literature documents [66], in which a model improved on its own metric can leave play unchanged or worse; no formal mapping between the two settings is claimed.

The inherited hypothesis was discharged separately and by re-measurement rather than by the new evaluator. The single underpowered cell on which it rested, one 4,000-game cell at $+0.08$ pp,

is the `depth=24` configuration; re-run on the training banks with v0.6's own leaf it gives +1.02 [−0.55, +2.62], and full-game search at `depth=12` gives +1.78 and +1.25 pp, pooled +1.52. Phase 1's +2.19 pp refuted the conditional in the endgame regime only, at `depth=12` with `maxq=26`; it was this full-game cell that left it binding for full-game search.

9.4.3 Per-decision training objectives

A further hypothesis held that a per-decision objective is necessary because the scoreboard is too noisy to resolve small effects. Its precision claim is supported and its fitting form is not, and the two halves are independent.

The design effect of deal clustering was measured for the first time in this cycle. A one-point-equivalent effect reaches a planning half-width of its own size in 284 to 560 games on the declaration channel and 244 to 299 on the allocation-error channel, against 9,604 on the scoreboard, a ratio of 17 to 39 times. That ratio compares planning sample sizes for different outcome channels, linked by the design register's assumed conversion factors between per-decision accuracy and win rate; the per-decision figures apply the measured clustering design effect and the scoreboard figure is the independent-game approximation of §7.4, so the ratio is a planning comparison rather than a measured precision gain on one estimand.

But all three self-oriented per-decision objectives, fitted at matched budget against a per-game control, produced worse policies: −2.26, −1.59 and −1.20 pp, each replicated in sign and two of them with intervals beyond the reference bar. Each fit moved its own proxy in the intended direction and lost anyway. The declaration-oriented fit increased declaration accuracy by +0.39 pp while reducing ask accuracy by −2.33 pp, and the ask-oriented fit did the reverse. The two per-decision channels exhibited a trade-off in these fits, so a fitter holding 55 coordinates improved one at the expense of the other. The results motivate further study of the estimator without establishing the tested proxy as a suitable training objective.

9.4.4 Learned action ranking and tie-breaking

In the v0.6 mirror 53.80% of contested ask decisions tie bit-for-bit, and the open question was whether anything can be *selected* within that set rather than merely randomised. The learned candidate this cycle built was not built for that question. It was the brief's wildcard, aimed at amortising test-time search, and it was fitted unrestricted over 88,502 searched decisions. Deployed unrestricted it scores −1.83 pp pooled over 48,000 games, negative on both banks [−2.22, −1.45]. Restricting it to the tied set is one added switch on the same fitted weight vector, and it is there that the tie-group question gets answered.

Restricted, against the deployed policy it scores +1.19 pp, not statistically resolved from the free public-history tie-break's +1.14 pp; head to head against that free tie-break at 48,000 games it is −0.01 pp ± 0.45 (95% half-width), replicated in sign. The restricted learned ranker did not demonstrate an advantage over public-history tie-breaking, and the appropriate control for any mechanism operating inside the tied set is the decorrelated policy rather than the deployed one.

The same instrumentation explains why. Setting every candidate's true advantage to exactly zero and feeding the capture's real standard errors through the engine's real deviation rule reproduces the search's observed deviation rate to a 0.4% relative error, on a quantity the null was never fitted to. The search's per-decision selection signal is approximately 84% winner's curse on its own Monte Carlo draw, and approximately 95.5% within the tied set, where the rule applies no shrinkage at all.

9.4.5 Forced-endgame improvement estimates

Closing the entire forced-endgame accuracy gap at the measured incidence, re-derived from this cycle's own capture, is estimated at 0.030 pp, which is the register's own figure corrected upward by a factor of two and still a fiftieth of the reference bar. That estimate holds for the incidence observed in the measured play. The route by which the estimate could grow is an adversary driving the target's incidence upward, and the one construction measured does not do that: a team that never declares voluntarily raises its *own* forced-endgame rate, not its opponent's, and pays 6 to 8 pp for it. No evaluated intervention increased the target's forced-endgame incidence, so this route remains untested rather than closed (Appendix E.2 records the earlier report's unsupported figure for it).

9.5 Configuration selection and freezing

15 configurations were scored against the deployed policy during development. One was selected, its specification written to an artifact whose reconstruction is executed rather than asserted, and the configuration frozen. Nothing about it was changed afterwards, and no gradient or population fit was applied to it. §11.2.3 quantifies what that selection could be worth under the null.

10 Experimental design

Figure 2 summarises development, configuration freezing, and holdout evaluation: what was developed on training material, what was frozen, and what was measured against sealed material.

10.1 Evaluated configurations

Table 3 defines the evaluated configurations. The paper's central distinction depends on two of its rows. SESTINA v1.0 is compared against `F-cheap`, a configuration that existed before this cycle, and improves on it. It is also compared against the phase-2 composite, which was assembled *during* this cycle by the adversary-generation phase, and that comparison is not statistically resolved. Those are different claims about different objects and §11.2.1 keeps them apart.

10.2 Evaluation data and integrity checks

All 7 bank digests reproduce, at 24,000 deals each (Table 4). The sealed adversary half decodes to 14 configurations whose plaintext SHA-256 matches the commitment recorded at sealing. The frozen configuration's artifact verifies, its specification rebuilding character for character from its own ordered option map, and its round trip with search enabled is identical over 800 games. The engine's soundness audit reports 0 violations in 6,737,436 checks and its self-test passes on 424,986 checks.

The homogeneity controls of §7.10 calibrate as specified: each planted violation fails exactly the tests named for it and no others, identically on both banks, over 12 cells (Table 5).

10.3 Behavioural validation results

SESTINA v1.0 passes all eight rules of Table 6: 0.05820% provably unsuccessful asks against a 0.10% bar, longest dead run 1, mirror tail maximum 134 with 99th percentile 130, and 0 seat-isolation mismatches over 542,483 audited decisions. Reported but not gated: 99.805 events per

Table 3: Every configuration this paper compares. Configurations marked pre-existing were available before the v0.7 programme began; the phase-2 composite was assembled during it. Only SESTINA v1.0 was frozen for holdout evaluation.

Name	Definition	Origin	Role in this paper
SESTINA v1.0	the evaluated configuration of §6	frozen in this cycle	the subject; evaluated on sealed holdout material
v0.6 deployed	the previous release, search disabled	pre-existing	lineage baseline and reference opponent for ablations
`F-cheap`	the v0.6 policy with its endgame search enabled at the search settings SESTINA v1.0 uses; the least expensive searching member of the v0.6 frontier	pre-existing	the registered primary comparison target
`F-mid`	a more expensive frontier configuration	pre-existing	secondary frontier comparison
the phase-2 composite	v0.6 frontier search, plus the contestation weight, plus one switch	assembled in development phase 2	the configuration assembled before the architecture work; the comparison of §11.2.1
v0.5, v0.4, v0.3	earlier lineage releases	pre-existing	lineage baselines and panel members
scripted archetypes	thirteen hand-built playstyles, including deception archetypes	pre-existing	opponent panel members
sealed adversaries	fourteen configurations sealed before this cycle	sealed in phase 2	opponent panel members, decoded only at evaluation

Table 4: The deal banks. Each digest was recomputed before any cell was played, by folding the six dealt hands and the dealer of every deal into a rolling hash without constructing a policy or playing a game. The committed digests had been computed by a different build of the engine at sealing time, so agreement is between two binaries as well as two dates.

Bank	Deals	Digest reproduced	Matches commitment
7090001	24,000	`896dbc89be124d85`	✓
7090002	24,000	`0b6e40d834ac0ca1`	✓
7090003	24,000	`863bea69baf6e73c`	✓
7090004	24,000	`54f257c3f8ae9fab`	✓
7090005	24,000	`268a1dae71a31713`	✓
7091001	24,000	`958ada042cc26900`	✓
7091002	24,000	`5c39af3b5e0bd9a0`	✓

game, 50.716% ask hit rate and 2.556% misdeclaration. The negative control fails 5 rules, namely G1, G2, G3, G4, G5, which are exactly the rules the protocol names for it in advance.

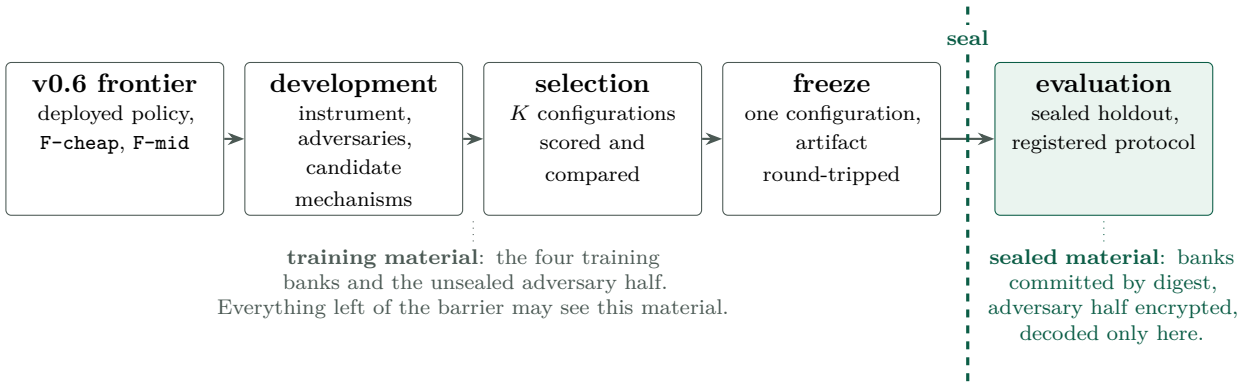


Figure 2: The development and evaluation protocol. Configurations were generated, scored and selected using training material only: the training banks 7030001–7030004 and the unsealed half of the adversary population. Exactly one configuration was then frozen, and evaluated against material whose digests had been committed before any candidate existed, under a protocol registered in advance (§7.6). The seal is enforced by the engine's match runner: a scored run naming a sealed seed fails unless an explicit variable is set, which was set once and recorded; Appendix F (D6) lists the diagnostic and fitting commands that are not gated.

Table 5: Calibration of the side-channel controls on holdout material. *Must fail* is the set the protocol names for each planted violation in advance. It is not one test each: the seed cheat is visible to two of them. A control failing a test outside its set, or missing one inside it, would mean the gate does not discriminate.

Planted channel	S3	S4	S5	S6	Must fail	As required
`v07x:cheat=seed`	PASS	FAIL	FAIL	PASS	S4, S5	✓
`v07x:cheat=shared`	PASS	PASS	PASS	FAIL	S6	✓
`v07x:cheat=conv`	FAIL	PASS	PASS	PASS	S3	✓

Table 6: The soundness gate, eight rules, run on training material by design. The negative control is a configuration constructed to be rejected.

Rule	SESTINA v1.0	v0.6	`F-cheap`	Negative control
provably-dead asks $\leq 0.10\%$	ok	ok	ok	**fail**
longest dead run ≤ 5	ok	ok	ok	**fail**
games with a run $\geq 6 = 0$	ok	ok	ok	**fail**
action-limit games $= 0$	ok	ok	ok	**fail**
mirror tail max < 220, p99 ≤ 150	ok	ok	ok	**fail**
declarations at/after event $220 = 0$	ok	ok	ok	ok
S3/S4/S5 certified, zero tolerance	ok	ok	ok	ok
S6 $= 0$ at `--threads=1 --freshagents`	ok	ok	ok	ok
verdict	**PASS**	**PASS**	**PASS**	**FAIL**

10.4 Preregistered evaluation protocol

The protocol fixed, before any holdout deal was played: the primary comparison target (`F-cheap`, for the reason in §9.1); the primary decision rule; seven secondary claims each with its own threshold; and seven conditions that would constitute non-confirmation. Appendix C reproduces them with their recorded verdicts and an uncertainty-aware reading of each.

The primary decision rule is stated here because the paper names what was satisfied rather than using the protocol's label *certified*. The registered rule was that the comparison against `F-cheap` would be regarded as a confirmed advance if the pooled edge had a lower bound above 1.53 pp, *and* the sign replicated on both banks, *and* the configuration passed the behavioural gate. The 1.53 pp threshold is the phase-2 calibration floor of the principal responder class, adopted by the protocol as a pre-committed reference bar for a directly measured paired difference (§7.9). A pooled edge positive with a lower bound below 1.53 but above zero, with the sign replicating, would be recorded as a measured but unconfirmed advance; an interval containing zero, a sign that did not replicate, or a gate failure would be recorded as no advance.

The protocol also fixed in advance the sentence to be used if the partner-regime comparison retained a particular ordering, so that its direction could not be chosen after the fact, and recorded that on training material that comparison ran *against* SESTINA v1.0. §11.5 reports the outcome and uses the registered wording.

10.5 Experiment inventory

428 scored cells over 4,322,400 games (Table 7). No registered cell was dropped; 18 were added, and Appendix F records why.

Table 7: Every registered battery against what was measured. *Registered* is the count the protocol names, not the count that ran: two deviations add a reference cell and sixteen control cells, because without them three of the protocol's own questions are not computable from the cells it lists.

Battery	What it measures	Registered	Added	Measured	Status
B2	headline strength against the frontier	10	0	10	complete
B3	the shared 31-member panel	248	0	248	complete
B4	a fresh adversary search	16	0	16	complete
B5	the attribution lattice	22	2	24	complete (D1)
B6	the partner-regime table	64	0	64	complete
B7	cross-play between independent runs	24	0	24	complete
B8	the rule-dialect table	16	0	16	complete
B9	the negative controls	10	16	26	complete (D2)
B10	the S6 residual	16	0	16	complete

Scored cells carry no soundness-audit evidence, and this paper does not imply otherwise. The match harness does not run the audit, so the audit check count is 0 in all 428 of them, and a report of zero violations over a scored cell would be zero out of zero. The engine-wide audit figure quoted in §5.6 and §10.2 comes from the dedicated verification run and is not restated as a property of the battery.

11 Results

Every holdout measurement in this section is produced from the artifacts by the generator described in Appendix G and emitted into this document mechanically. The figures those measurements are read against are not measurements of this holdout: registered expectations, earlier cycles' results, and structural counts. Appendix G records where each comes from. The section answers RQ1 in §11.1, RQ3 in §11.4, and RQ2 in §11.5 through §11.8, before reporting the evaluation controls and the computational cost.

11.1 Baseline comparisons

Against `F-cheap`, the registered primary baseline, SESTINA v1.0 achieved a win-rate edge of +3.33 pp relative to parity (95% CI [+2.88, +3.78]) over 48,000 games; this edge corresponds to a 53.33% head-to-head win rate against `F-cheap`, not to a change from a separately measured baseline win rate. The two bank estimates were +3.67 and +2.99 pp. The pooled lower bound exceeded the preregistered 1.53 pp threshold, both bank estimates were positive, and the configuration passed the specified behavioural gate in its designated test mode (§10.3). The registered primary decision rule was therefore satisfied, and the comparison satisfies the registered primary criterion. The scored-mode isolation limitation is reported in §11.9. Figure 3 places this comparison beside the others, and Table 8 gives the cells behind it.

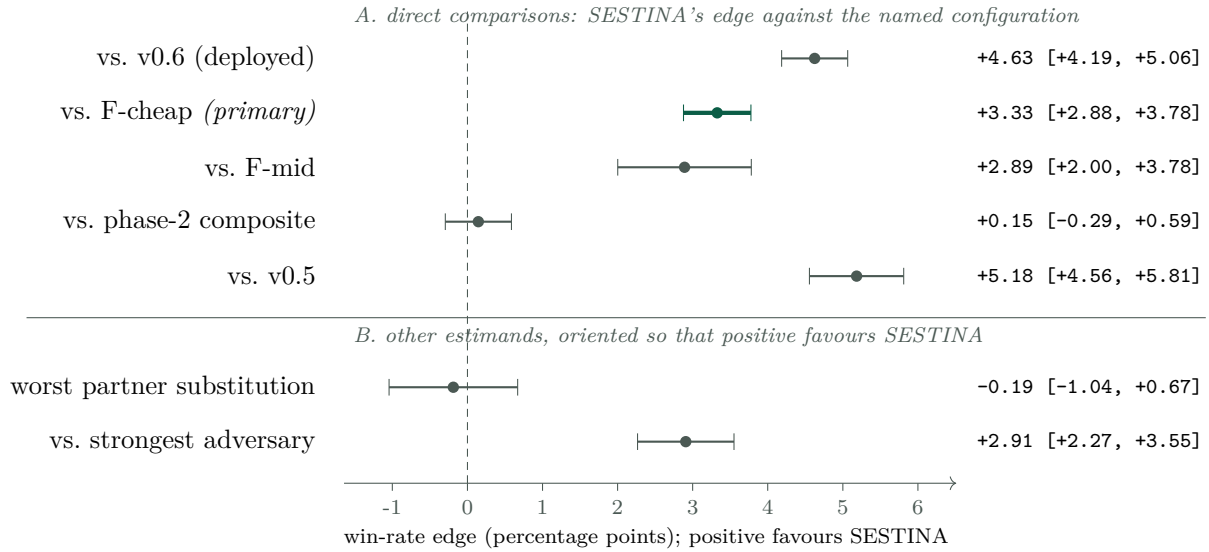


Figure 3: Principal comparisons for SESTINA v1.0. Each point is a pooled win-rate edge in percentage points over both holdout banks (§7.3: cross-bank pooling of per-bank deal-clustered bootstrap intervals; the bars are 95% intervals). Both panels are oriented so that positive values favour SESTINA v1.0. The upper panel shows the five direct head-to-head comparisons, with the registered primary comparison highlighted. The lower panel shows two different estimands: the worst partner-substitution row of §11.5, which is a difference between two upgrade effects (SESTINA v1.0's upgrade over v0.6 minus v0.6's upgrade over v0.5, with the named partner) rather than a head-to-head edge, and SESTINA v1.0's edge against the strongest of the 8 adversarial responses of §11.8, shown here as the negation of the adversary's edge printed in Table 22, whose convention is the adversary's edge against SESTINA v1.0.

Table 8: Strength against the v0.6 frontier and lineage baselines. Each row is SESTINA v1.0's win-rate edge in percentage points against the named configuration, paired duplicate blocks, pooled over both holdout banks with the per-bank estimates beside it; intervals are 95% intervals as constructed in §7.3.

Cell	Opponent	Edge (pp)	95% CI	Games	Sign replicates
B2.1	v0.6	+4.63	[+4.19, +5.06]	48,000	yes
B2.2	`F-cheap`	+3.33	[+2.88, +3.78]	48,000	yes
B2.3	`F-mid`	+2.89	[+2.00, +3.78]	12,000	yes
B2.4	the phase-2 composite	+0.15	[−0.29, +0.59]	48,000	yes
B2.5	v0.5	+5.18	[+4.56, +5.81]	24,000	yes

Against the lineage baselines, SESTINA v1.0 gains +4.63 pp over the deployed v0.6 policy [+4.19, +5.06], +2.89 pp over `F-mid` [+2.00, +3.78], and +5.18 pp over v0.5.

Against the phase-2 composite it gains +0.15 pp [−0.29, +0.59], an interval containing zero. §11.2.1 treats that result, which the protocol had named in advance as a condition of non-confirmation.

Descriptive declaration accuracy. The declaration-family floor applies to the declaration-accuracy column. The harness emits declaration accuracy as a bare rate with no interval of any kind, so no interval can be quoted and no claim about the declaration channel is made. SESTINA v1.0 runs 0.970 to 0.982 across the five cells against opponents' 0.976 to 0.986. That is a diagnostic.

11.2 Secondary criteria and diagnostic comparisons

The protocol prespecified seven non-confirmation conditions. Two arose, and they are reported here rather than in a discussion of caveats; the remaining diagnostic comparisons follow.

11.2.1 Phase-2 composite comparison

SESTINA v1.0's comparison with the phase-2 composite was +0.15 pp (95% CI [−0.29, +0.59]) over 48,000 games. Both banks are positive, so the sign replicates, but the interval includes zero, so the evaluation does not resolve an additional benefit over that composite.

The registered interpretation is explicit, and this paper adopts it. The increment that an earlier phase had reported over the composite, +1.42 [+0.18, +2.68] on a single bank, did not replicate on holdout.

The arithmetic makes the cell interpretable. SESTINA v1.0 minus the phase-2 composite is exactly the tie-break repair, the urgency removal and the stall detector added, with one further switch removed. §11.4 measures the stall detector's contribution as identically zero, and the removed switch's leave-one-out drop was zero on training material. The protocol states that group's training value in advance as +0.78 [+0.33, +1.22]; on holdout it is +0.15 [−0.29, +0.59].

This result does not alter the positive comparison with `F-cheap`, which is a different baseline (Table 3): `F-cheap` is a configuration of the v0.6 frontier that existed before this cycle, and SESTINA v1.0 improves on it by a replicated margin. What the result does is limit the attribution of further improvement to the modifications introduced after the composite was assembled; it does not establish that those modifications contribute nothing.

11.2.2 Component-attribution criterion

The protocol's third non-confirmation condition is that the lattice cannot locate the gain: if removing every single component in turn costs less than a third of the whole, the configuration has a gain it cannot explain. Read per bank, as §7.5 requires, the two banks disagree (Table 9).

Table 9: The attribution location test, read per bank. *Whole* is the measured whole, *one third* the protocol's threshold, and *largest single drop* the biggest leave-one-out drop the lattice contains.

Bank	Whole	One third	Largest single drop	Verdict
7090001	4.883	1.628	`L-r12` +2.350	LOCATED
7090003	4.575	1.525	`L-r12` +1.025	NOT LOCATED

The two banks' largest drops differ by 1.325 pp against a half-width on that difference of comparable size, so the split is real on the point estimates and is not itself statistically resolved. Under the replication criterion the pooled verdict is not replicated (no), and the supported interpretation is that this battery does not settle whether the gain is located by the protocol's one-third rule. Note that the disagreement is between verdict labels rather than between signs, which is itself a recorded extension of the criterion. The sub-additivity claim of §11.4 passes on both banks; it is the location test specifically that splits.

11.2.3 Selection-bias diagnostics

15 configurations were scored against the deployed policy during development. At the lattice cell size the per-cell standard deviation is 0.3163 pp, so the quantity the protocol registers as the expected maximum of 15 draws under the null that none differs from the reference is $\sigma\sqrt{2\ln K} = 0.74$ pp, which is an upper bound on that expectation rather than the expectation itself.

The held-out deals were not used for configuration selection, so the selection term on holdout material is 0.00 pp: this removes direct selection on those outcomes, subject to the documented separation procedure of §7.6, and it is a property of the design rather than a measurement. Table 10 sets the four registered quantities against their holdout measurements.

Table 10: The four quantities the registered selection check tabulates, against their holdout measurements. Three reproduce or exceed; one falls short. These are not every figure the protocol states in advance, and two of the partner-regime predictions in §11.5 also come in low.

Quantity	Training	Holdout	Shortfall
B2.4 — `rtie` + urgency-off + `stall`, as a group	+0.78	+0.15	−0.63
B6 self-play row, opponent `v05`	+2.94	+4.49	+1.55
B7 diagonal (self-play)	+4.51	+4.55	+0.04
B7 off-diagonal (cross-play)	+4.48	+4.55	+0.07

3 of the four reproduce or exceed their training values. The largest shortfall, 0.63 pp against the registered 0.74, falls on the composite cell, which is precisely the quantity measuring the group of components the configuration was selected over. That is the pattern the check was written to look for and it appears where selection occurred. One quantity cannot establish that the group's training value was entirely selection effect, and the check is not a formal correction (§7.7); what it provides is a consistency test that the data pass in the direction §11.2.1 indicates.

11.2.4 Preregistered non-confirmation criteria

Table 11 answers each of the seven in the protocol's own order. The recorded verdicts apply the protocol's rules as written; the third column states the reading each rule uses, because the rules are not uniform: conditions 1 and 5 are decided on interval bounds, condition 2 on whether the interval contains zero, condition 3 per bank on point estimates, and conditions 4 and 7 on point estimates against thresholds that the deciding intervals cross.

Two of the seven are met or unresolved, being condition 2 outright and condition 3 by non-replication. The other five are not met on the recorded reading; conditions 4 and 7 are not resolved against their thresholds once uncertainty is taken into account.

Table 11: The seven registered non-confirmation conditions against what the holdout returned. *Recorded* is the verdict under the protocol's rule as written; *reading* states the statistic that rule uses and, where the deciding interval crosses the threshold, says so. Appendix C reproduces the criteria with the same distinction.

#	Condition	Recorded	Reading
1	the frontier comparison	not met. $+3.33$ $[+2.88, +3.78]$, sign replicates	lower bound above 1.53; resolved
2	the composite comparison	met. $+0.15$ $[-0.29, +0.59]$	interval contains zero; the comparison is unresolved rather than negative
3	attribution: the location test	does not replicate. LOCATED on one bank, NOT LOCATED on the other	per-bank point estimates against one third of the whole; not resolved
4	brittleness under partner change or cross-play	not met. Minimum -0.19; cross-play off-diagonal minus diagonal $+0.01$	point estimates; the deciding partner interval $[-1.04, +0.67]$ crosses the -1.00 threshold, and the cross-play contrast carries no interval
5	a successful adversary	not met. No search found a positive edge; largest upper bound -2.27	upper bounds below the class floor; resolved for the tested searches
6	the soundness gate	not met. All 8 rules pass	deterministic checks on training material
7	the validity controls	not met. Every control behaves as specified on the point-estimate reading	point estimates; on the lower-bound reading the `0.08` rung is not recovered and the sub-floor rung's upper bound exceeds the floor (§11.9)

11.3 Opponent-panel performance

248 cells. Four configurations were scored against the same 31-member panel (28 distinct policies, §7.11) on the same two banks: 19 near-class members at 6,000 deals per bank, 10 far-class archetypes at 1,500, and 2 expensive members at 1,200. All four carry a cell for every member, which is what makes the empirical regret of §7.11 defined. Table 12 gives each configuration's worst cell. The extrema and rankings in this subsection are computed from pooled point estimates and are descriptive; the interval beside a worst cell is that cell's own interval.

Table 12: Worst panel cell per configuration: the panel member against which the configuration's pooled win-rate edge (percentage points, both banks) is lowest, with that cell's 95% interval. Per-bank values are given in bank order, 7090001 then 7090002.

Configuration	Worst cell	Edge (pp)	95% CI	Per bank
SESTINA v1.0	phase-2 composite	-0.04	$[-1.41, +1.33]$	$+1.67$ / -1.75
v0.6 deployed	`SEALED:X01xC3f`	-4.24	$[-4.86, -3.62]$	-3.86 / -4.63
`F-cheap`	phase-2 composite	-2.08	$[-3.48, -0.69]$	-2.21 / -1.96
phase-2 composite	phase-2 composite	$+0.00$	$[+0.00, +0.00]$	$+0.00$ / $+0.00$

SESTINA v1.0's worst cell is -0.04 pp $[-1.41, +1.33]$ against the deployed policy's -4.24 $[-4.86, -3.62]$, satisfying the registered secondary claim that the worst case not be catastrophic.

Two qualifications belong with that number.

First, it does not replicate in sign. SESTINA v1.0's worst cell is +1.62 on the first bank and −1.75 on the second, attained at a different panel member on each (`SEALED:X01xC3f` and `composite`); on one bank SESTINA v1.0 has no negative cell anywhere on the panel. Under the criterion of §7.5 the pooled value is not replicated. What survives replication is the comparison the protocol actually gated on, since SESTINA v1.0's worst cell is far above the deployed policy's on both banks, rather than the pooled value itself.

Second, it is not the best worst case of the four. Configurations that are themselves panel members meet themselves in a mirror cell that is exactly zero, which can only lower their worst case (§7.11), so the worst case is printed both ways; the phase-2 composite is ahead of SESTINA v1.0 on both readings, at +0.00 with its self-cell and +1.23 without.

Table 13: Empirical minimax regret over the shared panel (§7.11), in percentage points of win-rate edge on pooled point estimates, best first, with the panel member at which each configuration's largest shortfall from the best of the four is attained. No interval is attached to this statistic. The protocol sets no threshold on it and requires only that it be reported.

Configuration	Minimax regret (pp)	Attained against
`F-cheap`	4.00	v0.3
phase-2 composite	4.08	`withholder`
SESTINA v1.0	4.53	`withholder`
v0.6 deployed	5.67	`SEALED:X01xC3f`

SESTINA v1.0 is 3rd of four on empirical minimax regret (Table 13), at 4.53 against F-cheap's 4.00 and the phase-2 composite's 4.08. The two statistics rank the configurations differently: the configuration with the best worst case is not the configuration with the best regret. The regret difference is concentrated in the far archetypes (Table 14), opponents every configuration already beats by between 24.72 and 50.00 pp, which is a reason to weight it less but not a reason to discount it.

Table 14: The far archetypes, where the minimax-regret differences are concentrated.

Far archetype	SESTINA v1.0	v0.6	`F-cheap`	Composite
v0.3	+28.72	+25.47	+24.72	+28.40
v0.2	+34.32	+31.68	+32.67	+33.17
`lockout`	+30.78	+28.65	+28.97	+29.07
`detective`	+29.65	+28.08	+27.98	+28.32
`hunter`	+46.93	+48.58	+48.65	+47.10
`diversifier`	+43.40	+44.90	+45.18	+43.32
`bluffer`	+49.87	+49.98	+49.95	+49.88
`silent`	+32.75	+34.23	+34.05	+30.92
`withholder`	+25.35	+29.88	+29.55	+25.80
`random`	+50.00	+50.00	+50.00	+50.00

Against the hardest sealed adversary the ordering reverses: on `SEALED:X01xC3f`, SESTINA v1.0 scores +1.43 [+0.80, +2.06] where the deployed policy scores −4.24 and `F-cheap` scores −1.25. 2 of the four configurations beat that member with a lower bound above zero, namely FROZEN, composite.

Descriptive panel means. Over the 31 shared cells, which comprise 28 distinct policies since three are named twice and are therefore double-counted by any mean, the figures are SESTINA v1.0 +14.94, the phase-2 composite +14.35, `F-cheap` +13.28 and the deployed policy +11.99. The reporting rule of §7.11 forbids leading with this, and it points the opposite way from minimax regret. Appendix D gives every cell.

11.4 Component ablations

The registered prediction was that the contestation weight of §6.3 carries the gain, checkable as a leave-one-out drop of roughly two percentage points. On that check it holds: the holdout drop is +1.69 pp [+0.80, +2.57], with the training figure +2.10 inside the interval, and an add-one-in of +2.18. It is the largest observed leave-one-out drop of the five, which is a descriptive ordering rather than a statistically established largest contribution, since the drops' intervals overlap. The stronger statement, that this coordinate is where the gain lives, is not supported at this precision: the drop's lower bound is +0.80, and §11.2.2 reports the protocol's own location test failing to replicate. Table 15 is the lattice.

Table 15: The attribution lattice, in percentage points of win-rate edge against the deployed v0.6 policy on the two lattice banks. Add-one-in is the edge of the deployed policy plus the named component; leave-one-out is the drop from SESTINA v1.0's edge when the component is removed, a between-cell contrast whose interval is propagated as in §7.3. Reference cell SESTINA v1.0 against the deployed policy on the lattice banks: +4.73 [+4.10, +5.36]. Harness identifiers are given in parentheses.

Component	Add-one-in	95% CI	Leave-one-out	95% CI
half-suit contestation (`r12=25`)	+2.18	[+1.55, +2.81]	+1.69	[+0.80, +2.57]
guarded search (`s1=1`)	+1.91	[+1.38, +2.44]	+0.57	[−0.33, +1.46]
public-history tie-break (`rtie=1`)	+0.71	[+0.08, +1.34]	+1.08	[+0.20, +1.97]
urgency escalation off (`urgency-off`)	+1.37	[+1.12, +1.63]	+0.64	[−0.25, +1.53]
stall detector (`stall=12`)	+0.00	[+0.00, +0.00]	+0.00	[−0.89, +0.89]
rescoring switch (`m2=0`; not in the freeze)	+0.77	[+0.66, +0.88]	—	—

The stall detector behaves exactly as predicted. It is worth +0.00 [+0.00, +0.00] added and +0.00 removed, bit-identical on holdout in win rate and in events per game alike: the rule does not fire.

The naive sum of the five components SESTINA v1.0 carries is +6.18 pp against a measured whole of +4.73 pp, a sub-additivity ratio of 0.766 [0.599, 0.933], which contains the phase-2 composition figure of 0.83. Quoting the sum would overstate the configuration by 1.45 pp. The registered attribution claim, that components are sub-additive and no single leave-one-out drop exceeds the whole, is satisfied.

One component was measured but is not carried by SESTINA v1.0. The `m2` switch is worth +0.77 [+0.66, +0.88] added to the blueprint, an interval excluding zero and larger than the tie-break's add-one-in. Its zero leave-one-out drop was measured on training material and is not re-measured here, because SESTINA v1.0 does not carry the switch and the protocol directs that it not be added back; there is no leave-one-out cell for it and there could not be.

11.5 Partner substitution and single-seat upgrades

6 of 7 changed-partner rows against a v0.5 opponent are positive (Table 16). The minimum is −0.19 [−1.04, +0.67], which is above the registered −1.00 collapse threshold on the point estimate

but *not resolved against it*, since the interval's lower bound falls below the threshold. This battery does not separate a pass from a failure on the row that decides the claim, and reading a threshold from a point estimate is a move this paper declines elsewhere (§11.9) and declines here.

Table 16: Partner-regime table against a v0.5 opponent. Each row is the difference, in percentage points of win-rate edge, between the edge of a team consisting of SESTINA v1.0 plus two copies of the named partner and the edge of a team consisting of v0.6 plus the same two partners, both against three copies of the opponent; the interval is a between-cell contrast as in §7.3. Per-bank values are given in bank order, 7090001 then 7090002.

Partners	SESTINA − v0.6	95% CI	Per bank	Repl.
itself	+4.49	[+3.61, +5.38]	+4.18 / +4.81	yes
v0.6	+1.93	[+1.04, +2.82]	+2.07 / +1.79	yes
v0.5	+0.78	[−0.09, +1.66]	+0.95 / +0.62	yes
v0.4	+1.34	[+0.46, +2.22]	+1.30 / +1.38	yes
v0.3	+1.40	[+0.56, +2.25]	+1.08 / +1.72	yes
`detective`	+1.83	[+0.97, +2.68]	+1.35 / +2.30	yes
`withholder`	−0.19	[−1.04, +0.67]	+0.00 / −0.38	**no**
`lockout`	+1.89	[+1.04, +2.74]	+1.39 / +2.39	yes

Against a v0.6 opponent all four rows are positive and clear zero (Table 17), and this is where the measurement disagrees with the registered expectation. The protocol states in advance +4.82 in self-play and +2.50 to +2.88 under partner change. Measured: +4.55 in self-play, and of the three changed rows two fall below the stated range, namely +1.46 against `v03`, more than a point below its floor, and +2.34 against `detective`.

Table 17: Partner-regime table against a v0.6 opponent. Per-bank values are given in bank order, 7090001 then 7090002.

Partners	SESTINA − v0.6	95% CI	Per bank	Repl.
itself	+4.55	[+3.92, +5.18]	+4.88 / +4.21	yes
v0.6	+2.75	[+2.14, +3.37]	+2.97 / +2.54	yes
v0.3	+1.46	[+0.63, +2.29]	+1.25 / +1.67	yes
`detective`	+2.34	[+1.49, +3.19]	+2.42 / +2.26	yes

Table 18: Transfer across partners: SESTINA v1.0 over v0.6 against v0.6 over v0.5, on the same eight settings and the same banks. The two training rows are quoted from the registered protocol; the protocol records that its v0.6-over-v0.5 training row is computed on three of seven changed settings, so that row is not directly comparable with the holdout row above it. Reported and not gated.

Comparison	Min	Median	Self	Median/self
SESTINA over v0.6 — holdout	−0.19	+1.40	+4.49	0.313
v0.6 over v0.5 — holdout	−0.40	+0.26	+0.69	0.380
SESTINA over v0.6 — training, stated in advance	−0.15	+1.26	+2.94	0.428
v0.6 over v0.5 — training, stated in advance	−0.61	+0.64	+1.35	0.472

The protocol stated in advance that if the ordering in Table 18 survived, the appropriate summary is that SESTINA v1.0's advantage transfers across partners about as well as v0.6's did and, on the

median-to-self-play ratio, slightly less well. The ordering survived (0.313 against 0.380), so that is the paper's summary. The absolute figures belong beside the ratio: SESTINA v1.0's changed-partner median is +1.40 against v0.6's +0.26, and the ratio is smaller mainly because the self-play row it divides by is more than six times larger.

A one-seat upgrade is worth +1.93 pp against a v0.5 opponent (registered expectation +1.26) and +2.75 pp against a v0.6 opponent (registered +2.88). The single-seat upgrade produced a positive gain, which is consistent with benefits beyond identical-agent coordination; because interacting team effects need not be additive, the ratio of the one-seat gain to the three-seat gain is descriptive and does not establish what share of the advantage is individual.

11.6 Cross-play across training runs

Diagonal mean +4.55, off-diagonal mean +4.55, off-diagonal minus diagonal +0.01 pp against a per-cell half-width of 0.63; no interval is computed for the off-diagonal-minus-diagonal contrast itself, so the comparison is descriptive rather than a formal equivalence test. The worst off-diagonal cell (+3.80) is 0.40 pp below the weakest diagonal cell (+4.20), and the best off-diagonal (+5.15) exceeds every diagonal cell. The registered claim that the advantage is not a private convention between identical fits is satisfied on the point estimate against a threshold of 1.50 pp. For scale, the coordination literature documents convention collapses under cross-play that are large relative to self-play performance [39, 42], in score units that are not directly comparable with win-rate points; no similarly large cross-play decrease was observed in these runs. Table 19 is the cross-play matrix and Table 20 the three pairs head to head.

Table 19: Cross-play. Three runs of the same architecture, trained independently on disjoint fitting banks with different search trajectories and, for one, a different starting basin; pairwise parameter distance 6.3 to 9.6 over 55 coordinates.

Run \ partners	xp1	xp2	xp3
xp1	+4.54 [+3.91, +5.17]	+3.80 [+3.17, +4.42]	+4.55 [+3.92, +5.19]
xp2	+4.93 [+4.29, +5.56]	+4.91 [+4.29, +5.54]	+4.83 [+4.20, +5.46]
xp3	+5.15 [+4.52, +5.78]	+4.06 [+3.44, +4.69]	+4.20 [+3.56, +4.83]

Table 20: The three independently trained runs head to head. 2 of the three pairs separate from zero, which is what establishes that these are distinguishable policies rather than one policy three times.

Pair	Edge (pp)	95% CI
xp1 vs xp2	+0.20	[−0.43, +0.83]
xp1 vs xp3	+0.70	[+0.08, +1.32]
xp2 vs xp3	+1.05	[+0.43, +1.68]

11.7 Rule-variant sensitivity

Every row keeps its sign, every row replicates, and the largest excursion from the default row is +0.78 pp on arb=turn, inside the registered 2.00 pp tolerance. The advantage retains a positive estimate under 48-card Literature, all three arbitration orders, both declaration-permission changes and the full legacy variant. Table 21 gives every row.

Table 21: The rule-variant table: SESTINA v1.0's win-rate edge in percentage points against `F-cheap` under each variant, both configurations playing the same variant in every row, with 95% intervals as in §7.3. Per-bank values are given in bank order, 7090003 then 7090001. This is the first sweep of its kind in the corpus and has no training-bank counterpart.

Dialect	Edge (pp)	95% CI	Per bank	vs default
default	+4.73	[+4.10, +5.36]	+4.57 / +4.88	+0.00
no-out-of-turn	+5.25	[+4.63, +5.87]	+4.72 / +5.78	+0.52
no-cardless-declare	+4.90	[+4.27, +5.52]	+4.71 / +5.08	+0.17
maxasks=360	+4.73	[+4.10, +5.36]	+4.57 / +4.88	+0.00
arb=high	+4.85	[+4.21, +5.49]	+5.03 / +4.67	+0.12
arb=turn	+5.51	[+4.88, +6.14]	+5.26 / +5.77	+0.78
sets=8	+4.13	[+3.52, +4.73]	+3.77 / +4.48	−0.60
legacy	+5.12	[+4.50, +5.75]	+4.62 / +5.63	+0.40

Two readings belong with the table. The `maxasks` row is bit-identical to the default (yes), because the 400-ask cap is never reached in this population, so the cap change produced no observed behavioural effect in this sample. Consequently the legacy residual, which the protocol assigns to the forced-endgame willingness ladder for want of a flag of its own, is arithmetically legacy minus two effective components rather than three; it is −0.292 [−1.55, +0.96]. That interval is a quadrature propagation over cells that share banks and deals with unit coefficients, so it is not a valid 95% interval and is wider than a paired difference would be (Appendix E.2); it is 8.6 times the size of the residual, and the data are consistent with the ladder contributing nothing.

11.8 Adversarial-search results

8 of eight searches produced a weight vector and all eight were evaluated. None of the eight searches found a positive edge against SESTINA v1.0 at the tested budgets. Every one loses, by between 2.91 and 10.70 pp; every sign replicates; and the largest upper bound anywhere is −2.27, which is 3.80 pp below the class floor. The registered claim is satisfied. Table 22 gives every search.

Table 22: 8 independently constructed adversarial searches against SESTINA v1.0. Each row is the adversarial response win-rate edge in percentage points: the fitted adversary's edge against SESTINA v1.0, so a positive value would mean the adversary beats the target. Each was fitted on a deal stream disjoint from the banks it is evaluated on. Per-bank values are given in bank order, 7090001 then 7090002.

ID	Class	Objective	Edge (pp)	95% CI	Per bank	Repl.
Z01	C1 in-class	win	−2.91	[−3.55, −2.27]	−2.76 / −3.06	yes
Z02	C2 extended	win	−3.90	[−4.52, −3.28]	−3.80 / −4.00	yes
Z03	C1	declerr	−10.70	[−11.32, −10.09]	−10.65 / −10.76	yes
Z04	C1	events	−9.64	[−10.27, −9.01]	−9.83 / −9.46	yes
Z05	C1	forced	−7.14	[−7.76, −6.52]	−7.17 / −7.11	yes
Z06	C2 wide sigma	win	−5.25	[−5.88, −4.62]	−4.93 / −5.57	yes
Z07	C1 v05 basin	win	−5.21	[−5.84, −4.58]	−4.85 / −5.58	yes
Z08	C5 white-box	win	−3.56	[−4.19, −2.92]	−3.95 / −3.17	yes

The searches control all three opposing seats, run in the synchronised correlation regime, and use a budget of 8 generations of a 12-member population at 120 deals per candidate. They maximise four

different objectives, being win rate, declaration error, event count and forced-endgame incidence, and two of them depart deliberately from the template, one using a wide search variance and one seeded in the v0.5 basin, so that the set is not one search repeated. The declaration-objective arm is read against the declaration-family floor of 2.13 rather than the general floor.

These are observed response edges within the tested classes and budgets; they do not provide an upper bound on best-response value. §12.4 states what would be required to convert them into a statement about SESTINA v1.0.

11.8.1 Transcript-inversion adversary

`Z08` is the adversary class the adversary model of §8 was written for: an opponent that reads a model of the target's deterministic policy and reconstructs a policy-aware deal posterior from the public transcript, testing whether public policy specification among identical deterministic teammates exposes exploitable behavioural information. It reaches -3.56 pp $[-4.19, -2.92]$, a replicated loss.

Three limits belong with that cell, since it is the one a sceptical reader should weigh most heavily. The target model is a surrogate: the fixed transcript-inversion component in the base agent queries a v0.6 model of the target rather than the frozen configuration itself, a setting carried forward from the phase whose target *was* v0.6, so the likelihood is computed against a policy that does not search, leaves the urgency escalation on, and carries neither the tie-break nor the contestation coordinate; this is an inverter using a v0.6 surrogate, not an exact reconstruction of SESTINA v1.0's policy, and the corpus does not measure how far the surrogate's choice diverges from the frozen target's, so the size of the resulting degradation is unquantified. The inverter also accumulates per-card evidence additively across the game rather than carrying the joint conditioning of (25) (§6.8). 18 of its fitted coordinates are inert, because the white-box base derives from the v0.6 agent and reads only that policy's 37 coordinates while the search extends its vector to 55; the effective search is therefore over 37 live coordinates plus the fixed transcript-inversion component in the base agent. And the budget is small. The result is evidence that this inverter, at this budget, does not exploit SESTINA v1.0; it is not evidence that no white-box attack can.

Training and evaluation discrepancy. The in-class arm reaches $+6.67$ pp of paired margin in its final generation on its own fitting stream, and $+8.33$ pp at its best generation, before landing at -2.91 pp on disjoint banks. Most of that difference is not the change of bank: the same run re-scores its selected vector on the fitting stream and already falls below parity there, because the generation figure is a maximum over candidates at 120 deals each rather than a measurement of the vector that was retained. Re-evaluation closes most of the gap and disjoint banks close the rest.

11.9 Evaluation controls

These controls assess selected failure modes of the evaluation procedure. Table 23 is the planted-edge recovery.

Two of the three supra-floor rungs are recovered with a lower bound above the 1.53 pp floor. The third, the `0.08` rung, recovers $+2.38$ $[+1.50, +3.26]$, above the floor on the point estimate but with an interval straddling it, so on the lower-bound reading this paper applies to its own primary claim it is not recovered. Its planted cost, $+1.59$ $[+1.05, +2.13]$, straddles the floor as well, so that rung is supra-floor only on a point estimate. Recovered size tracks planted size monotonically across all four rungs.

Table 23: Planted-edge recovery, in percentage points of win-rate edge, pooled over both banks with 95% intervals. *Planted cost* is the handicap's cost to the target against SESTINA v1.0; *recovered excess* is the responder's edge against the handicapped target minus its edge against the unhandicapped one, a between-cell contrast. The sub-floor rung must *not* be recovered.

Planted handicap	Planted cost	vs. handicapped	vs. SESTINA	Recovered excess
0.05 *(sub-floor)*	+0.78 [+0.25, +1.32]	−3.62 [−4.26, −2.99]	−4.32 [−4.94, −3.69]	**+0.70** [−0.20, +1.59]
0.08	+1.59 [+1.05, +2.13]	−2.29 [−2.92, −1.67]	−4.67 [−5.30, −4.05]	**+2.38** [+1.50, +3.26]
0.11	+8.52 [+7.93, +9.12]	+2.52 [+1.89, +3.16]	−6.90 [−7.52, −6.28]	**+9.42** [+8.54, +10.31]
0.15	+11.44 [+10.85, +12.03]	+7.30 [+6.68, +7.93]	−5.34 [−5.97, −4.71]	**+12.64** [+11.75, +13.53]

The sub-floor rung is not recovered, at +0.70 [−0.20, +1.59]: the sub-threshold control did not show a statistically resolved recovered excess, which is the required outcome, although its upper bound lies above the floor. Two qualifications belong with it: phase 2 measured the excess at this same handicap rung at +0.25 [−0.19, +0.68], so this cycle reaches the same conclusion from a point estimate a little under half a point higher, and the upper bound here grazes the floor. The protocol had named a different phase-2 row for this comparison; Appendix E.2 records the correction.

The identity control is exact: SESTINA v1.0 against itself is 0.500000 with a zero-width interval on both banks over 12,000 games each. The responders fitted for these controls also lose to unhandicapped SESTINA v1.0 in all four rows, a second sample of §11.8's finding from a different bank and against a handicapped target; they maximise win rate as five of the eight adversarial searches do, so this is a second sample rather than an independent method. Table 24 is the seat-isolation residual.

Table 24: The seat-isolation residual on holdout material, 16 cells over 19,200 audited deals: decisions the S6 reconstruction could not reproduce from own hand, public record, rules and reset seed, over decisions audited. The right column is the registered zero-tolerance condition (one worker thread, agents rebuilt per deal); the left column is one worker thread with reused agents. Neither column is the scored mode, which used 13 worker threads with reused agents and was not audited on holdout.

Configuration	Bank	`--threads=1`	`--threads=1 --freshagents`
SESTINA v1.0	7090001	0 / 813,512	**0** / 813,512
SESTINA v1.0	7090002	3 / 816,049	**0** / 816,016
v0.6 deployed	7090001	0 / 791,530	**0** / 791,530
v0.6 deployed	7090002	0 / 792,734	**0** / 792,734
`F-cheap`	7090001	1 / 792,140	**0** / 792,140
`F-cheap`	7090002	2 / 795,218	**0** / 795,224
phase-2 composite	7090001	2 / 814,702	**0** / 814,726
phase-2 composite	7090002	0 / 816,225	**0** / 816,225

The gate column is zero for every configuration on both banks (0 nonzero cells), so no inter-seat channel is detected under the certifying condition. The fresh-agent, single-thread configuration

passed the specified isolation tests. The scored configuration retained a cross-deal state residual in searching agents, so these test results do not establish strict seat isolation or exact reproducibility for the scored mode. The left column reproduces the known engine defect of §6.8 on holdout material and reproduces its shape: 4 cells are nonzero and every one belongs to a *searching* configuration, with the non-searching blueprint zero on both banks. The audited decision counts move between the two columns in 3 of the cells, which is the measured form of the statement in §7.10 that rebuilding agents per deal changes play. The residual's effect on a scored win rate is not bounded by a control cell; the largest discrepancy observed between identical scored commands is one game in 12,000 (Appendix E.2), and a small observed residual does not establish a bound on possible bias across all experiments.

11.10 Computational cost

The record priced the frontier's search at 3.2× the blueprint, a figure that describes `F-cheap`. On a like-for-like basis the observed whole-match slowdown is 4.10× for `F-cheap` and 4.52× for SESTINA v1.0, measured against the cheapest opponent. All three figures exceed 3.2×, so the record understates the cost. The whole-panel column is lower because expensive panel members place a large common term on both sides of the ratio, and it is the most diluted of the three. Table 25 gives the figures. The execution environment is recorded in Appendix G.

Table 25: Observed whole-match slowdown relative to the blueprint. The harness reports whole-match throughput, which carries the opponent's cost on both sides of the ratio, so these ratios are diluted by the opponent's cost and are closest to the agent-only cost against the cheapest opponent; a strict lower bound on agent-only cost would require assuming that a policy does not change game length or the opponent's work, which is not established. No figure carries an interval, these being wall-clock timings taken across a battery that ran over two days, and the aggregation is a ratio of medians, which differs from a median of per-bank ratios by at most 0.041 across the six median cells.

Configuration	vs `random` (tightest)	Far archetypes (median)	Whole panel (median)
SESTINA v1.0	4.52×	4.72×	2.72×
`F-cheap`	4.10×	4.36×	2.63×
phase-2 composite	4.20×	4.51×	2.69×

12 Discussion

12.1 Observed performance gains

Against `F-cheap`, the registered baseline, SESTINA v1.0 gains +3.33 pp of win-rate edge (95% CI [+2.88, +3.78]) over 48,000 games of material sealed before it existed, with positive estimates on both banks; the registered primary decision rule was satisfied, and the unresolved secondary criteria are listed in §11.2.4. It gains +4.63 pp on the deployed v0.6 policy and +5.18 pp on v0.5.

The evidence is consistent with an advantage that is not an artefact of self-play coordination, which is the failure mode the cooperative-play literature warns about [39, 42]. A single seat's upgrade is worth +1.93 pp against the +4.49 pp three-seat gain, a positive gain consistent with benefits beyond identical-agent coordination; cross-play between three independently trained runs moves the result by 0.01 pp on the point estimate; and the advantage retains a positive estimate under eight rule variants with a largest excursion of +0.78 pp.

On RQ3, the largest observed leave-one-out drop belongs to one component, a half-suit contestation weight, at +1.69 pp [+0.80, +2.57]. The stronger statement, that this coordinate is where the gain lives, is not supported at this precision (§11.4), and the registered location criterion did not replicate across banks (§11.2.2). That component originated in a coordinate sweep during adversary generation rather than in architecture work; its value was chosen as the argmax of a measured dose response, and the mechanism it turned out to express is not the one it was designed for (§6.3). Its provenance matters for the next subsection.

12.2 Null results and attribution limits

Against the phase-2 composite, SESTINA v1.0 gains +0.15 pp [−0.29, +0.59], an interval containing zero (§11.2.1). The composite is not an external baseline; it was assembled during this cycle's own development, from the v0.6 frontier's search, the contestation weight and one switch. The account of this cycle's progress that the evidence supports is that the gain over `F-cheap` is replicated, that the components assembled during adversary generation account for the part of it the evaluation can resolve, and that the evaluation does not resolve an additional benefit from the modifications introduced afterwards.

Three further results are consistent with that account. The frozen configuration is 3rd of four on empirical minimax regret over the shared panel. Its headline worst cell does not replicate in sign. And of the four quantities the registered selection check tabulates, the largest shortfall against an advance-stated training value falls on precisely the group of components the configuration was selected over.

Four candidate mechanisms were tested and did not demonstrate an additional benefit (§9.4); each result constrains a previously plausible direction under the tested conditions. The declaration-allocation channel, ranked second in the project's design register, produced no difference in the tested comparison, because the two allocators agreed at every recorded decision; the posited mechanism was not observed, which is an empirical finding about these implementations rather than an identity. A leaf evaluator improved 74-fold in sample in the only statistic its consumer can use did not produce a resolved change in play, because the deviation rule that consumes it is nearly leaf-invariant, a mismatch between surrogate metric and playing performance related to the abstraction pathology of [66]. Each of the three self-oriented per-decision objectives tested produced a worse policy at matched budget while moving its own proxy in the intended direction; the two channel-oriented fits lost because the ask and declaration channels traded against each other in these fits. And a learned re-ranker aimed at amortising test-time search loses outright deployed unrestricted at −1.83 pp; restricted to the exchangeable tied set, it did not demonstrate an advantage over a free public-history tie-break. Inside that tied set approximately 95.5% of the search's own selection signal is winner's curse on its Monte Carlo draw, against approximately 84% for the search as a whole.

12.3 Scope of the policy-class findings

On RQ4, the tested candidate modifications did not demonstrate a replicated additional gain under the evaluated settings. This finding does not establish local optimality or uniform performance across the policy class. Improvements from untested modifications, different training procedures, larger search budgets, or combinations of small effects remain possible: the evaluation resolves effects of about a point and does not resolve effects of a quarter of a point, so a family of small mechanisms that individually sit below that resolution and compound is consistent with everything reported here, and neither this cycle nor the last tested that hypothesis. The precision of direct

policy comparisons, which is a property of the cell sizes, should be distinguished from the sensitivity of the calibrated adversarial searches, which is a property of the responder classes and budgets and did not improve with evaluation games in the one comparison made (§7.9).

The one mechanism that did convert was found by sweeping a coordinate, not by designing an architecture. That is a weak signal about where remaining value lies, but it is the signal the data provide.

12.4 Limits of adversarial evaluation

None of the eight adversarial searches found a positive edge against SESTINA v1.0 at the tested budgets, in the tested classes, controlling all three opposing seats in the synchronised correlation regime. The result is conditional on the specified response classes, budgets, and correlation regime, and it is a set of observed response edges rather than a bound on best-response value in either direction.

A stronger response outside the tested classes, or the same classes at a larger budget, could give a larger figure. The white-box transcript inverter, which is the class this domain's adversary model was written for, effectively searched 37 live coordinates at 8 generations of 12, against a v0.6 surrogate of the target rather than the target itself (§11.8.1). Its failure is evidence about that search. This is the reading error the v0.4 record in this corpus committed, having reported a null adversarial result as a demonstration of robustness, and naming it here is how this paper avoids repeating it.

Converting any of this into a statement about SESTINA v1.0 rather than about the searches would require a bound in the other direction, and §12.7 explains why none exists.

12.5 External validity

Three limitations bound how far any of these results generalise.

The opponent panel is a modelling choice. Its deception archetypes are stylised implementations of manoeuvres one person reported from live play, and they are not people. A gain against one of them is evidence about robustness to a class of behaviour, not a measurement of play against a person. The partner-substitution columns have the same limitation from the other side, since a foreign partner here is another bot.

There is no human evaluation of any kind in this corpus. The project's standing practice is never to present a self-play configuration as the one a person would encounter, because bot-only evaluation does not establish performance with human players.

Every number in this paper was produced by one engine, written by this project, audited by tests this project wrote. The strongest available substitute for external replication was used, in the form of material sealed before the configuration existed, digests computed by a second build, and a protocol registered in advance, and it remains a substitute. In particular, there are no independently authored six-player Fish agents to compare against, which is why §1.2 confines the paper's comparative claim to this project's own lineage.

12.6 Implementation and evaluation limitations

Cross-deal state persistence. Agents are constructed once per worker thread and reused across deals, and some per-agent state survives the inter-deal reset, reachable only under the search. The carrier was identified as the rollout engine, the specific field was not, and the repair was deliberately deferred because it changes the play of every searching configuration by about one deal in 800 and applying it after freezing would defeat the purpose of freezing. §11.9 measures it on holdout material:

4 of 16 audited cells show a nonzero residual with reused agents on one thread, every one of them a searching configuration, and the zero-tolerance gate condition is clean. The residual's effect on a scored win rate is bounded only by incidental observations (§11.9), not by a control cell. The frozen results are preserved as measured; a separately identified corrected implementation with sensitivity reruns of the key comparisons is additional research, listed in §12.7.

Validation-mode mismatch. The isolation tests' zero-tolerance form requires one thread and agents rebuilt per deal, and rebuilding agents per deal changes play. The scored cells ran with 13 worker threads and reused agents, a mode in which the strict isolation test was not run on holdout material, and the residual is not removed from it.

Shared training ancestry. The three independently trained runs of §11.6 all begin from or near the v0.6 vector. A convention all three inherited from that common ancestor would appear on the diagonal and off it alike and would be invisible to the test. What the test rules out is a convention formed during each run's own fit.

Unused reserved deal banks. The fitting driver derives its own per-generation deal stream from its seed rather than constructing the named bank's deals, so three banks reserved for fitting are verified and then unused as deals; a fourth is unused because the sealed panel members were scored on the two primary banks so that minimax regret is computed over one shared pair. The material accounting should be read as three evaluation banks played plus a derived fitting stream.

Inactive action-cap intervention. The ask-cap row is bit-identical to the default, so the cap change produced no observed behavioural effect in this sample, and the legacy residual it feeds is arithmetically legacy minus two effective components rather than three, with a propagated interval 8.6 times the size of the residual itself and not a valid 95% interval (§11.7).

Termination limitations. No configuration here reaches the action cap in any scored cell but one, in 12,000 games, and the stall detector's separation between ordinary play and deadlock is a measurement over 800 mirror games. What is not proved is that the stall rule together with the stopping rule terminates on every reachable position.

12.7 Future work

Two claims a future release might wish to make, that a Fish agent is near-optimal or that it is the strongest known, are unsupported by anything here. Both are reachable in principle, and the evidence each requires is specific enough to enumerate.

Exploitability upper bounds. Every adversarial figure in this corpus is the observed edge of a particular response found by a search. A near-optimality claim requires a bound in the other direction, showing that no adversary in a stated class can exceed some ε. None exists here for any class, and producing one even for a restricted class would convert every adversarial sentence in this corpus from a statement about searches into a statement about the agent.

Team local-best-response methods. The standard cheap exploitability probe rests on a passive continuation [38]. Fish has none, since a player must ask or declare (§2.3). Until a defensible rollout policy for that role is named and calibrated the probe cannot be ported, and the literature's own caution that choosing a value function is the difficult part applies directly [24].

Ex-ante correlated adversaries. The adversary model names three correlation regimes and makes the ex-ante correlated one the principal case. What has been built and measured is the synchronised regime, plus one correlated fit reaching +0.70 pp, below that class's 2.31-point calibration threshold. Whether such a responder is buildable for a game of this size is open; the principal regime has in practice fallen back to a weaker one.

Composition of small improvements. This is the one hypothesis both this cycle and the previous one explicitly declined to test. A one-point planning half-width on the scoreboard requires roughly 9,604 games per cell, and a quarter of a point sixteen times that, while the per-decision estimator of §9.4.3 reaches a comparable planning half-width for a one-point-equivalent effect in 284 to 560 games under the design register's conversion factors. That estimator is supported as an estimator and unused, because fitting against it failed. Using it as an estimator while continuing to fit against the scoreboard is the obvious untried experiment. A joint bootstrap of between-cell contrasts, which would require re-running cells with per-deal capture (§7.3), belongs with it.

External-agent and human evaluation. A comparative claim beyond this lineage requires at least one independently authored opponent and, for any claim about play against people, a duplicate-style human battery with enough deals to resolve a point. Neither exists, and the six-seat duplicate design for the latter has not been settled.

Cross-deal state reset. The defect of §12.6 should be repaired in a separately identified implementation and the key comparisons re-measured as sensitivity reruns, before rather than after any subsequent configuration is frozen; the frozen results reported here are preserved as measured.

13 Conclusion

SESTINA v1.0 combines constraint-based inference, a linear action policy, public-history tie-breaking, half-suit contestation, stall detection, and guarded search for six-player Canadian Fish. Against the preregistered `F-cheap` baseline, it achieves a win-rate edge of +3.33 pp relative to parity (95% CI $[+2.88, +3.78]$) over 48,000 holdout games. Its comparison with the phase-2 composite is unresolved, at +0.15 pp $[-0.29, +0.59]$, and the component-attribution criterion is not consistently satisfied across banks.

The evaluation includes partner substitution, cross-play, rule variants, and adversarial-response searches. These tests provide evidence within the evaluated configurations and budgets, with unresolved partner-threshold uncertainty and known implementation limitations: the partner row that decides the registered coordination criterion is −0.19 pp $[-1.04, +0.67]$, above the −1.00 threshold on the point estimate but not resolved against it (§11.5); over the shared panel SESTINA v1.0 is 3rd of four on empirical minimax regret with a worst cell that does not replicate in sign (§11.3); and every scored cell ran in an execution mode that retains a known cross-deal state residual in searching agents (§12.6). None of the eight tested responses, including a transcript-inversion adversary using a v0.6 surrogate of the target, achieves a positive edge, but the study does not bound the performance of stronger or differently constructed adversaries. Four candidate mechanisms did not demonstrate an additional improvement under the tested conditions, and each result is reported with the measurement and the limitation that scope it (§9.4).

The main outcomes are a documented improvement over the specified baseline, an exact reference computation for the constraint-conditioned belief model, and an evaluation procedure that separates observed gains from unresolved differences. The transferable design elements are the

constraint factorisation and count program, the duplicate-deal design with held-out banks and a sign-agreement criterion, and the calibration of adversarial searches against planted handicaps. Priorities for further work are correction and re-evaluation of cross-deal state persistence, stronger adversarial searches, testing whether small improvements compound, and comparison with independently developed agents (§12.7).

References

[1] J. McLeod, "Literature," *Pagat*, 2006–2026, last updated 1 July 2026. https://www.pagat.com/quartet/literature.html

[2] M. Develin, "Canadian Fish," chapter 9 of *The Ten Best Card Games You've Never Heard Of*, author's online manual. https://web.archive.org/web/20170720075756/http://bantha.org/~develin/cardgames.html#ch9

[3] D. Dorsa, "Literature Game Strategy," Deposit Genius, 2018. https://depositgenius.com/literature-strategy-canadian-fish/

[4] Canadian Fish Project, "Strategy notes," accessed 27 August 2026. The site is a single-page application that serves one shell for every path, so it carries no publication date. https://canadian-fish.vercel.app/strategy

[5] Wikipedia, "Literature (card game)," online encyclopedia entry. https://en.wikipedia.org/wiki/Literature_(card_game)

[6] N. Somani, "literature: card game implementation and learning bot," GitHub repository, MIT licence, Python, 2018–. https://github.com/neelsomani/literature

[7] N. Somani, "literature-server," GitHub repository, with a live deployment. https://github.com/neelsomani/literature-server https://literature.neelsomani.com/

[8] "Literature: Fish Card Game" (`com.cards.game.literature`), Google Play store listing, closed source. https://play.google.com/store/apps/details?id=com.cards.game.literature

[9] D. Zha, K.-H. Lai, S. Huang, Y. Cao, K. Reddy, J. Vargas, R. Wei, J. Guo, and X. Hu, "RLCard: A Toolkit for Reinforcement Learning in Card Games," arXiv:1910.04376, 2019.

[10] M. Goadrich, A. Morenville, and É. Piette, "Valet: A Standardized Testbed of Traditional Imperfect-Information Card Games," arXiv:2603.03252, 2026.

[11] D. Levy, "The Million Pound Bridge Program," *Heuristic Programming in Artificial Intelligence: The First Computer Olympiad*, Ellis Horwood, 1989.

[12] M. L. Ginsberg, "GIB: Imperfect Information in a Computationally Challenging Game," *Journal of Artificial Intelligence Research* 14:303–358, 2001.

[13] I. Frank and D. Basin, "Search in Games with Incomplete Information: A Case Study Using Bridge Card Play," *Artificial Intelligence* 100(1–2):87–123, 1998.

[14] I. Frank, D. Basin, and H. Matsubara, "Finding Optimal Strategies for Imperfect Information Games," *AAAI-98*, pp. 500–507, 1998.

[15] J. Long, N. R. Sturtevant, M. Buro, and T. Furtak, "Understanding the Success of Perfect Information Monte Carlo Sampling in Game Tree Search," *AAAI-10*, pp. 134–140, 2010.

[16] P. I. Cowling, E. J. Powley, and D. Whitehouse, "Information Set Monte Carlo Tree Search," *IEEE Transactions on Computational Intelligence and AI in Games* 4(2):120–143, 2012.

[17] J. Goodman, "Re-determinizing Information Set Monte Carlo Tree Search in Hanabi," arXiv:1902.06075, 2019.

[18] T. Cazenave and V. Ventos, "The $\alpha\mu$ Search Algorithm for the Game of Bridge," arXiv:1911.07960, 2019.

[19] S. Edelkamp, "Knowledge-Based Paranoia Search in Trick-Taking," arXiv:2104.05423, 2021.

[20] M. Buro, J. R. Long, T. Furtak, and N. R. Sturtevant, "Improving State Evaluation, Inference, and Search in Trick-Based Card Games," *IJCAI-09*, pp. 1407–1413, 2009.

[21] C. Solinas, D. Rebstock, and M. Buro, "Improving Search with Supervised Learning in Trick-Based Card Games," *AAAI-19*, 2019; arXiv:1903.09604.

[22] D. Rebstock, C. Solinas, M. Buro, and N. R. Sturtevant, "Policy Based Inference in Trick-Taking Card Games," *IEEE Conference on Games 2019*, 2019; arXiv:1905.10911.

[23] C. Solinas, D. Rebstock, N. R. Sturtevant, and M. Buro, "History Filtering in Imperfect Information Games: Algorithms and Complexity," *NeurIPS 2023*, 2023; arXiv:2311.14651.

[24] N. Brown, T. Sandholm, and B. Amos, "Depth-Limited Solving for Imperfect-Information Games," *NeurIPS 2018*, 2018; arXiv:1805.08195.

[25] B. H. Zhang and T. Sandholm, "Subgame Solving without Common Knowledge," *NeurIPS 2021*, 2021; arXiv:2106.06068.
https://proceedings.neurips.cc/paper/2021/hash/c96c08f8bb7960e11a1239352a479053-Abstract.html

[26] B. H. Zhang, L. Carminati, F. Cacciamani, G. Farina, P. Olivieri, N. Gatti, and T. Sandholm, "Subgame Solving in Adversarial Team Games," *NeurIPS 2022*, 2022.
https://proceedings.neurips.cc/paper_files/paper/2022/hash/aa5f5e6eb6f613ec412f1d948dfa21a5-Abstract-Conference.html

[27] S. Sokota, G. Farina, D. J. Wu, H. Hu, K. A. Wang, J. Z. Kolter, and N. Brown, "The Update-Equivalence Framework for Decision-Time Planning," *ICLR 2024*, 2024; arXiv:2304.13138.
https://openreview.net/forum?id=JXGph215fL

[28] A. Lerer, H. Hu, J. Foerster, and N. Brown, "Improving Policies via Search in Cooperative Partially Observable Games," *AAAI 2020*, 2020; arXiv:1912.02318.
https://ojs.aaai.org/index.php/AAAI/article/view/6208

[29] H. Hu, D. J. Wu, A. Lerer, J. Foerster, and N. Brown, "Learned Belief Search: Efficiently Improving Policies in Partially Observable Settings," arXiv:2106.09086, 2021.

[30] A. Fickinger, H. Hu, B. Amos, S. Russell, and N. Brown, "Scalable Online Planning via Reinforcement Learning Fine-Tuning," *NeurIPS 2021*, 2021; arXiv:2109.15316.

[31] D. Milec, O. Kubíček, and V. Lisý, "Continual Depth-limited Responses for Computing Counter-strategies in Sequential Games," arXiv:2112.12594, 2021.

[32] D. Milec, V. Kovařík, and V. Lisý, "Adapting Beyond the Depth Limit: Counter Strategies in Large Imperfect Information Games," arXiv:2501.10464, 2025.

[33] O. Kubíček and V. Lisý, "Look-ahead Reasoning with a Learned Model in Imperfect Information Games," *ICLR 2026*, 2026; arXiv:2510.05048, 2025.

[34] O. Kubíček, V. Lisý, and T. Sandholm, "Equilibrium Refinements Improve Subgame Solving in Imperfect-Information Games," arXiv:2601.17131, 2026.

[35] N. Brown, A. Bakhtin, A. Lerer, and Q. Gong, "Combining Deep Reinforcement Learning and Search for Imperfect-Information Games," *NeurIPS 2020*, 2020; arXiv:2007.13544.

[36] M. Schmid, M. Moravčík, N. Burch, R. Kadlec, J. Davidson, K. Waugh, N. Bard, F. Timbers, M. Lanctot, G. Z. Holland, E. Davoodi, A. Christianson, and M. Bowling, "Student of Games: A Unified Learning Algorithm for Both Perfect and Imperfect Information Games," *Science Advances* 9(46), 2023; arXiv:2112.03178 (preprint title "Player of Games").

[37] M. Moravčík, M. Schmid, N. Burch, V. Lisý, D. Morrill, N. Bard, T. Davis, K. Waugh, M. Johanson, and M. Bowling, "DeepStack: Expert-Level Artificial Intelligence in Heads-Up No-Limit Poker," *Science* 356(6337):508–513, 2017.

[38] V. Lisý and M. Bowling, "Equilibrium Approximation Quality of Current No-Limit Poker Bots," arXiv:1612.07547, 2016.

[39] N. Bard, J. N. Foerster, S. Chandar, N. Burch, M. Lanctot, H. F. Song, E. Parisotto, V. Dumoulin, S. Moitra, E. Hughes, I. Dunning, S. Mourad, H. Larochelle, M. G. Bellemare, and M. Bowling, "The Hanabi Challenge: A New Frontier for AI Research," *Artificial Intelligence* 280, 2020; arXiv:1902.00506.

[40] J. N. Foerster, H. F. Song, E. Hughes, N. Burch, I. Dunning, S. Whiteson, M. Botvinick, and M. Bowling, "Bayesian Action Decoder for Deep Multi-Agent Reinforcement Learning," *ICML 2019*, 2019; arXiv:1811.01458.

[41] H. Hu and J. N. Foerster, "Simplified Action Decoder for Deep Multi-Agent Reinforcement Learning," *ICLR 2020*, 2020; arXiv:1912.02288.

[42] H. Hu, A. Lerer, A. Peysakhovich, and J. N. Foerster, "'Other-Play' for Zero-Shot Coordination," *ICML 2020*, 2020; arXiv:2003.02979.

[43] H. Hu, A. Lerer, B. Cui, L. Pineda, N. Brown, and J. Foerster, "Off-Belief Learning," *ICML 2021*, 2021; arXiv:2103.04000.

[44] S. Sokota, E. Lockhart, F. Timbers, E. Davoodi, R. D'Orazio, N. Burch, M. Schmid, M. Bowling, and M. Lanctot, "Solving Common-Payoff Games with Approximate Policy Iteration," *AAAI 2021*, 2021; arXiv:2101.04237.

[45] H. Hu, S. Sokota, D. J. Wu, A. Bakhtin, A. Lupu, B. Cui, and J. N. Foerster, "Self-Explaining Deviations for Coordination," *NeurIPS 2022*, 2022; arXiv:2207.12322.
https://proceedings.neurips.cc/paper_files/paper/2022/hash/faa6276ea12d7afeb3e42b210c86f688-Abstract-Conference.html

[46] C. Cox, J. De Silva, P. DeOrsey, F. H. J. Kenter, T. Retter, and J. Tobin, "How to Make the Perfect Fireworks Display: Two Strategies for Hanabi," *Mathematics Magazine* 88(5):323–336, 2015.

[47] Y. Tian, Q. Gong, and T. Jiang, "Joint Policy Search for Multi-Agent Collaboration with Imperfect Information," *NeurIPS 2020*, 2020; arXiv:2008.06495.

[48] A. P. Jacob, D. J. Wu, G. Farina, A. Lerer, H. Hu, A. Bakhtin, J. Andreas, and N. Brown, "Modeling Strong and Human-Like Gameplay with KL-Regularized Search," *ICML 2022*, 2022; arXiv:2112.07544.

[49] H. Hu, D. J. Wu, A. Lerer, J. Foerster, and N. Brown, "Human-AI Coordination via Human-Regularized Search and Learning," arXiv:2210.05125, 2022.

[50] Meta Fundamental AI Research Diplomacy Team (FAIR), A. Bakhtin, N. Brown, E. Dinan, G. Farina, C. Flaherty, D. Fried, A. Goff, J. Gray, H. Hu, and others, "Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning," *Science* 378(6624):1067–1074, 2022.
https://www.science.org/doi/10.1126/science.ade9097

[51] T. Dizdarević and others, "Ad-Hoc Human-AI Coordination Challenge," *ICML 2025*, 2025; arXiv:2506.21490.
https://proceedings.mlr.press/v267/dizdarevic25a.html

[52] A. Nayyar, A. Mahajan, and D. Teneketzis, "Decentralized Stochastic Control with Partial History Sharing: A Common Information Approach," *IEEE Transactions on Automatic Control* 58(7):1644–1658, 2013; arXiv:1209.1695.

[53] A. Celli and N. Gatti, "Computational Results for Extensive-Form Adversarial Team Games," *AAAI 2018*, 2018; arXiv:1711.06930.

[54] G. Farina, A. Celli, N. Gatti, and T. Sandholm, "Connecting Optimal Ex-Ante Collusion in Teams to Extensive-Form Correlation: Faster Algorithms and Positive Complexity Results," *ICML 2021*, PMLR 139:3164–3173, 2021.

[55] B. H. Zhang, G. Farina, A. Celli, and T. Sandholm, "Team Belief DAG: Generalizing the Sequence Form to Team Games for Fast Computation of Correlated Team Max-Min Equilibria via Regret Minimization," *ICML 2023*, 2023; arXiv:2202.00789.

[56] L. Carminati, F. Cacciamani, M. Ciccone, and N. Gatti, "A Marriage between Adversarial Team Games and 2-Player Games: Enabling Abstractions, No-Regret Learning and Subgame Solving," *ICML 2022*, 2022; arXiv:2206.09161.

[57] L. Carminati, B. H. Zhang, G. Farina, N. Gatti, and T. Sandholm, "Hidden-Role Games: Equilibrium Concepts and Computation," *ACM Conference on Economics and Computation (EC) 2024*, 2024; arXiv:2308.16017.

[58] H. J. Ryser, *Combinatorial Mathematics*, Carus Mathematical Monographs 14, Mathematical Association of America, 1963.

[59] D. G. Glynn, "The Permanent of a Square Matrix," *European Journal of Combinatorics* 31(7):1887–1891, 2010.

[60] M. Jerrum, A. Sinclair, and E. Vigoda, "A Polynomial-Time Approximation Algorithm for the Permanent of a Matrix with Non-Negative Entries," *Journal of the ACM* 51(4):671–697, 2004.

[61] N. Linial, A. Samorodnitsky, and A. Wigderson, "A Deterministic Strongly Polynomial Algorithm for Matrix Scaling and Approximate Permanents," *Combinatorica* 20(4):545–568, 2000.

[62] Y. Chen, P. Diaconis, S. P. Holmes, and J. S. Liu, "Sequential Monte Carlo Methods for Statistical Analysis of Tables," *Journal of the American Statistical Association* 100(469):109–120, 2005.

[63] N. Bard, J. Hawkin, J. Rubin, and M. Zinkevich, "The Annual Computer Poker Competition," *AI Magazine* 34(2):112–118, 2013.

[64] M. Zinkevich, M. Bowling, N. Bard, M. Kan, and D. Billings, "Optimal Unbiased Estimators for Evaluating Agent Performance," *AAAI-06*, pp. 573–578, 2006.

[65] N. Burch, M. Schmid, M. Moravčík, D. Morrill, and M. Bowling, "AIVAT: A New Variance Reduction Technique for Agent Evaluation in Imperfect Information Games," *AAAI-18*, 2018; arXiv:1612.06915.

[66] K. Waugh, D. Schnizlein, M. Bowling, and D. Szafron, "Abstraction Pathologies in Extensive Games," *AAMAS 2009*, pp. 781–788, 2009.

[67] D. Balduzzi, K. Tuyls, J. Pérolat, and T. Graepel, "Re-evaluating Evaluation," *NeurIPS 2018*, 2018; arXiv:1806.02643.

[68] R. Coulom, "Whole-History Rating: A Bayesian Rating System for Players of Time-Varying Strength," *Computers and Games 2008*, 2008.

[69] R. Herbrich, T. Minka, and T. Graepel, "TrueSkill™: A Bayesian Skill Rating System," *Advances in Neural Information Processing Systems* 19, 2006.

[70] M. Van den Bergh, "Comments on Normalized Elo," Fishtest technical note. `https://cantate.be/Fishtest/normalized_elo_practical.pdf`

[71] Stockfish contributors, "Statistical Methods and Algorithms in Fishtest," Fishtest project documentation.

[72] N. Kelidari, M. Haghi, and M. Salmani, "A Gold-Standard Study of What Makes a Lightweight Game-Playing Agent Strong," arXiv:2607.06854, 2026.

A Rule specification

Six of the eight contested rule choices of §2.2 are engine switches, so their effect can be measured; the misdeclaration award is hardcoded, and the forced-endgame ladder has a threshold flag but not one the scored harness exposes, so within a scored match it moves only inside the `--legacy` bundle. This appendix lists each choice against the switch that changes it, so that a reader can re-implement the variant or vary it. The turn-transfer, declaration-timing, forced-endgame and action-cap rules that the table abbreviates are stated in full in §2.1 and §2.2, from the engine source.

Clause	Switch	Setting studied
Deck and half-suits	`--sets=8`	54 cards in nine half-suits, nine per player; `--sets=8` plays the 48-card, eight-half-suit variant with eight cards per player, in which a 4–4 game is recorded as a win for the team at seat index 0
Out-of-turn declaration	`--no-out-of-turn`	permitted; the switch forbids it
Cardless declaration	`--no-cardless-declare`	permitted; the switch forbids it
Simultaneous declarers	`--arb=high`, `--arb=turn`	lowest seat index first; the switches arbitrate by highest seat index first and by scanning from the current turn-holder. Arbitration by stated confidence is deliberately absent from the engine
Misdeclaration	—	the half-suit is awarded to the opposing team
Forced endgame	—	eight-level willingness ladder when one whole team is cardless: thresholds 0.995, 0.98, 0.95, 0.90, 0.80, 0.65, 0.50, then a compulsory best guess (§2.2)
Action cap	`--maxasks=N`	400 asks; residue adjudicated by physical majority, a 3–3 split going to the team holding the half-suit's lowest-index card
Legacy bundle	`--legacy`	the v0.3-era variant: no out-of-turn declaration, no cardless declaration, a 360-ask cap, and a two-level ladder (0.38, then a compulsory best guess)

§11.7 measures the paper's headline comparison over 8 rule-variant rows: the default, and seven variations drawn from six of these clauses. Every row keeps its sign and every row replicates; the largest excursion from the default row is +0.78 pp, on arb=turn.

Corrections to earlier rule interpretations.

1. **Action-cap adjudication.** An earlier report justified an unconditional forcing rule on the premise that an unclaimed half-suit is worthless. Under the dialect actually evaluated, residue at the action cap is awarded by physical majority, and on a wrong post-horizon declaration the declaring team holds a mean 4.08 of the six cards. The rule converts a probable award into a certain concession.
2. **Forced-endgame declaration order.** The argument is that restarting the willingness ladder after each declaration lets early, safer declarations resolve cards and so sharpen later ones. It is sound and, in the play it was measured on, inoperative: on every occasion a team went cardless with live half-suits — 566 such records at seed 31 in v0.4 mirror self-play, half of them duplicates,

because at each deal shift the two team orientations play identical policies and so record every event twice — exactly 1 half-suit was live, so there was no ordering to exploit. That is a strong tendency and not a universal: a second sample at seed 1234567 finds 60 *distinct* forced declarations, all but one of them with a single live half-suit. It is also policy-dependent: the source records that a policy made more patient about declaring would begin to produce multi-half-suit forced endgames, at which point the ordering would start to matter. SESTINA v1.0 is a change in that direction, since it disables the urgency escalation entirely (§6.5), and no measurement of live-half-suit multiplicity at forced-endgame entry exists for it.

B Configuration grammar and parameters

Configuration grammar

A configuration is named by a single string, and every policy in this paper, every arm, every panel member, every fitted adversary, is one of these strings. The grammar is

`base[:key=value(,key=value)*]`

where `base` names a policy family (`v03` … `v07`, a scripted archetype such as `withholder`, or an instrumented variant such as `v07i` for the white-box inverter), and each `key=value` sets one option on it. Two properties of the grammar matter for the evaluation:

- **It is total.** Any configuration this paper measures can be written as one string and reconstructed from it, which is what makes an ablation lattice mechanical rather than a set of hand-edited builds.
- **An explicit key beats a bulk vector.** A spec may carry `allparams=`$w_1|w_2|\ldots$, the entire parameter vector; explicit keys written alongside it are re-applied afterwards. Before this was fixed, a sentinel such as `pool=-1` was silently clamped away by the bulk vector, and three independently fitted configurations carried machinery their spec said was switched off.

Frozen configuration

```
v07:r12=25,rtie=1,pool=-1,oppfloor=-1,force=1000000,askfloor=-1,stall=12,s1=1,
det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26
```

Key	Value	What it controls
r12	25	half-suit contestation weight (oppCertDonate)
rtie	1	tie-break by a hash of the public event stream, not sort order
pool	-1	urgency pooling threshold; -1 disables
oppfloor	-1	urgency opponent-ownership floor; -1 disables
force	1000000	urgency forcing horizon in events; 10^6 disables
askfloor	-1	urgency ask floor; -1 disables
stall	12	escalate after K public events with no change in this seat's certificate hash
s1	1	enable the endgame-truncated determinized search
det	12	determinizations sampled per searched decision
cand	4	candidate actions carried into the search
kappa	2.5	shrinkage of the paired lower-confidence-bound deviation rule
rbelief	indep	belief model used to reconstruct continuation players
depth	12	search depth limit in plies
maxq	26	search runs only when at most this many cards remain unresolved

Three of these are switches that cannot be expressed as coordinates of the parameter vector at all (askfloor, pool, oppfloor), which is why the freeze artifact records the option map separately from the vector and why the round-trip check of §6 has to execute rather than compare strings.

Parameter vector

Block	Coordinates
nfeat	20
v05knobs	14
v06askTerms	3
v07responder	18
total	55

the flat allparams vector: NFEAT ask weights, the fourteen v0.5 knobs, the three v0.6 ask terms, then the eighteen v0.7 responder coordinates (v07_responder.hpp r7Name)

The first 37 coordinates are inherited unchanged from the previous release's fit. The freeze artifact records its provenance as

```
fitC.jsonl [final weights record] sha256:be4a4c31d9153208 gens=14 obj=minimaxregret
paired=1 panel=v05+v03+withholder+feint seed=20260824 deals=200x2 sigmaRel=0.04
```

The twenty ask features of (11) and their deployed weights, the first twenty coordinates of the frozen allparams vector, are listed in Table 26; the feature definitions are the features routine of engine/src/v05.hpp, and the weights are read from engine/fishbot_v07.json by the generator. The state-value function V of (20) is a linear function of sixteen public-state features, defined in the same source file: a constant, the score margin, the team's aggregate expected control of the live half-suits and a sharpened form of it, the counts of locked and of contested half-suits, the turn indicator, the card-count difference, the number of unresolved cards, the number of live half-suits,

the acting seat's and its weakest teammate's hand sizes, counts of half-suits the team probably and probably does not control, and two interaction terms of the turn indicator with control and with the unresolved count. Its sixteen weights are compiled constants of the v0.5 policy family and are not coordinates of the fitted vector.

Table 26: The twenty ask-score features and their deployed weights w, in the order of the feature vector $\phi(a)$; the short names are those used in the source. The features are functions of the approximate belief $\hat{\mu}$ and the public state, as described in §6.1.

Index	Feature	Weight
0	hit probability	+11.26561
1	squared hit	+4.18288
2	certain hit	+3.05733
3	own set progress	+3.80008
4	team control	+4.35318
5	lock completion	+6.42340
6	continuation	+1.36441
7	completion bonus	−0.38185
8	reply threat	−0.81545
9	information leak	−6.24532
10	target hand size	−2.62967
11	empties target	+1.49745
12	repeats set	+3.12582
13	known team cards	+3.96890
14	location entropy	+1.06205
15	team owns set	−1.62528
16	exposure on miss	+4.04349
17	trailing pressure	+1.01429
18	runway	+1.21342
19	leak magnitude	−0.79873

The remaining coordinates are the extended responder block, and they are all zero but one. That is the identity condition: with every responder coordinate at zero the extended class reproduces v0.6 bit for bit, which is the certificate §6 relies on. No gradient or population fit was run in this cycle. The single non-zero coordinate, `r12`, had its value chosen as the argmax of a measured dose response (§6.1, §6.3).

Inactive parameters and interventions

Not every fitted coordinate does anything, and a paper that presented all 55 as tuned quantities would be overstating the search. Known inert coordinates in this family:

- two v0.5 knobs unreachable under the deployed declaration branch, so setting them produces a bit-identical transcript;
- three v0.5 ask terms belonging to a re-scoring pass that the deployed configuration does not run;
- `m2`, whose leave-one-out drop is exactly zero and which was dropped from the freeze for that reason (§11.4);
- `stall`, which is armed and never fires: its mirror digest is byte-identical to the same configuration without it (§6.6);

- and, in the white-box adversary class specifically, 18 of the 55 fitted coordinates, because that base reads only the 37-coordinate v0.6 vector (§11.8.1).

C Preregistered criteria and recorded verdicts

The criteria below are taken from `docs/v07/PREREGISTRATION.md`, committed before any holdout bank had been played and before the sealed adversary half had been decoded. The verdict labels and thresholds are the protocol's own; the rule wording in the tables is condensed from the protocol, and the source document carries the exact text. Three things are kept distinct: the *registered rule*, which is a quotation or close paraphrase of the protocol; the *recorded verdict*, which applies that rule as written and is the verdict of the phase-5 record; and the *reading with uncertainty*, which is later commentary added in this revision and not part of the registration. The verdicts are reported in §11 and §11.2.

Primary criterion

SESTINA v1.0 beats `F-cheap` on both primary holdout banks, pooled over 48,000 games.

Verdict	Condition
CERTIFIED advancement	pooled lower bound above 1.53, sign replicates on both banks, gate passes
measured advancement	pooled lower bound above 0 but under 1.53, sign replicates, gate passes
not an advancement	the pooled interval contains zero, or the sign does not replicate, or the gate fails

The verdict labels above are the protocol's own. This paper does not use the word *certified*, which suggests verification by a third party, and reports the top outcome instead as what was actually satisfied (§10.4). Recorded verdict: the comparison satisfies the registered primary criterion. Reading with uncertainty: the rule is stated on the pooled lower bound and is resolved.

Secondary criteria

#	Claim	Registered rule	Recorded	Reading with uncertainty
S1	no collapse under partner change	at least five of the eight changed-partner deltas positive, and the minimum not below −1.00 (registered as "about 1.5 cell half-widths")	PASS	point-estimate rule; the deciding row's interval $[-1.04, +0.67]$ crosses the threshold, so the verdict is not resolved against it
S2	not a private convention	the off-diagonal is within 1.50 points of the diagonal	PASS	point-estimate rule; no interval is computed for the contrast
S3	no fresh adversary exploits it	every arm's upper bound below its floor	PASS	upper-bound rule; resolved for the tested searches and budgets
S4	the gain is attributable	components sub-additive, no single drop exceeding the whole	PASS	point-estimate rule on the sum and the drops; the sub-additivity ratio's interval is $[0.599, 0.933]$, and the location criterion of §11.2.2 does not replicate
S5	it survives the rule variants	every row keeps its sign, none more than 2.00 pp below default	PASS	point-estimate rule on the excursions; every row replicates in sign
S6	the worst case is not catastrophic	SESTINA v1.0's worst panel cell no worse than the deployed v0.6 policy's	PASS	point-estimate rule; the two worst-cell intervals do not overlap, so the comparison is resolved, although the worst cell itself is not replicated in sign
S7	the instrument is intact	the planted edges recover (registered as "at or above 1.53 points"), the sub-floor rung does not, the cheats fail exactly their own tests	PASS	point-estimate rule as recorded; on the lower-bound reading the `0.08` rung is not recovered, and the sub-floor rung's upper bound exceeds the floor (§11.9)

The protocol's "eight changed-partner rows" counts the self-play row among them; §11.5 reports 6 of the 7 changed rows, excluding self-play. Both row counts give the same recorded verdict, and the phase-5 recording carries the ambiguity as deviation D9. The PASS entries for S3, S4, S6 and S7 are the phase-5 record's verdicts and are not computed by this paper's generator; S1, S2 and S5 are recomputed from the artifacts.

A failure of S1, S2 or S6 would have been a substantive failure, and the paper would report SESTINA v1.0 as brittle in that named way. A failure of S7 would have been procedural and the evaluation would have stopped.

Non-confirmation criteria

The seven conditions are answered in §11.2.4, with the reading each uses. Two are met or unresolved. The protocol's own closing provision is quoted in full, because this paper relies on it:

> "And the honest all-negative path is a legitimate deliverable. If phase 5 returns 'no advancement over the frontier', the v0.7 report is a barrier report: the four ledger entries phase 3 closed with measurements, the two corrections to the corpus's own record, the mechanical side-channel gate with its three calibrated positive controls, and the S6 residual this document commits. That is a real contribution and the report should not apologise for it."

D Complete opponent-panel results

Every cell of the shared panel, four arms against 31 named members, pooled over both primary banks, printed in full to support inspection and recomputation of the panel summaries of §11.3, including the empirical minimax regret of Table 13. The 31 named members are 28 distinct policies: three sealed aliases duplicate members that also appear under their own names, and the duplicated cells were run as named and serve as a determinism check (Appendix F, D15).

Class `near` is 6,000 deals a bank, `far` 1,500, `expensive` 1,200. A positive number means the arm beats that member.

Panel member	Class	SESTINA v1.0	v0.6	`F-cheap`	Composite
`F-fast`	near	+4.55	+0.00	+1.83	+4.17
`F-cheap`	near	+3.38	−1.83	+0.00	+2.37
v0.5	near	+5.18	+0.69	+1.84	+4.45
v0.4	near	+4.66	+0.99	+2.09	+3.86
`feint`	near	+6.07	+3.32	+4.61	+4.93
`SEALED:R-v04`	near	+4.66	+0.99	+2.09	+3.86
`SEALED:S-archetype-0`	near	+6.07	+3.32	+4.62	+4.93
`SEALED:S-ask-1`	near	+4.62	−0.53	+1.01	+3.29
`SEALED:S-ask-3`	near	+4.08	−0.22	+1.55	+3.85
`SEALED:S-decl-0`	near	+4.06	−0.72	+1.10	+3.68
`SEALED:S-invert-0`	near	+3.93	−1.74	+0.90	+1.90
`SEALED:S-reference-0`	near	+4.55	+0.00	+1.83	+4.17
`SEALED:S-search-1`	near	+3.43	−1.81	+0.07	+2.97
`SEALED:X01`	near	+3.38	−0.75	+1.64	+3.35
`SEALED:X01xC3f`	near	+1.43	−4.24	−1.25	+1.23
`SEALED:X02`	near	+3.60	−1.37	+1.17	+3.04
`SEALED:X13`	near	+4.84	−0.30	+1.82	+4.12
`SEALED:X18`	near	+3.37	−0.46	+1.75	+3.22
`SEALED:X20`	near	+13.36	+11.62	+13.54	+13.40
v0.3	far	+28.72	+25.47	+24.72	+28.40
v0.2	far	+34.32	+31.68	+32.67	+33.17
`lockout`	far	+30.78	+28.65	+28.97	+29.07
`detective`	far	+29.65	+28.08	+27.98	+28.32
`hunter`	far	+46.93	+48.58	+48.65	+47.10
`diversifier`	far	+43.40	+44.90	+45.18	+43.32
`bluffer`	far	+49.87	+49.98	+49.95	+49.88
`silent`	far	+32.75	+34.23	+34.05	+30.92
`withholder`	far	+25.35	+29.88	+29.55	+25.80
`random`	far	+50.00	+50.00	+50.00	+50.00
`F-mid`	expensive	+2.15	−3.15	−0.10	+2.00
phase-2 composite	expensive	−0.04	−3.69	−2.08	+0.00

E Corrections to prior reports

Earlier work in this lineage contains measurement and implementation errors that later cycles found. The project's convention is that these are stated as corrections rather than silently patched into the artifacts they correct, and the itemised register is placed here so that it does not compete with the subject of the paper. §4 summarises the consequences that bear on SESTINA v1.0: earlier evaluation lacked the power to resolve the effects it reported on, one exploitability interpretation was substantially too strong, an exact-belief implementation was unsound in multi-agent configurations, and the cost of the frontier's search was mischaracterised by two orders of magnitude. Those discoveries are why the evaluation framework of §7 exists in the form it does.

E.1 Corrections to earlier FishBot reports

Search-cost estimates. The v0.6 paper recorded the searching configuration at "three orders of magnitude" the blueprint, mixing an all-threads timing with a single-thread one; phase 1 of this cycle re-timed it on one basis at 242×. Both figures describe the *unrestricted* configuration, which is not the point of the frontier anything was measured at. The endgame-restricted operating point was re-measured at ∼3.2× during phase 3, and that correction was carried into the protocol. **This report corrects the correction**: measured on the shared panel at a fixed thread count, `F-cheap`, the configuration the 3.2× describes, costs at least 4.10× the blueprint, and SESTINA v1.0 at least 4.52× (§11.10). Every figure in that table is an observed whole-match slowdown, diluted by the opponent's cost, and none carries an interval. The record has now been wrong in both directions and the reason is the same both times, a ratio quoted without the basis it was computed on.

Interpretation of adversarial-response results. The v0.4 findings document reports that a policy fitted specifically against v0.4 fails to reach parity with it; the same study's own table records the response reaching 51.19% with an interval spanning parity. The v0.6 study corrected this. It is repeated here because this cycle's own principal adversarial result, §11.8, eight searches, all losing, is a number of exactly the same kind, and the correction is what stops this paper making the same claim.

Declaration-allocation comparison. The register ranked it second by priority, at 0.48, and estimated it at roughly two percentage points. §9.4.1 reports that the two allocators selected the same allocation at every one of ∼210,000 recorded declarations (counted over cells that include re-runs of the same seeds), that the exact-joint alternative performed worse on the declaration decision, and that most of the residual error class is an information limit under the constraint model rather than a defect; the tested mechanism therefore produced no difference, which is an empirical finding rather than an identity.

Forced-endgame improvement estimate. Re-derived from this cycle's capture, closing the entire accuracy gap is worth 0.030 pp at the measured incidence, the register's figure corrected *upward* by a factor of two, and still a fiftieth of the reference bar. The estimate is small either way; §9.4.5 records that the route by which it could grow was not tested.

Local-best-response citation correction. The vetted entry for the local-best-response result annotated it as showing "every ACPC 2016 entrant" exploitable above the stated bound. The source says *each evaluated bot*, and states that the competition's winning submission was not evaluated; it also describes the search as looking one action ahead, where the annotation said two. Both are corrected in §3.3 and in the bibliography entry, whose header records the edit.

Detection calibration across evaluation budgets. Phase 1 measured 1.68 and predicted the floor would fall as the inverse square root of evaluation games. At exactly four times the games the

smallest recovered rung was 1.53, and the next rung down remained unrecovered, with a phase-2 excess centred near zero rather than merely wide. For that responder class and budget, a planted effect below about a point and a half was not recovered; this does not establish that such effects are absent. §11.9 reports the corresponding controls on holdout material.

E.2 Corrections to development and evaluation records

Repository-history rewrite. A per-decision search capture from the development phase, 139,267,276 bytes, exceeded the repository host's file-size limit and was removed from git history so the work could be published. The file is retained on disk with its SHA-256 recorded in a sidecar (`bb655aac7cd413d2...`) and is regenerable from the capture harness; no measurement in this paper depends on it. The rewrite changed every commit hash from phase 3 onward, so hashes recorded inside the phase-5 artifacts no longer resolve. Appendix G gives the mapping and confirms the seal commit is unaffected.

Freeze-verification baseline. The freeze checker reads its source-drift baseline out of the freeze artifact's own provenance field, and that artifact was rewritten twice after the freeze commit, each rewrite refreshing the baseline to the then-current tree. The check therefore compared the tree against a baseline taken from the tree, and the sources the protocol predicted would be listed as changed no longer differed from it. Measured against the actual freeze commit, 9 of the 78 engine sources differ, `arena.hpp`, `factory.hpp`, `fish.hpp`, `game.hpp`, `human.hpp`, `main.cpp`, `table.hpp`, `tuner.hpp`, `v07_side.hpp`, of which 4 arrived in the commits that landed while the battery ran and 5 already differed before it began. **What is not weakened is the policy identity**: the round-trip assertions and the mirror digest are computed by *running* the engine, not by comparing file hashes, and all of them round-trip. Because the hash-based check was vacuous it was replaced by an executed one, the engine was rebuilt from the tree as it stood at the end of the battery and the mirror digest recomputed with the fresh binary. It is `5f81f440fc9c272a87e87c05fecc7b74`, identical to the frozen artifact and to the binary that played every cell.

Scored-run reproducibility. Before any holdout material was spent, one training cell was checked to be bit-identical over 5 runs across three thread counts, and it was (yes). **That check was too small.** The panel contains three policies twice, so twenty-four pairs of byte-identical commands were run twice each; one pair disagreed, by one game in 12,000. That is the cross-deal agent residue of §11.9 reaching the *scored* mode, which four hundred deals was not large enough to expose. The magnitude is roughly 76× below a cell half-width and no verdict in this paper turns on it, but the earlier claim should be read as "reproducible at four hundred deals", not "reproducible".

Experiment-count correction. The phase-5 recording stated 409 scored match cells in three places; the figure was hand-typed in the reduction script rather than computed. Counting rows that carry a match object across every scored battery gives 428, and the count was corrected from 409 to 428. Nothing turned on it, it was the denominator of an observation about a single action-limit hit. The generator for this paper computes it, and the phase-5 recording has since been corrected to agree.

Partner-comparison uncertainty. The phase-5 results document printed the partner deltas as bare point estimates with a replication flag. Every underlying cell carries a deal-clustered interval, and the delta arithmetic is fixed by the protocol, so the intervals were computable and had simply been compressed out of a markdown table. Table 16 carries them, and the phase-5 recording has since been corrected to carry them too, at which point the row that decides S1 turned out to

have an interval crossing its own threshold (§11.5). The recorded verdict is unchanged; the added intervals expose uncertainty not visible in the original point-estimate table.

Cross-play sign convention. The phase-5 document reports the gap as −0.01, which is diagonal minus off-diagonal. The claim it settles asks whether the *off*-diagonal collapses, so the informative direction is off-diagonal minus diagonal, which is +0.01. *Neither sign is resolved*: the magnitude is 0.01 pp against a per-cell half-width of 0.63 and a 1.50-point threshold, so no verdict moves and this paper reads no lift out of it either. What the correction is worth is only this: the sign as printed *implies* a small collapse, and the data support no direction at all.

Bibliographic metadata corrections. Every one of the seventy-one references this paper cites was checked against its source rather than against the earlier cycle's bibliography, which is where all of them were inherited from, and eight carried a metadata error. Three author lists were wrong: one entry carried the adjacent entry's five names where the paper it cites has four, one printed nine names from a later paper by overlapping authors where the cited preprint has seven, and one added an author the cited work does not have. The others were a repository start year a year early, a page range, the title of a containing work, a web article's site printed as its author, and a posting year. All eight are corrected in this document's reference list and none of them touches a claim in the text. This revision additionally moved evaluative annotations out of five entries and split two entries that referred to distinct documents; the reference list's header records those edits.

Rule-variant residual uncertainty. The phase-5 document forms the residual's half-width over five rule-variant cells with unit coefficients, one of which is the `--maxasks=360` row. That row is bit-identical to the default — the same win rate and the same interval on both banks — so it carries no variance, and its own document says so two lines later. Excluding it, and giving the default row the coefficient the two-component residual actually gives it, the interval is $[-1.55, +0.96]$ rather than the recorded $[-1.95, +1.37]$. This paper prints the narrower of the two. The two documents also express the width differently, this one as the full interval over the residual and the recording as the half-width over it, so their multiples are not comparable as printed and are omitted here. No verdict moves: S5 is decided by the largest excursion, and the residual is consistent with zero under either. Neither figure is a true 95% interval, because all the cells share banks and deals and the ask-cap identity proves the correlation is large.

Calibration-rung correspondence. The preregistration introduces B9.3's sub-floor rung as "`hstr=0.05`, the rung phase 2 measured at +0.88" and directs the phase-5 excess to be read against phase 2's −0.01 $[-0.45, +0.42]$. Phase 2 measured `hstr=0.05` at a planted cost of +0.67 and an excess of +0.25 $[-0.19, +0.68]$; +0.88 and −0.01 are its `hstr=0.08` row, and phase 1 pins the same split independently. The phase-5 recording inherited the pairing. §11.9 quotes the rung-matched figure. The verdict does not move — the control is whether the planted edge is recovered, and it is not on either pairing — but the margin it is compared against does, from seven tenths of a point to a little under half a point.

Live-ask filtering statistics. The phase-2 mechanism characterisation captures the *target* arm, so every rate in its table is measured on v0.6. Its rate column comes from a 4,000-deal mirror cell and its `n` column from the 3,000-deal treatment cell, so the pair 0.00994 over 260,987 decisions is not any one cell. The quantity also does not carry over: the gate acts on the argmax of an ask score that SESTINA v1.0 extends with a coordinate v0.6 does not have, so the rate at which it displaces that argmax is a different number. §6.1 now quotes a capture of the frozen configuration from its own seats on the same diagnostic bank, which binds at 1.4% of 269,423 ask decisions, half a point above the incumbent's rate.

Duplicate forced-endgame records. The second sample that refutes the one-live-half-suit universal (§A) is reported in its source as 118 of 120 forced declarations. That source's own preceding section establishes that a mirror run at `--rotations=6` plays identical policies in both team orientations and so records every declaration twice, and re-runs this very seed at `--rotations=3` for 60. The distinct evidence is one occasion in 60, not two in 120. The rate is unchanged; the sample size is halved. The same doubling applies to the earlier run quoted in the same sentence, which is why it is described there as records rather than as occasions.

Unsupported adversarial-cost estimate. The phase-2 document, closing the forced-endgame induction route, states that the construction which raises the *target's* incidence "was measured at −17 points". No cell in the results tree produces it: no recorded configuration string describes that construction, and the only measurement near −17 in the cycle is an unrelated belief ablation. What was measured is the other construction, a team that never declares voluntarily, which raises its own incidence and not its opponent's, at 6 to 8 pp on two independent cells. §9.4.5 now says that, and says that the route is untested rather than closed by a price; the phase-2 document still carries the unsupported figure.

Source attribution and leaf-evaluation metrics. The 242× cost of the unrestricted search is phase 1 of this cycle's own timing, produced expressly to replace the v0.6 paper's mixed-basis "three orders of magnitude"; the protocol presented it as an inherited record that this cycle corrected. And the leaf evaluator's 74-fold gain in between-candidate R^2 is an in-sample gain, described in its source as replicating out of sample; out of sample the same pair is 0.00664 against 0.08424, so the direction replicates and the multiple does not. Neither error changes a verdict: the cost figure is corrected again by §11.10 in any case, and the leaf's conversion failure is what kills it.

E.3 Cross-version comparability

Two things about this cycle's evidence standard differ from earlier cycles', and a reader comparing across versions should have them. Every claim here is measured on material sealed before the configuration existed, under a protocol committed before any of it was played, with a stated failure condition, which no previous study in this corpus did. And every claim is replicated across two disjoint banks with the sign rule of §7.5 applied, which is why §11.2.2 reports a non-replication that a pooled-interval convention would have reported as a pass.

F Protocol deviations and amendments

The protocol requires a deviation for anything the evaluation did that it does not specify, recorded next to the affected result rather than in a footnote. There are 27. This register is read out of `docs/v07/FINAL-RESULTS.md` by the generator rather than retyped, so it cannot drift from the results document it reports; long entries are elided with an ellipsis and the source document carries them in full. Entries are printed in the order of the source register, in which D24 follows D27; the record does not state why, and the ordering is preserved rather than corrected here.

ID	Kind	Deviation, addition or correction
D1	Added Cell	B5 REF-FROZEN. The eleven cells B5 names cannot yield a leave-one-out DROP: the drop is FROZEN's edge minus the variant's, and FROZEN's own edge on the lattice banks is not one of the eleven. A twelfth cell was added at the lattice size on both lattice banks. B5 therefore cost 288,000 games rather than the preregistered 264,000.
D2	Added Cells	B9 planted cost and responder-vs-unhandicapped. B9.2 requires that "the recovered size tracks the planted size", which needs the planted size; B9.3 speaks of the planted excess "resolving TO zero", which is the responder's edge against the handicapped target MINUS its edge against the unhandicapped one. Four rungs x 2 banks x 6,000 deals were added for each, 192,000 games in total.
D3	Unspecified Bank	B8. Section 2.1 assigns 7090003 the role "the ablation lattice and the dialect table" but names no replicate for the dialect rows. 7090001 was used, by symmetry with B5, which the protocol does specify as 7090003 primary and 7090001 replicate.
D4	Unspecified Bank	B3 sealed rows. Section 2.1 gives 7091001 the role "evaluation bank for the sealed adversary half", but section 4/B3 puts every panel member on the same two banks as every other arm, and section 3 makes 7090001/7090002 the primary and replicate for every claim. The sealed rows were run on 7090001/7090002 with the rest of the panel, so the whole panel is scored on one shared pair of banks, as minimax regret over a shared panel requires. 7091001 was . . .
D5	Protocol Note	what `tune --seed=<bank> --shard=s/4` actually plays. Section 4/B4 describes the eight searches as drawing "480 deal indices of the bank". They do not. `tune` uses `--seed` only as the CEM root: it derives a fresh per-generation seed (`tuner.hpp:310`) and from that a per-opponent match seed (`tuner.hpp:258`), so the deals are a derived stream and the named bank's own deals are never constructed. Banks 7090004, 7090005 and 7091002 are therefore . . .
D6	Protocol Note	the seal is not enforced on every command. Section 2.1(b) records that `pathology`, `v7bits` and `v6probe` bypass `seedUsable`. `tune` and `ablate` also bypass it, for the same reason as D5: the seed `runMatch` sees is derived, not the registered one. Phase 5 set FISH_UNSEAL_PHASE=5 once and used the seven registered seeds and no others.
D7	Observation, Superseded By D19	B0.3 printed no changed-source NOTE. The protocol expected three engine sources to be listed as changed since the freeze (`arena.hpp`, `main.cpp`, `v07_side.hpp`). Zero differ from the baseline the check actually uses. D19 explains why that baseline is not the freeze's and gives the correct figure – nine of 78 sources differ from the freeze commit. This entry is left standing rather than deleted, because the first reading of B0.3 was reported before it was . . .

continued overleaf

ID	Kind	Deviation, addition or correction
D8	Added Check	reproducibility, before the material was spent. `runMatch` schedules deals by work-stealing, so the deal-to-thread assignment is not fixed run to run even at a fixed thread count, and the frozen configuration carries the cross-deal agent residue of section 5.3. A scored cell was checked to be bit-identical over three runs at 13 threads and across 1, 2 and 13 threads, on training material, before any holdout bank was played.
D9	Reading	S1's row count. Section 5.2 says "five of the eight changed-partner rows"; B6 has seven changed settings plus self-play, and the protocol's own worked table counts eight including self-play. Both counts are printed and the verdict states which reading it used.
D10	Reading	S3's pass and fail conditions are not complements. "Every arm's upper bound below 1.53" and "any arm clears 1.53 with a replicated sign" leave a band in which neither holds. Both are computed and reported; if the band is occupied the verdict is reported as UNDETERMINED rather than resolved by choosing a reading.
D11	Reading	the 2.13 declaration-family floor. Section 3 correction 4 applies it to "B4's Z03, and the declaration-accuracy column of B2". Z03 is read against 2.13. B2 has no declaration-accuracy column with an interval – `match --json` emits declaration accuracy as a bare rate – so the accuracy is printed as a diagnostic and PHASE 5 MAKES NO CLAIM ABOUT THE DECLARATION CHANNEL, which is what discharges the rule.
D12	Arithmetic	S4's naive sum. The freeze carries five components; `m2=0` was dropped from it. The sum S4 tests is over those five. The six-term sum including A-m2 is printed beside it and labelled, because B5 preregisters the A-m2 cell even though the freeze does not carry the key.
D13	Note	every delta in this document is two independent intervals combined in quadrature, not a paired difference. The harness gives a paired delta only within one `ablate` invocation over a shared panel, and no cell of this battery is of that form. Section 5.2 fixes this arithmetic and calls it conservative.
D14	Note	`ablate` was not used anywhere. It derives its own per-opponent seeds by `mixSeed(seed, i*7919+3)` (`main.cpp:589`), so `ablate --seed=7090003` would not play bank 7090003. Every lattice cell is a `match` cell on the named bank instead.
D15	Note	the panel of 31 NAMED members contains three policies twice: R-v04 is v04, S-archetype-0 is feint, S-reference-0 is v06. The duplicated cells are run as named and used as a determinism check.
D16	Note	Z08 fits 55 coordinates of which 18 are inert. The protocol specifies `--base=v07i` for the white-box class; `v07i` derives from V06Agent and reads only the 37 v0.6 coordinates (`factory.hpp`), while the CEM extends the vector to 55 for any v07* base. The search is over 37 live coordinates plus the inverter the base fixes.

continued overleaf

ID	Kind	Deviation, addition or correction
D17	Note	thread counts. Every scored cell ran at `--threads=13`, which section 5.3(2) fixes for the preregistered batteries. The S6 gate condition ran at `--threads=1 --freshagents`, and the S6 residual column at `--threads=1` alone.
D18	Correction To D8	a scored cell is NOT bit-reproducible for a searching arm at panel sizes. The B0 check found five runs of a 400-deal cell bit-identical. The panel-duplicate check, on 1,500- and 6,000-deal cells, found one pair of identical commands disagreeing by one game in 12,000 – the cross-deal agent residue of section 5.3 reaching the scored mode. The magnitude is 0.008 points against a 0.63-point half-width. Reported in full under "Panel duplicates" above.
D19	Correction To D7	B0.3's changed-source NOTE could not fire, and reporting zero as "conservative" was wrong. `freeze_config_v07.py --verify-only` reads its drift baseline out of `engine/fishbot_v07.json`'s own `provenance.srcSha256_16`. That artifact was REWRITTEN after the freeze commit 0fa4a5f – at d1c6b35 and again at d8c554b – and each rewrite refreshes the baseline to the then-current tree. B0.3 therefore compared the tree against a baseline taken from the . . .
D20	Disclosure	five commits landed in the repository DURING the battery, and four engine sources now differ even from the refreshed baseline: `fish.hpp`, `game.hpp`, `human.hpp`, `table.hpp`, from the "web play" commits b8cb227, 2e829c2, 46f3515, 72fa936 and e0da8fd. They touch the interactive browser table, not the policy. The decisive fact is that engine/fish7 was NEVER REBUILT: sha256 `cf6d5ea2c1f0e9e3896b`..., mtime 2026-08-25 13:19, identical at the start of B0 and at the . . .
D21	Ordering	B0 was performed before anything else, but its ARTIFACT was written later. The five B0 checks were run interactively at the top of the session, before the commit gate and before any holdout deal was played. `engine/p5_b0.py`, which re-runs all five and records them as `P5-B0.json`, was written and run about fifteen minutes into B2. Every B0 check is deterministic and reproduced identically, so the artifact records what was checked first – but its file . . .
D22	Observation	one action-limit hit in 428 scored cells: the phase-2 composite against SEALED:S-ask-1 on bank 7090001, one game in 12,000 (limitHitRate 8.33e-05). G4 gates the mirror pathology run and not scored cells, so this is not a gate violation, but the corpus standard is zero and it is recorded. NOTE that scored cells carry NO audit evidence: `auditChecks` is 0 in all 428 of them, because `match` runs the audit only under `--audit` and no preregistered cell . . .
D23	Correction	the throughput columns were printed with median_high (`v[len(v)//2]` on an even-length list) rather than the median. Corrected to the true median; every figure moves by at most 0.11x and every one still exceeds the 3.2x the protocol records.

continued overleaf

ID	Kind	Deviation, addition or correction
D25	Reading	section 6 item 3 read per bank. The protocol states the one-third location rule without a per-bank clause, and section 3's replication rule is written for "a claim whose SIGN does not replicate". Here the sign of the drop does not change between banks; the VERDICT LABEL does – LOCATED on 7090001, NOT LOCATED on 7090003. Applying section 3's rule to a threshold crossing rather than to a sign is an extension, and section 7 requires a deviation for . . .
D26	Substitution	B11's third clause is not computable as written. It asks phase 5 to confirm "that the holdout estimate is not systematically smaller than the training estimate by about that amount", where the estimate B11 is about is the measured holdout gain – but the protocol supplies no training figure for B2.1 or B2.2. Phase 5 substituted the three quantities the protocol DOES state in advance (the B2.4 group at +0.78, the B6 self-play row at +2.94, and B7's . . .
D27	Added Artifact	`research/v07/results/P5-drift.json`. D19's executed replacement for the vacuous hash-based drift check was performed but not initially recorded, which made it an assertion rather than evidence. It is now an artifact: the build command, both binaries' SHA-256 and byte counts, the pathology argv, the two recomputed digests, and the source sets differing from the artifact's own baseline and from the freeze commit.
D24	Correction	per-bank columns were printed unlabelled and sorted by seed NUMBER, which inverts the attribution for B5 and B8, whose primary bank is 7090003 and whose replicate is 7090001. Every per-bank column in `P5-TABLES.txt` is now printed as `seed:value`. In THIS document the bank order is given in each table's first row and the pairs that follow keep it: section 5's B4 table is (7090001, 7090002) and section 9's B8 table is (7090003, 7090001). A reader . . .

G Reproducibility and artifact availability

Availability

The engine, every experiment artifact, the protocol, the generator that produces this paper's numbers and the provenance checker that audits them are in the project repository, `https://github.com/dylann29/fish-optimization`; the release corresponding to this paper is the annotated tag `sestina-v1.0`. Three identities are kept separate below: the frozen policy configuration, which is the specification string and parameter vector of Appendix B together with the mirror digest `5f81f440fc9c272a87e87c05fecc7b74`; the evaluation binary, identified by its SHA-256; and the repository tree, identified by commits that changed during the run. The evaluation of §11 ran at the commit whose current short hash is `4d8cec9`; the full hashes of every commit named in this appendix resolve in the repository. Two caveats attach to that hash and the subsections below give both: the repository's history was rewritten after the evaluation, and no single commit describes the tree for the whole run in any case.

Repository-history changes

After the evaluation was complete, one artifact of the development phase, a per-decision search capture of 139,267,276 bytes, exceeded the hosting limit and was excised from version-control history so the repository could be published. That rewrote every commit from phase 3 onward.

Three consequences, stated because a reader checking this paper against the repository will otherwise be misled. First, **the commit hashes recorded inside the phase-5 artifacts no longer resolve**: the verification block records `d920f5b` and the freeze artifact's drift check keys on `0fa4a5f`, and neither object exists in the published history. Every hash printed in this paper is the post-rewrite one, recovered by matching commit subject lines, so the freeze commit is `ed321cc` and the evaluation commit is `4d8cec9`. Second, the seal predates the rewrite and is unaffected: the commit at which the holdout material was sealed, `f4581da`, still resolves, so the commitment chain of §7.6 is intact and independently checkable. Third, the excised file is retained on disk and its digest recorded beside it in `research/v07/results/K5-capture-7030004.sha256`, SHA-256 `bb655aac7cd413d2...`; it is regenerable from the capture harness and no measurement in this paper depends on it.

Evaluation binary and source provenance

Five commits landed in the repository while the battery was running, so no single commit describes the tree for the whole of it, and a rewrite has since changed all of their hashes. Every one of the 428 scored cells was played by `engine/fish7`, SHA-256 `cf6d5ea2c1f0e9e3896bc51f2a366834e7d60303eaf85acc553346c97359f4e3`, 1,242,312 bytes, and that binary was never rebuilt between the first bank digest and the last cell. The five later commits touch the interactive browser table and not the policy; §E.2 records the executed check, rebuilding the engine from the tree as it stood at the end of the battery and recomputing the mirror digest, that establishes the frozen policy is unmoved by them for the tested mirror, which is evidence for the tested cases rather than a proof of global program equivalence.

Computing environment

The scored battery ran on one Apple M5 Pro desktop with 15 logical cores under macOS, with every scored cell at 13 worker threads (Appendix F, D17) and the isolation audits at one; the throughput figure of §2.3 was taken on the same machine at fifteen threads. The engine was built with `clang++` at `-std=c++20 -O3 -march=native -funroll-loops -fno-math-errno`, the evaluation build adding `-DFISH_NO_SERVE`. The compiler version and whether the machine was otherwise loaded during the battery are not recorded, and the cost figures of §11.10 are wall-clock ratios taken under those conditions.

Experiment inventory

The registered batteries and the cells measured for each are given in Table 7 (§10.5); deviations D1 and D2 add a B5 reference cell and sixteen B9 cells, because without them "no single leave-one-out drop exceeds the whole", "the recovered size tracks the planted size" and "the planted excess resolves to zero" are not computable from the cells the protocol lists. No cell was dropped.

Raw evaluation artifacts

Under `research/v07/results/`, digested in `research/v07/results/MANIFEST-P5.json`: 29 artifacts, of which 29 re-verify against their recorded SHA-256 and byte count on disk at the time this document was generated.

- `P5-B0.json`, the verification block: seven bank digests, the sealed plaintext hash, the freeze round-trips, the seed registry, the engine self-audit, and the reproducibility runs.
- `P5-gate.jsonl`, the commit gate for all four configurations, rule by rule.
- `P5-B2.jsonl` ... `P5-B9.jsonl`, one JSON row per cell, each carrying the literal argument vector it was produced by and the complete match object the harness emitted.
- `P5-B10.jsonl`, one row per audited cell. These are seat-isolation audits rather than matches and carry a `side` object with the mismatch count and its denominator, not a match object.
- `P5-Z01.jsonl` ... `P5-Z08.jsonl`, `P5-B9-h*.jsonl`, the fitting traces: header, one record per generation, and the final weight vector each adversary spec was rebuilt from.
- `P5-TABLES.txt`, the phase-5 reduction, produced by `engine/p5_analyse.py`.

Every row carries its own argument vector, so any single cell can be re-run by hand from the artifact without reference to this document or to the driver that produced it.

Numerical generation and provenance

1. `engine/build_tables_v07.py --paper` reads the artifacts above and writes `paper/numbers_v07_generated.tex` and `paper/tables_v07/*.tex`. Every macro it emits carries a comment naming the artifact and field it came from, and re-emitting a name with a different value raises rather than silently winning.
2. `paper/check_provenance.py --version v07` audits the result: a number used in a section must either be generated from an artifact or sit under a comment header naming an artifact that exists on disk. Of the numbers the sections use, 250 are emitted by this cycle's generator, all but the registered training figures of Table 10 computed from the phase-5 artifacts and those four carried in the generator as constants of the protocol; 24 are generated from earlier cycles' artifacts by those cycles' generators; and 153 are transcribed under a header naming a source document.
3. `paper/inline.py` flattens the modular source into a single self-contained file.

Sections 2–4 quote results from this programme's earlier cycles. Those numbers are not retyped here either: the earlier cycles' generators emitted them from their own artifacts, and this document loads those generated files directly. The split is therefore 250 generated by this cycle, 24 generated by an earlier one, and 153 transcribed under a header naming the report or artifact that records them, which `paper/check_provenance.py` verifies exists on disk, and whose printed split this paragraph is required to match.

The generated file is `\input` after the placeholder file, so a value in the placeholder can never reach a PDF built after the battery has been run. Generated on 2026-09-21.

Sample sizes

The project's sample-size planning rule is a 95% half-width of $98/\sqrt{N}$ pp at parity for N independent games, computed by the harness and emitted with every cell (§7.4); it is a planning approximation, not a power calculation. At 48,000 games the primary claim's pooled half-width is 0.45 pp; the frontier cells are sized so that the primary claim's half-width is small against the 1.53-point reference

bar rather than against the effect. Far-class panel members are sized at a quarter of the near class deliberately: they are decided by 24.72 to 50.00 pp and do not need the resolution.